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(54) FUSION STRUCTURE, INFLATABLE PRODUCT HAVING THE FUSION STRUCTURE, MANUFACTURING METHOD THEREOF, AND RAPIDLY PRODUCED INFLATABLE PRODUCT HAVING COVERING LAYER SIMULTANEOUSLY POSITIONED WITH INFLATABLE BODY

(71) Applicant: **Team Worldwide Corporation**, Taipei City (TW)

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Related U.S. Application Data

(63) Continuation-in-part of application No. 16/612,353, filed on Nov. 8, 2019, filed as application No. PCT/CN2018/085922 on May 8, 2018, now Pat. No. 12,304,155.

(30) **Foreign Application Priority Data**

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(51) **Int. Cl.**

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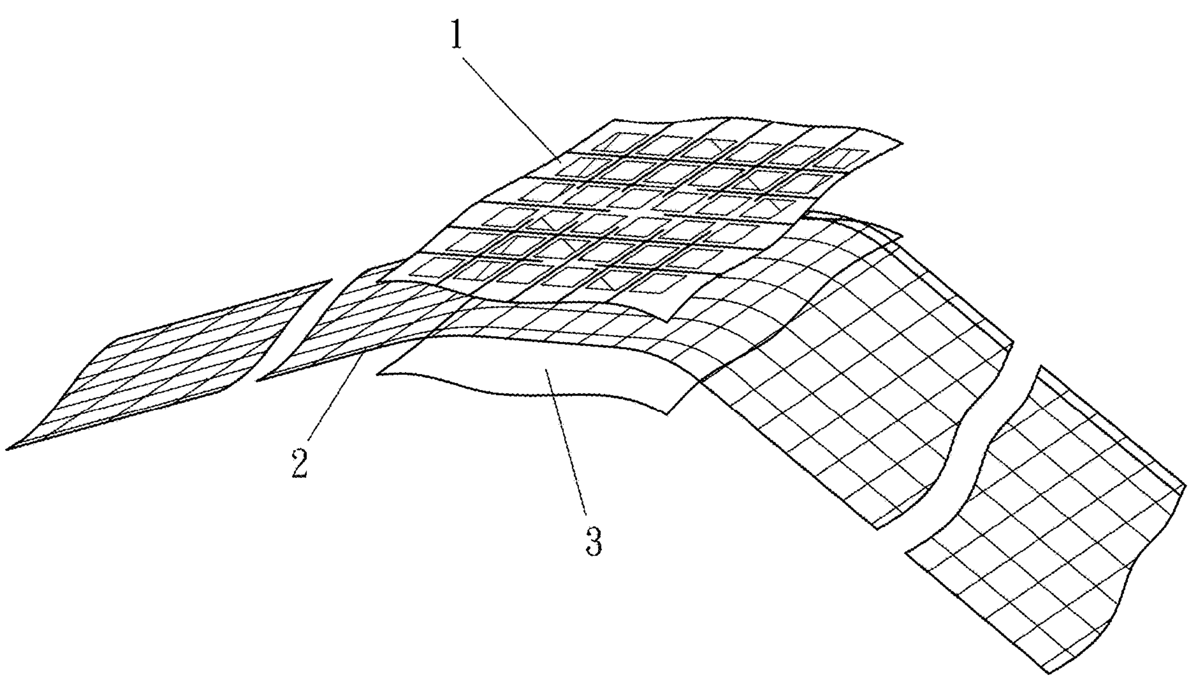
B32B 7/09	(2019.01)
B32B 27/30	(2006.01)

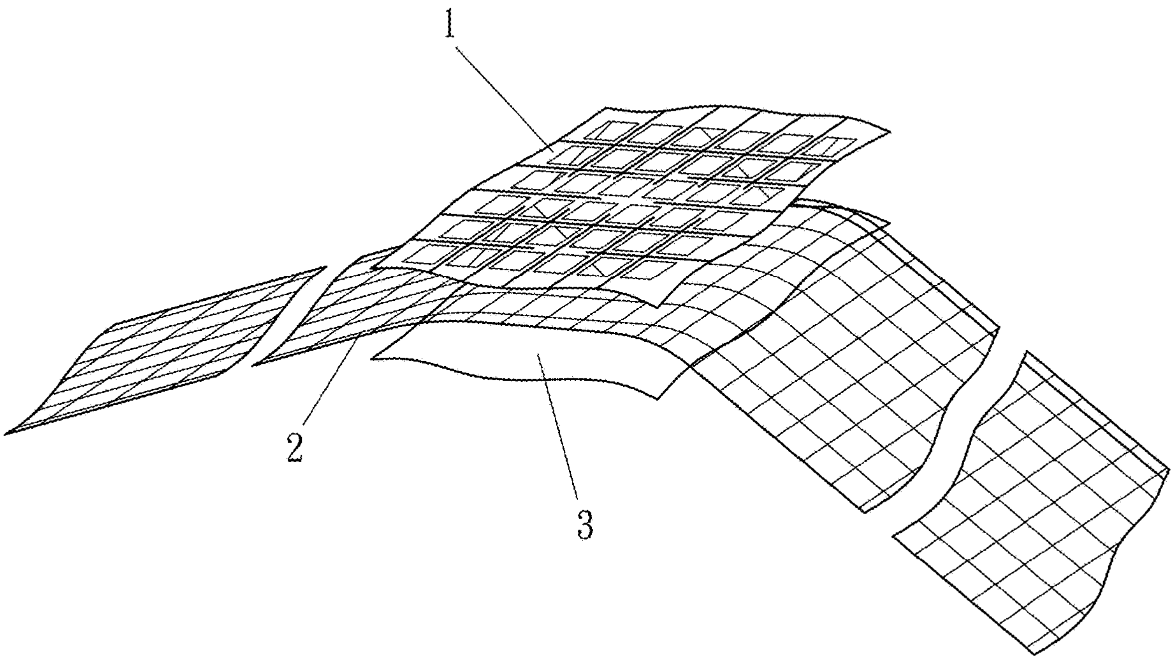
(52) **U.S. Cl.**

CPC ***A47C 27/008*** (2013.01); ***A47C 27/081*** (2013.01); ***B32B 3/266*** (2013.01); ***B32B 7/09*** (2019.01); ***B32B 27/304*** (2013.01); *B32B 2262/101* (2013.01); *B32B 2601/00* (2013.01)

(57) **ABSTRACT**

An inflatable product includes an inflatable body and a fiber-contained reinforcement element. The fiber-contained reinforcement element is connected to the inflatable body so as to increase mechanical strength of the inflatable product at a location where the fiber-contained reinforcement element is connected to the inflatable body.





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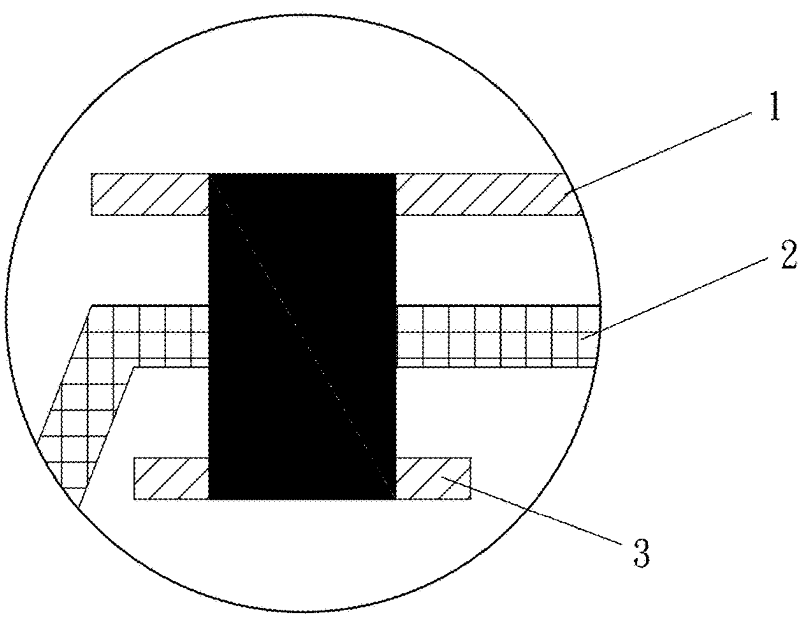


Fig. 1B

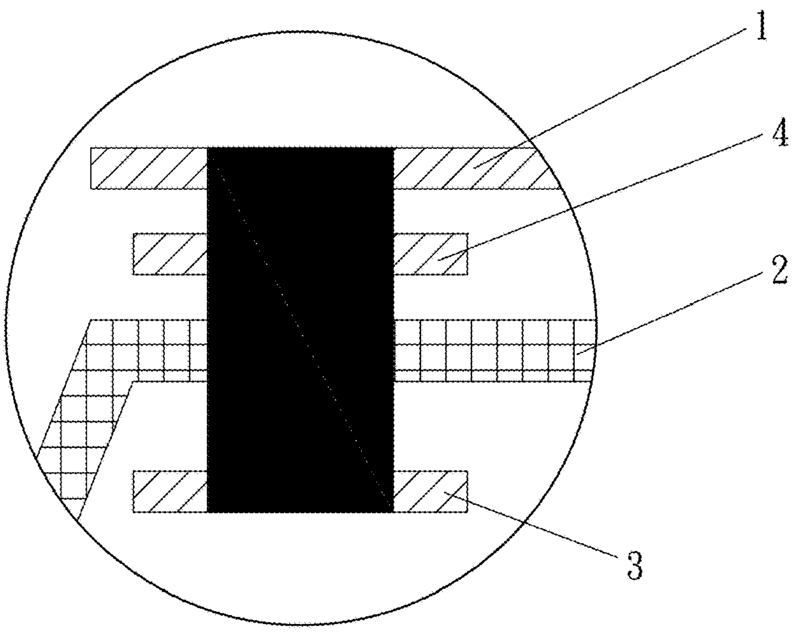
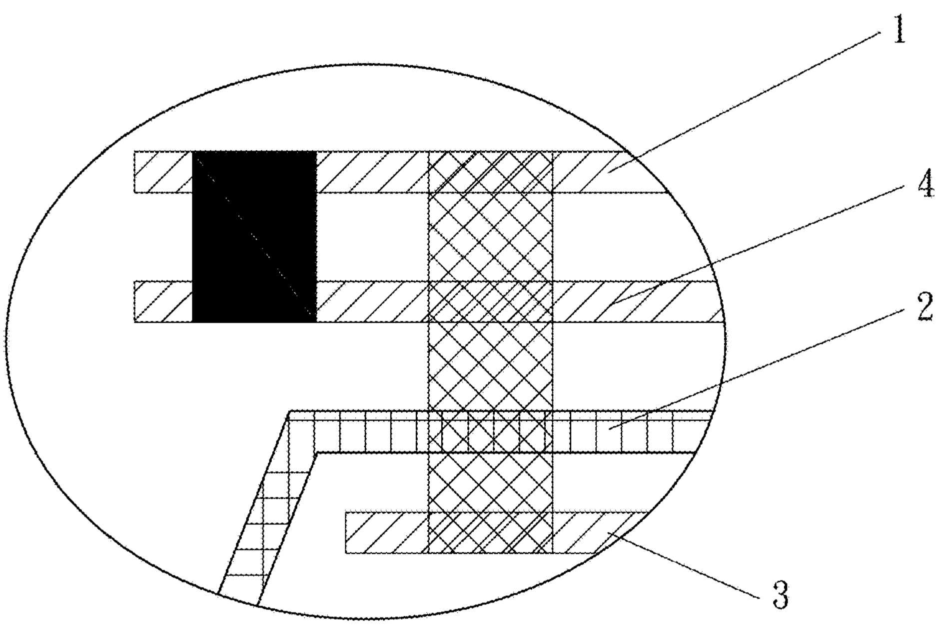


Fig. 2



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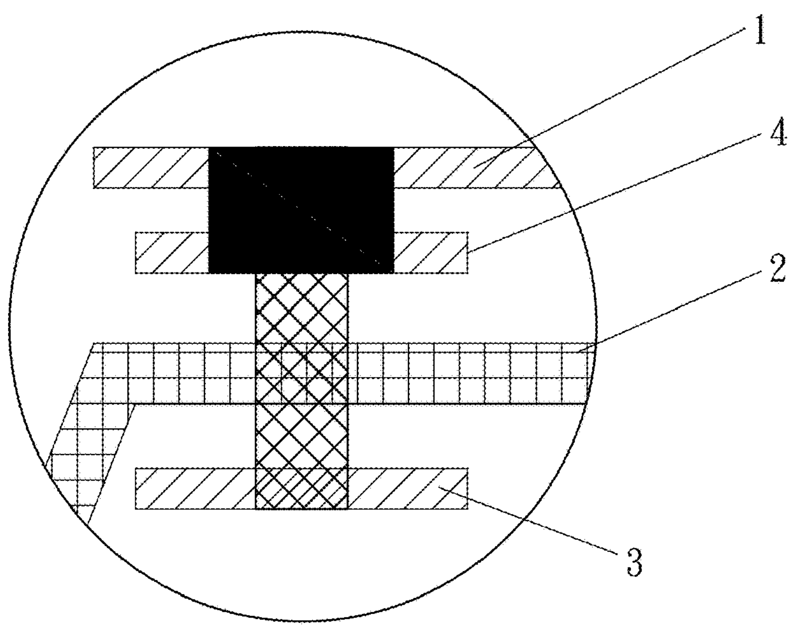
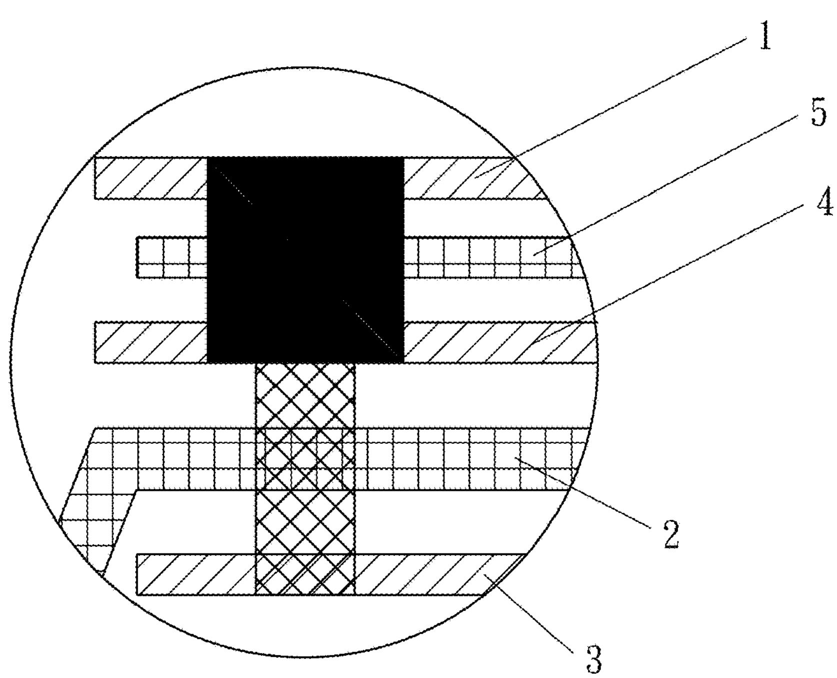


Fig. 4



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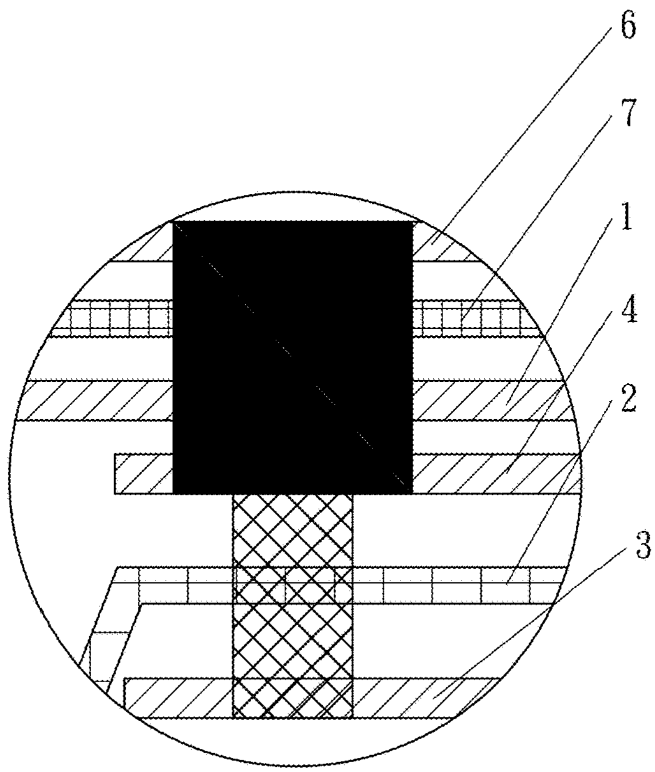

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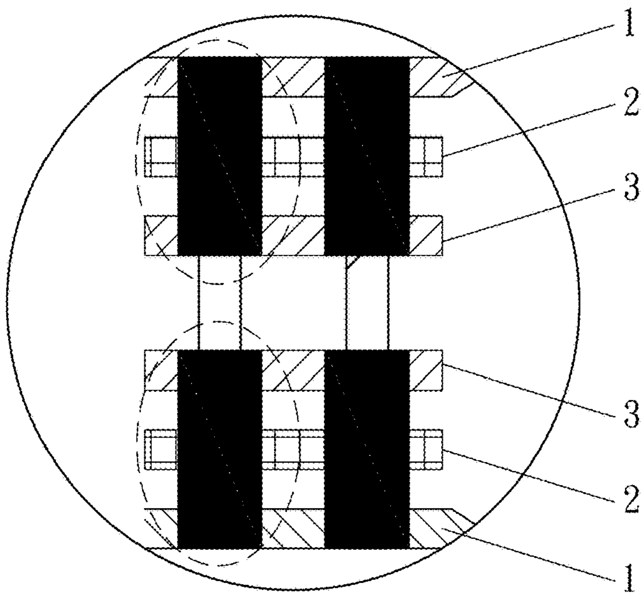

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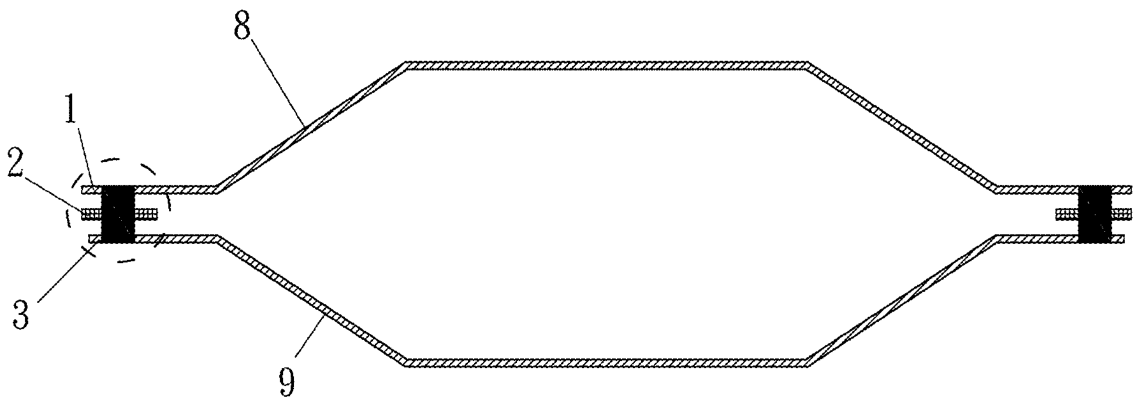
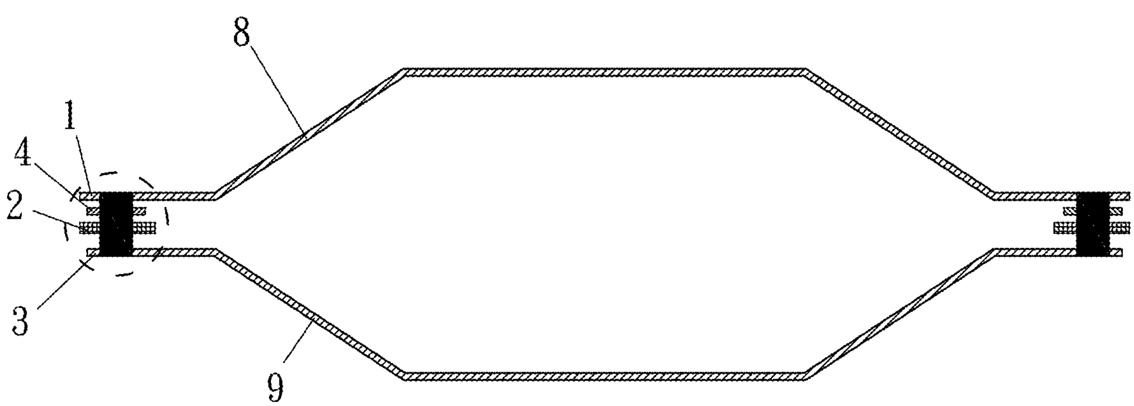


Fig. 8A



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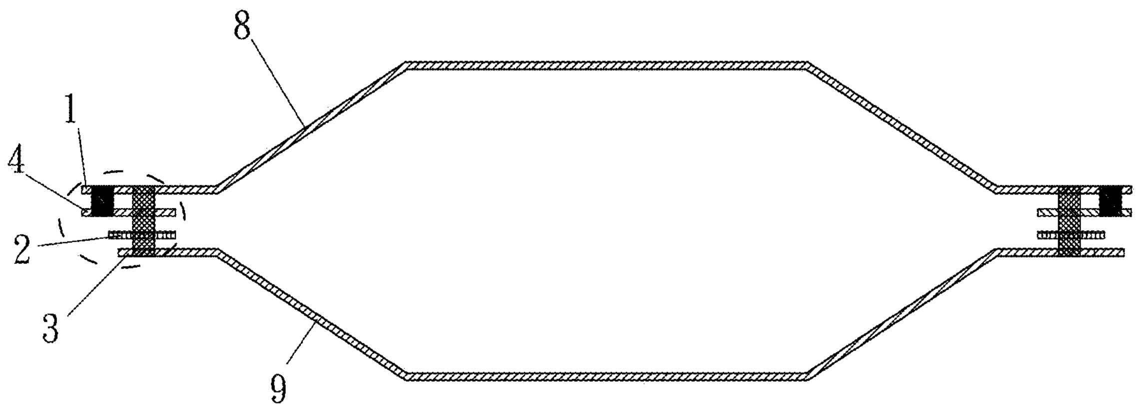

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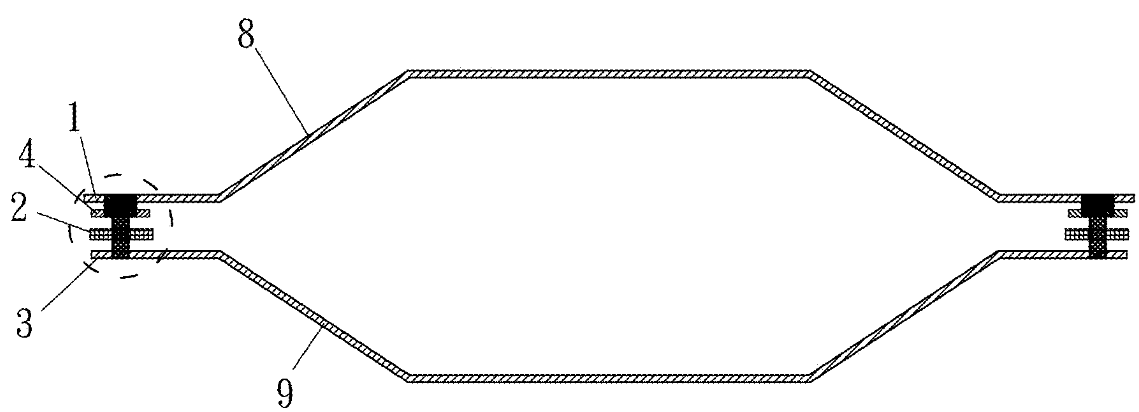

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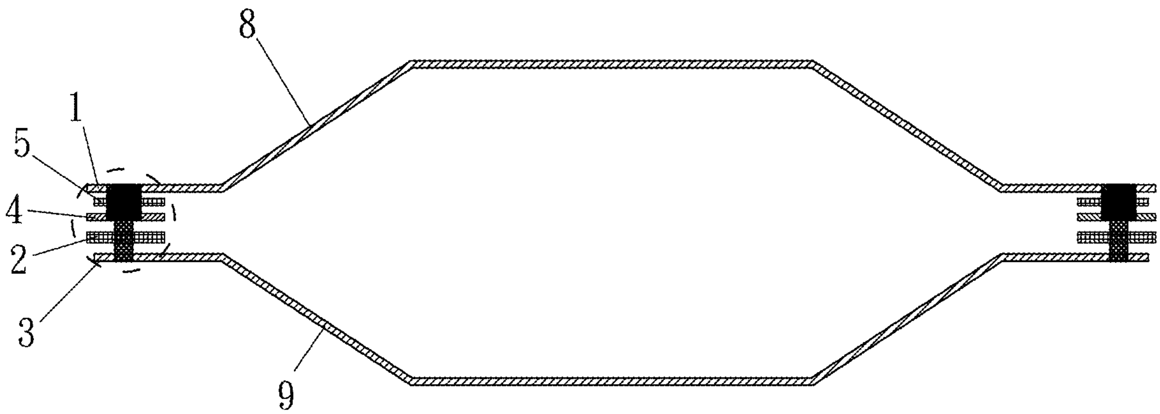

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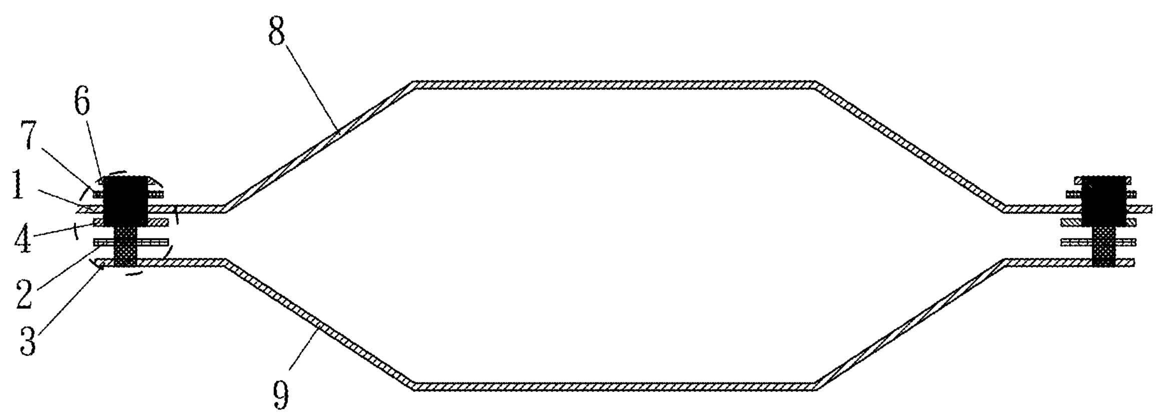

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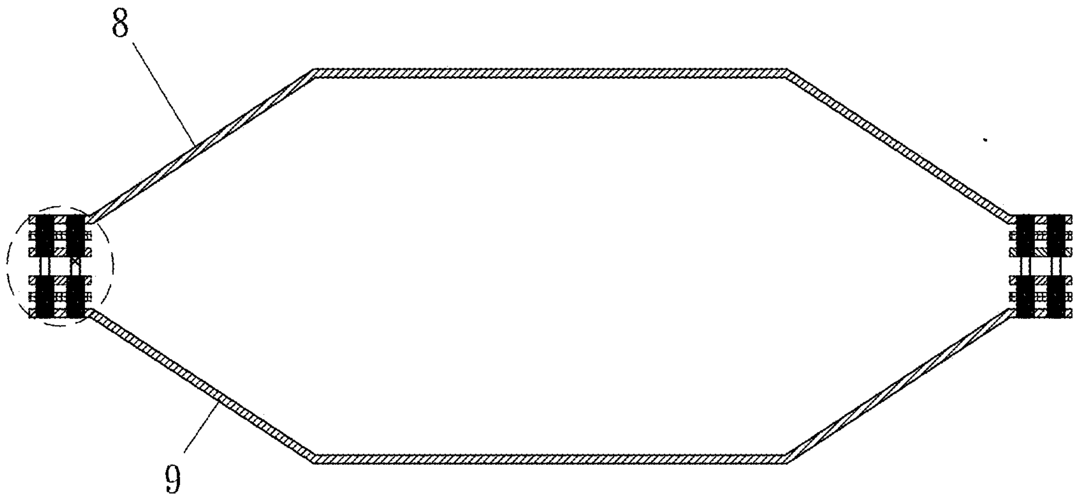

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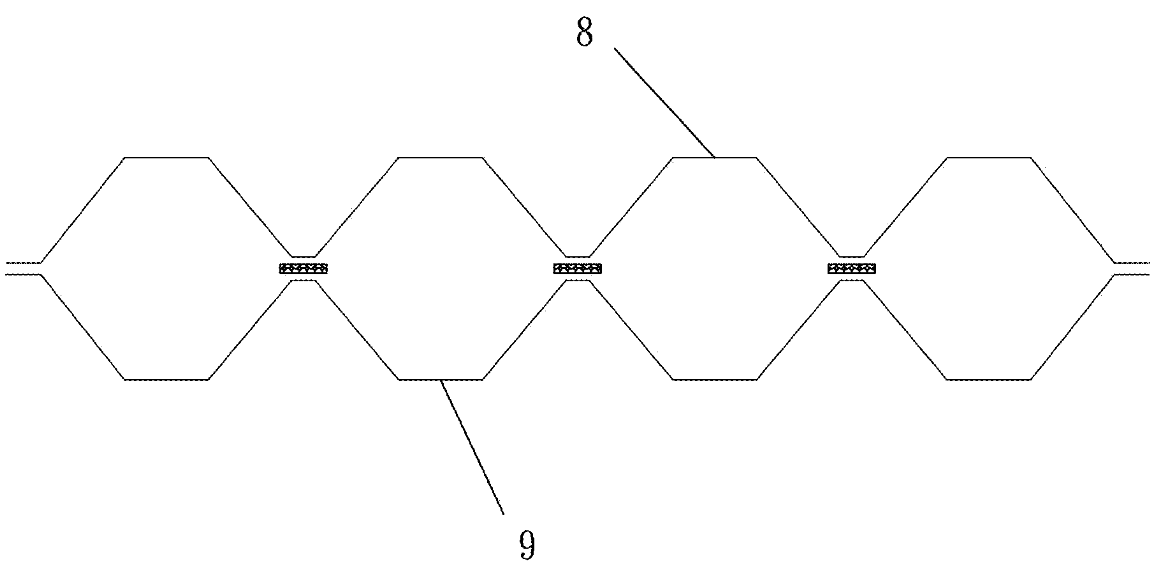

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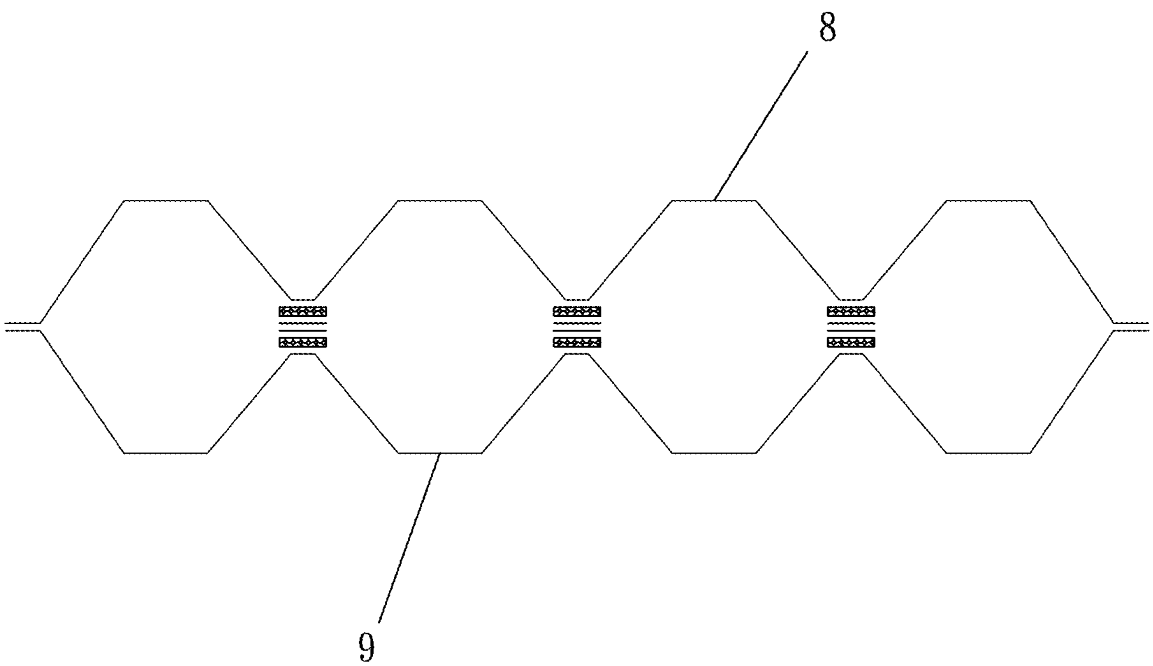

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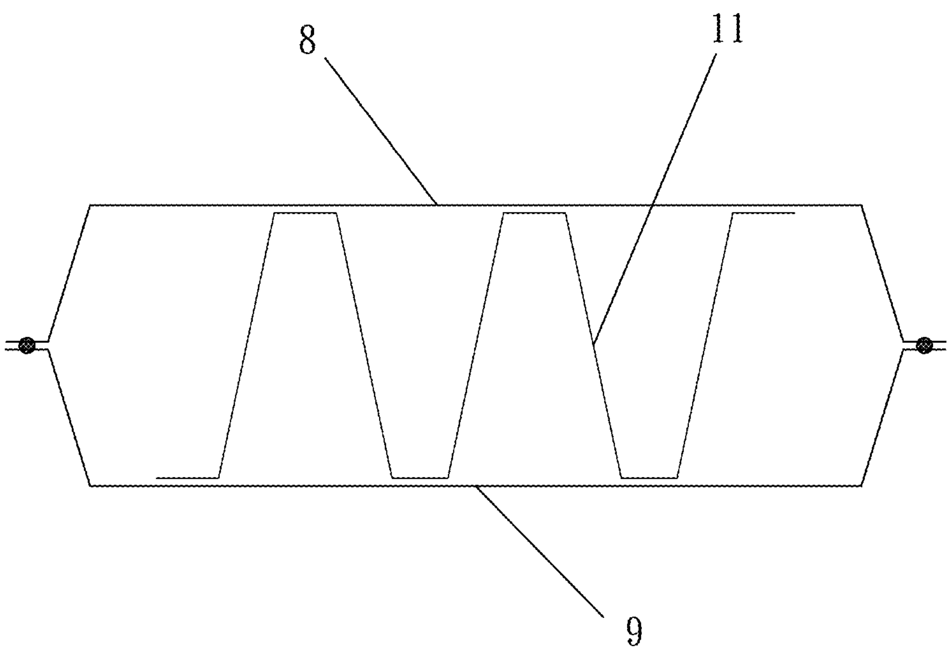

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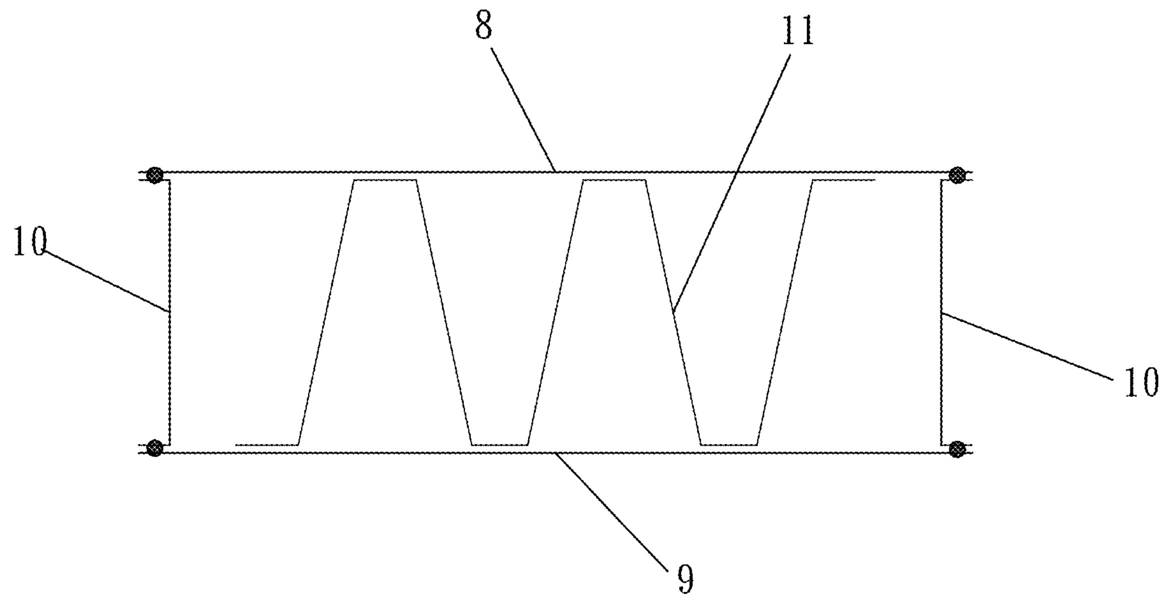

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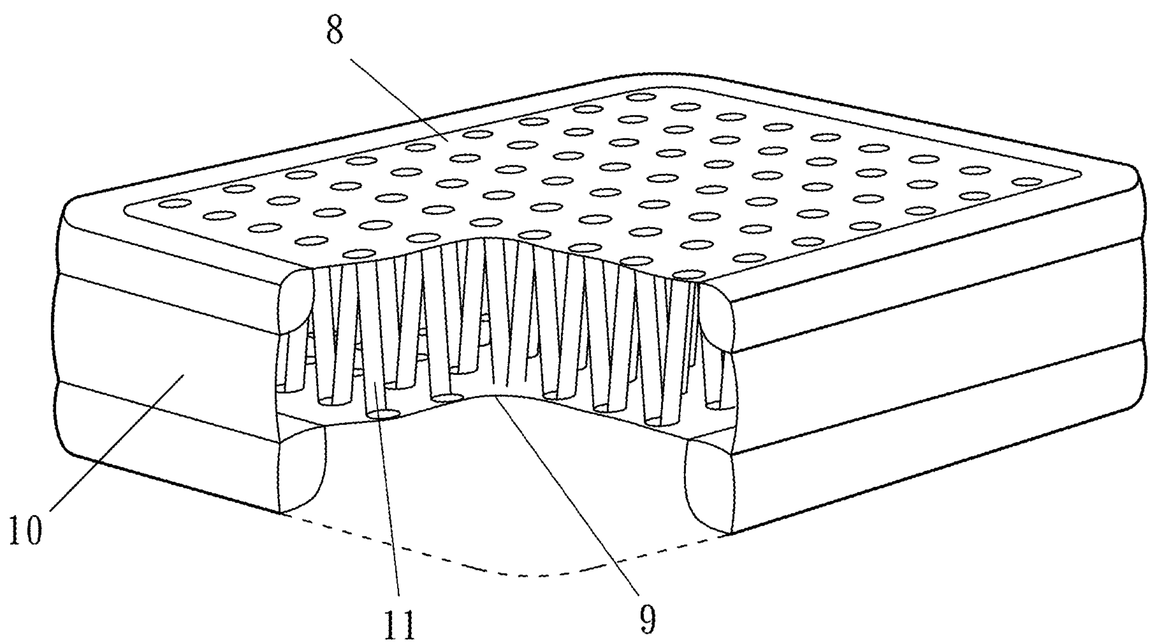

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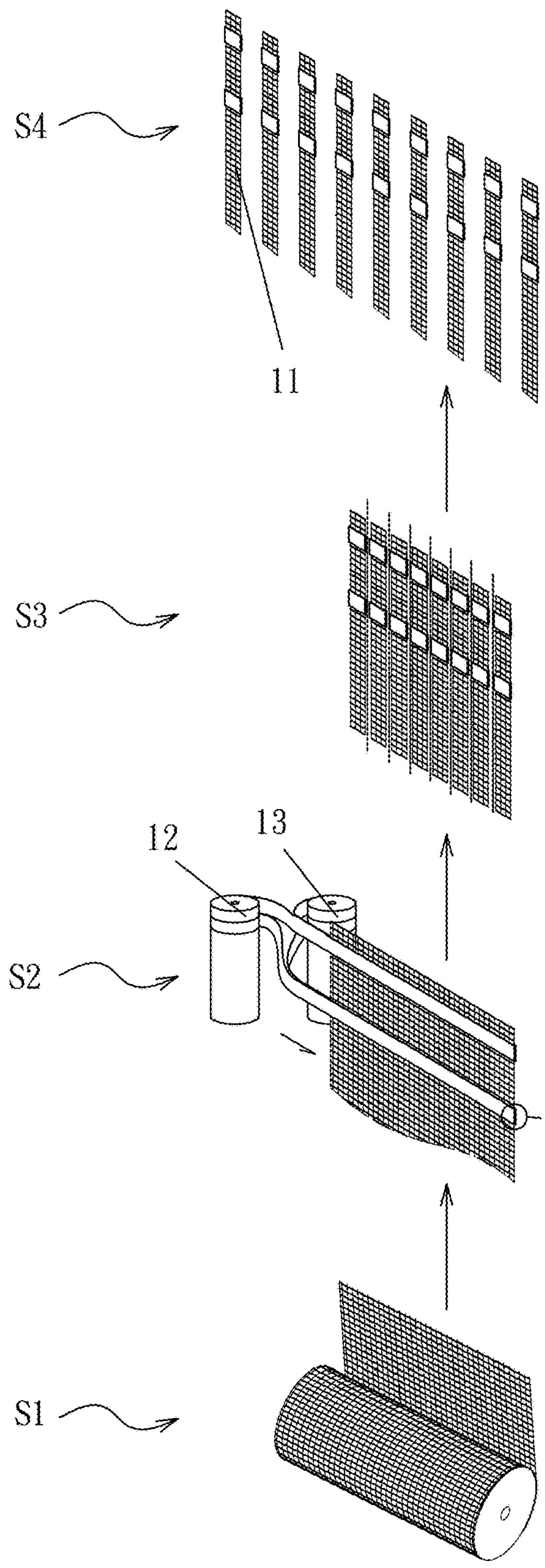

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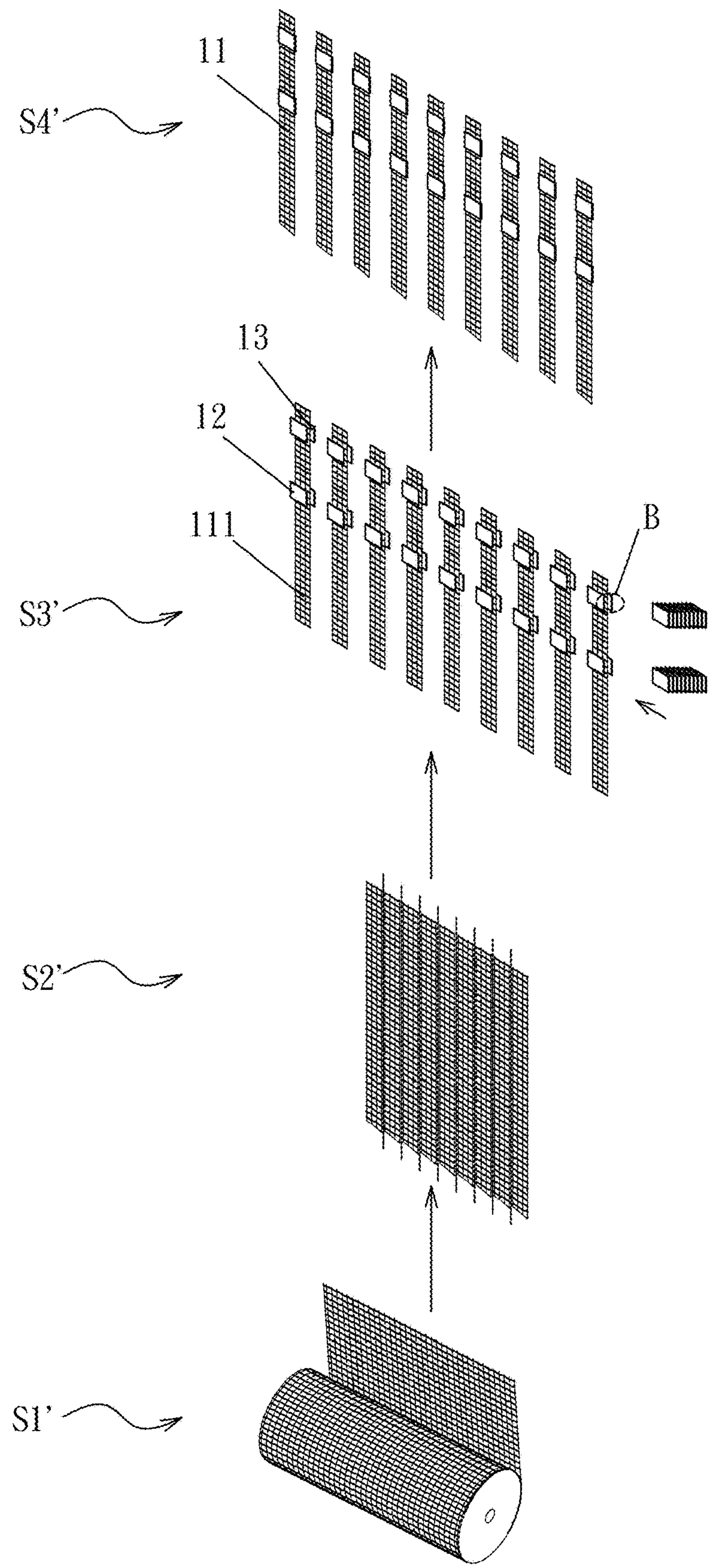

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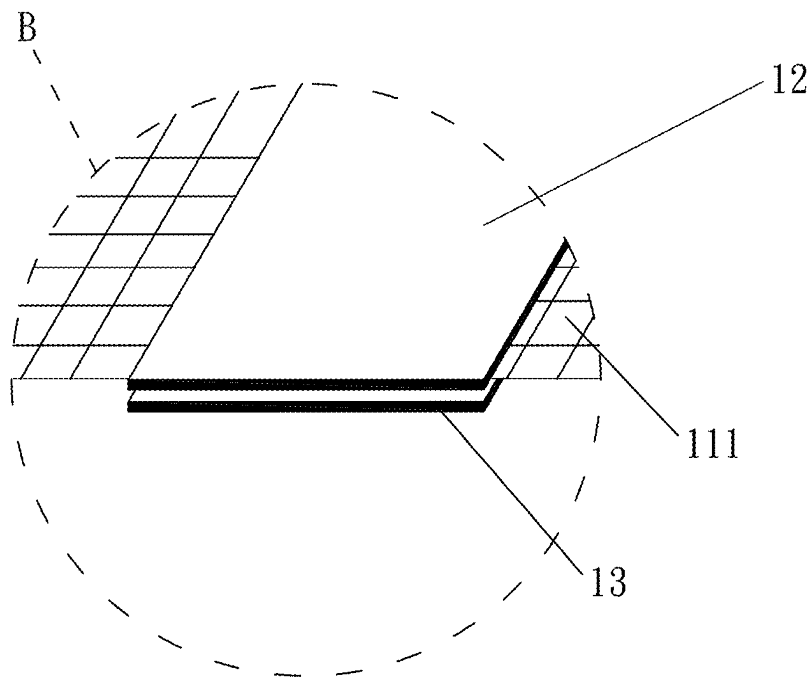

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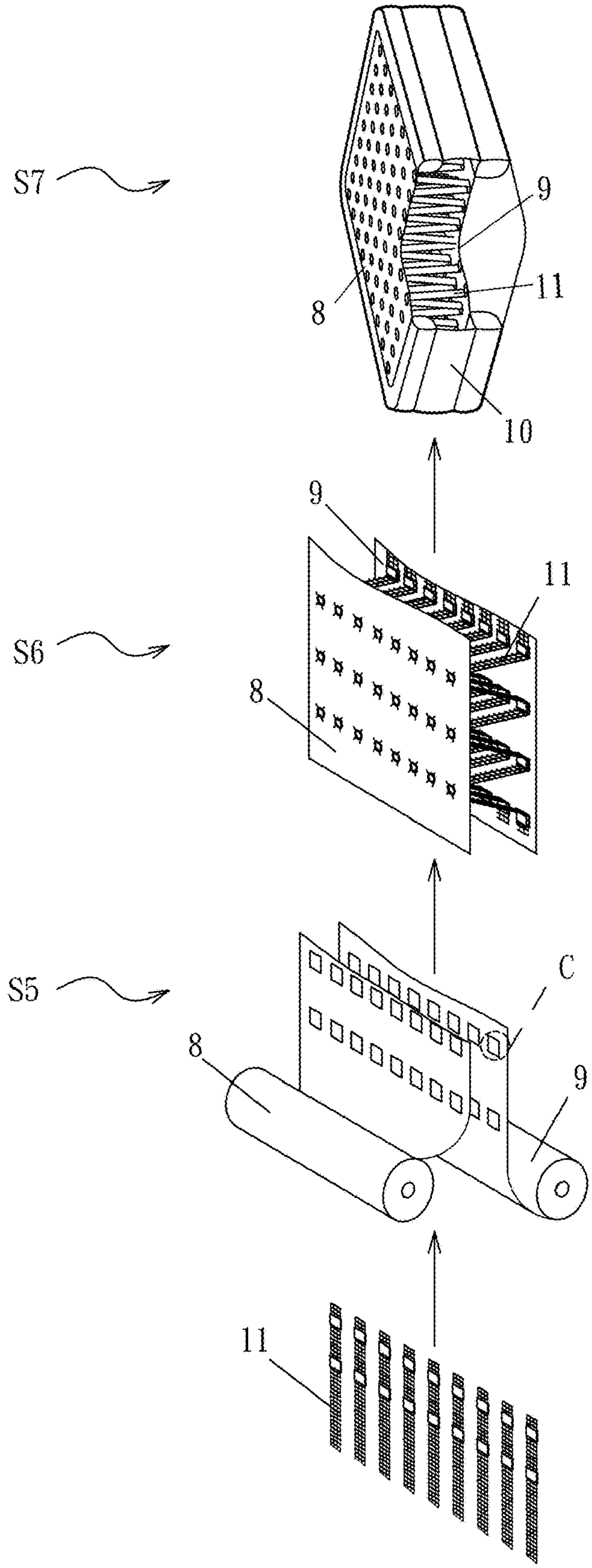

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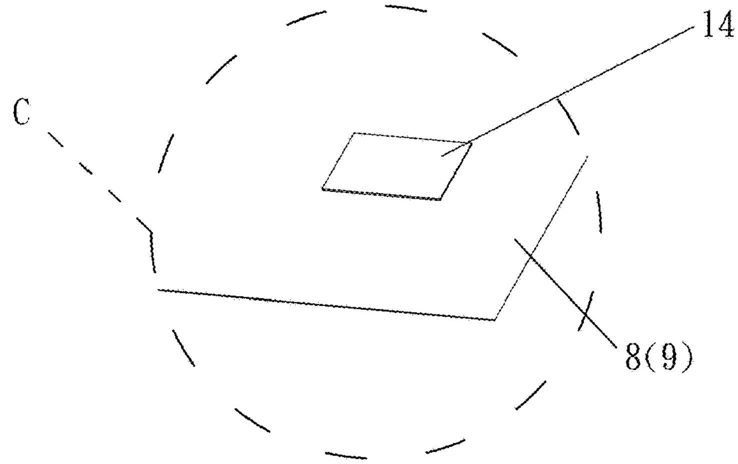

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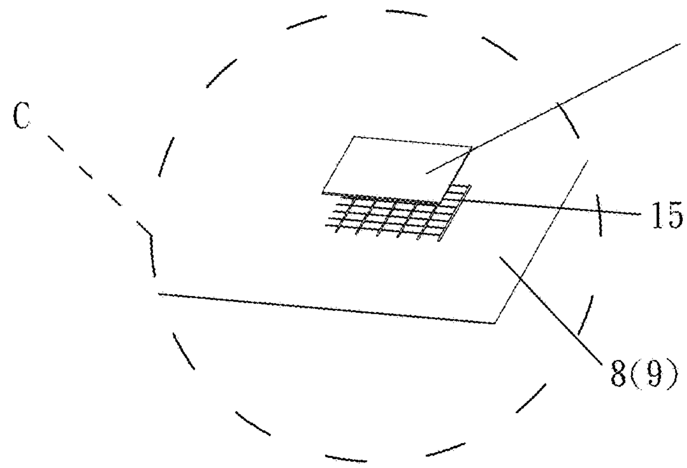

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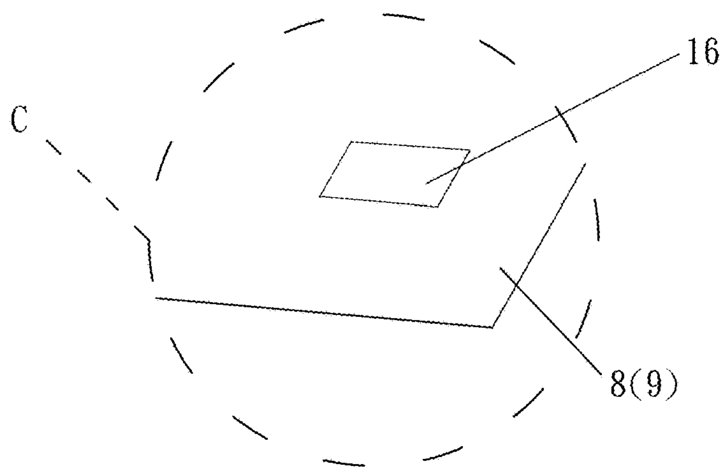

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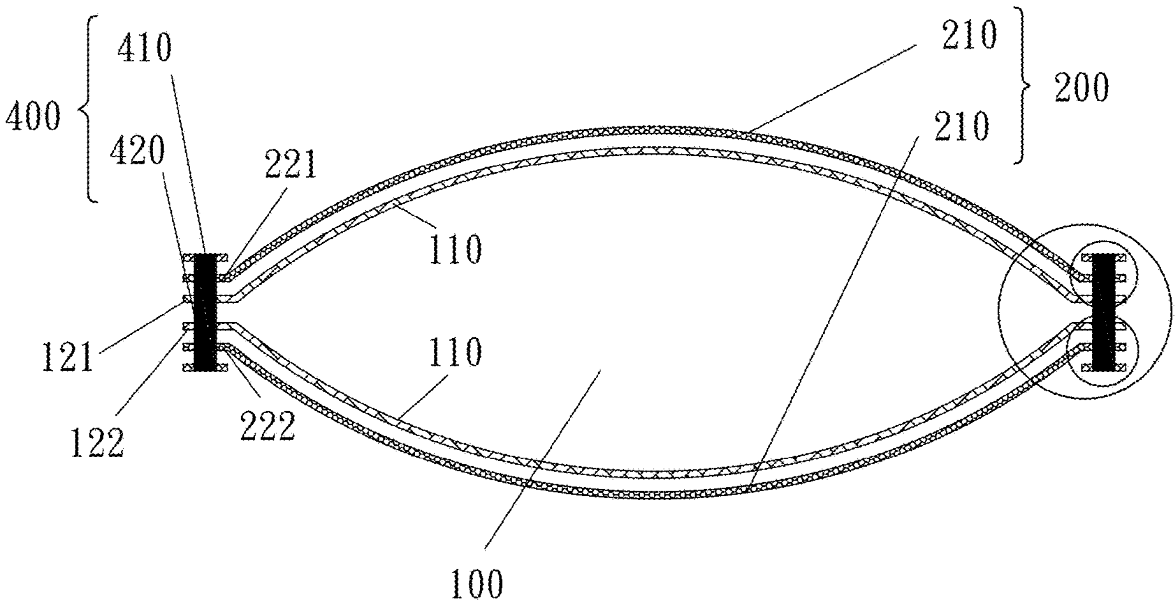
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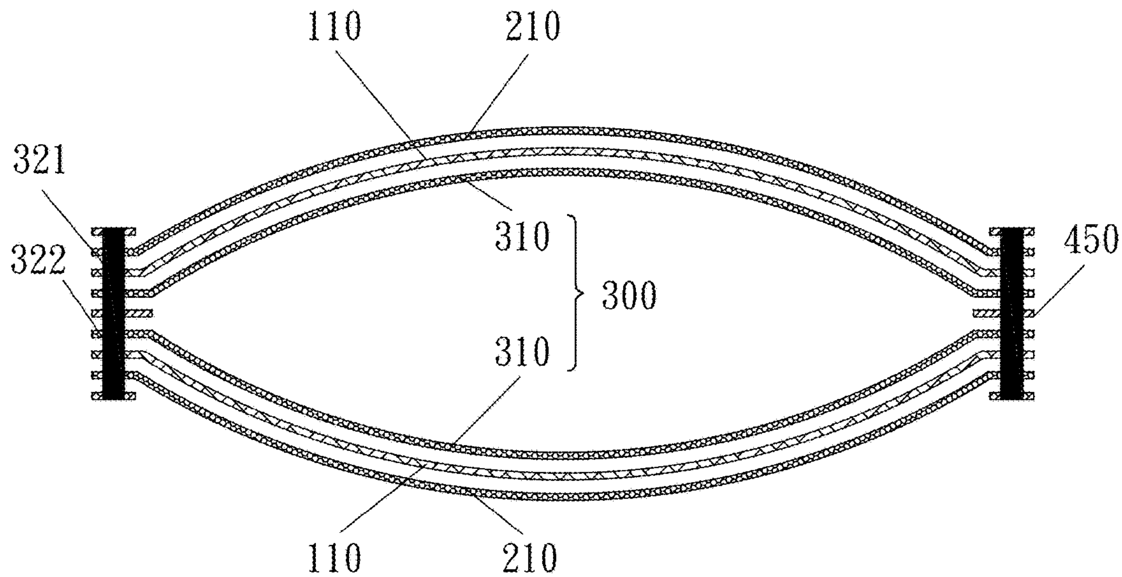
Fig. 22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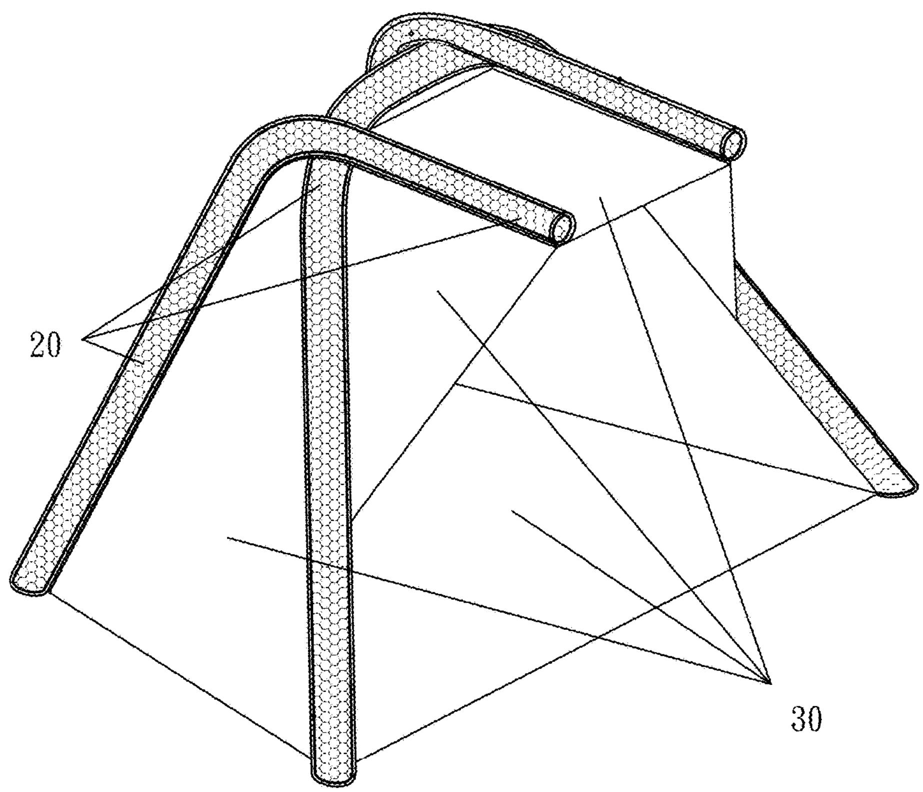

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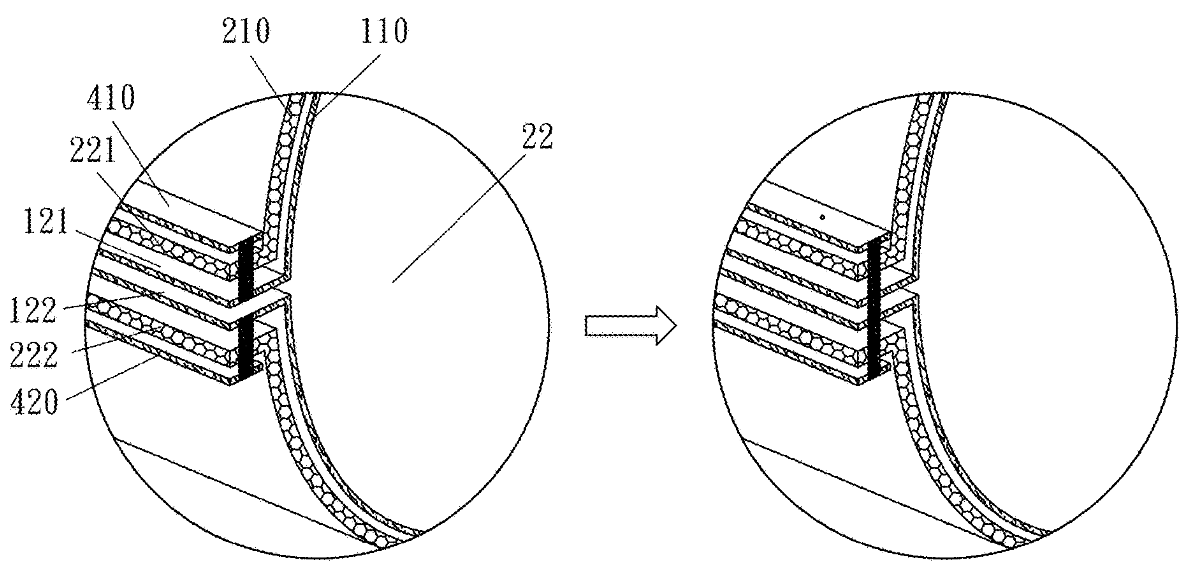

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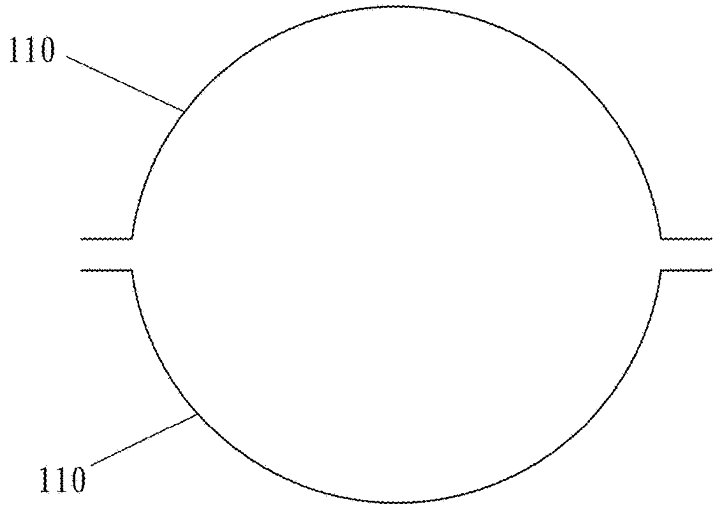

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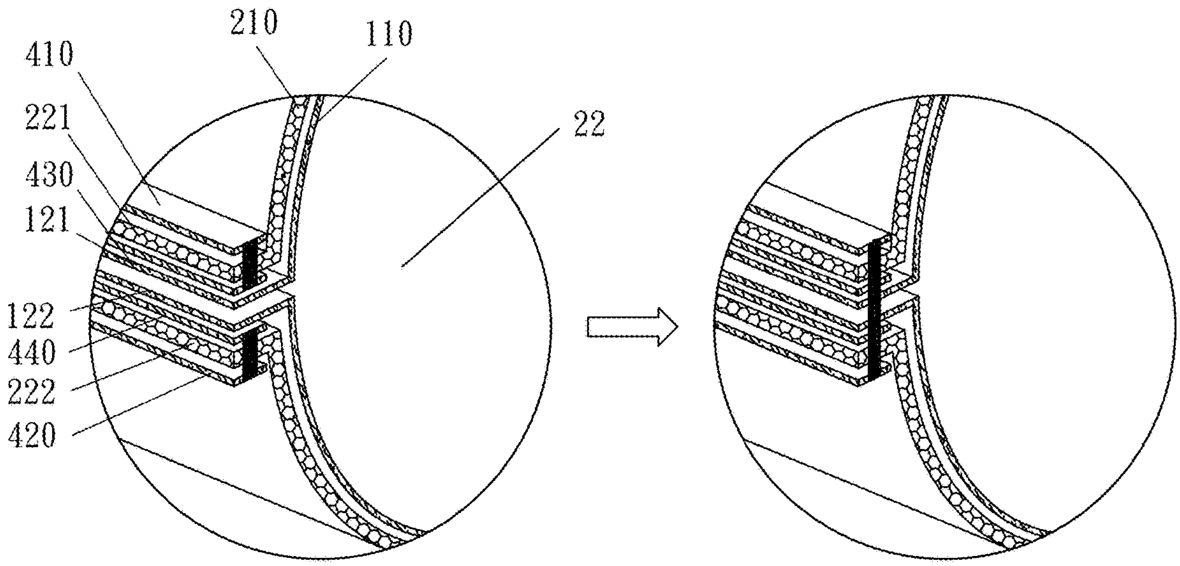

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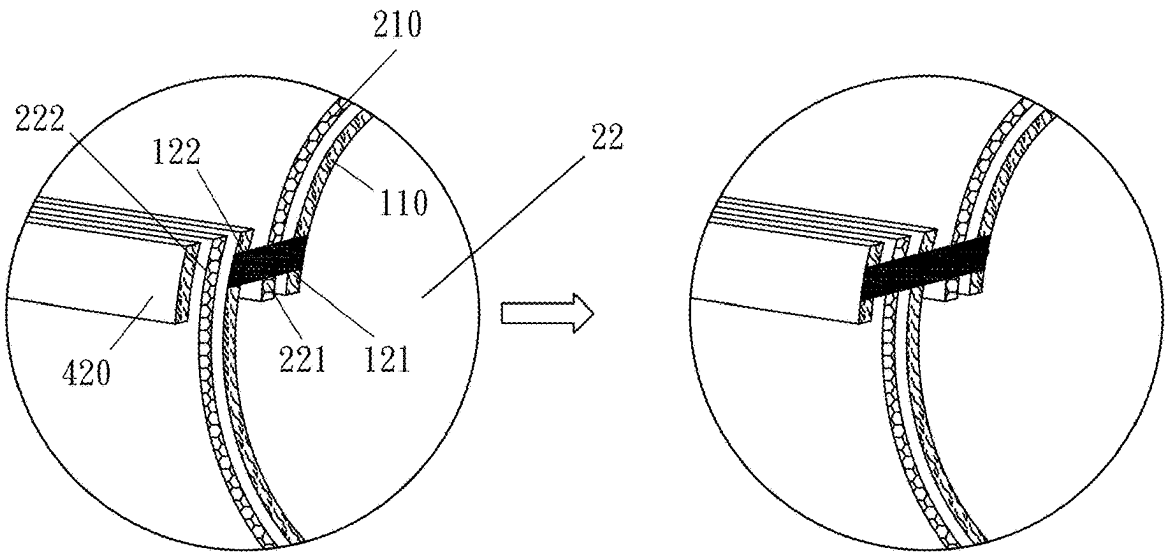

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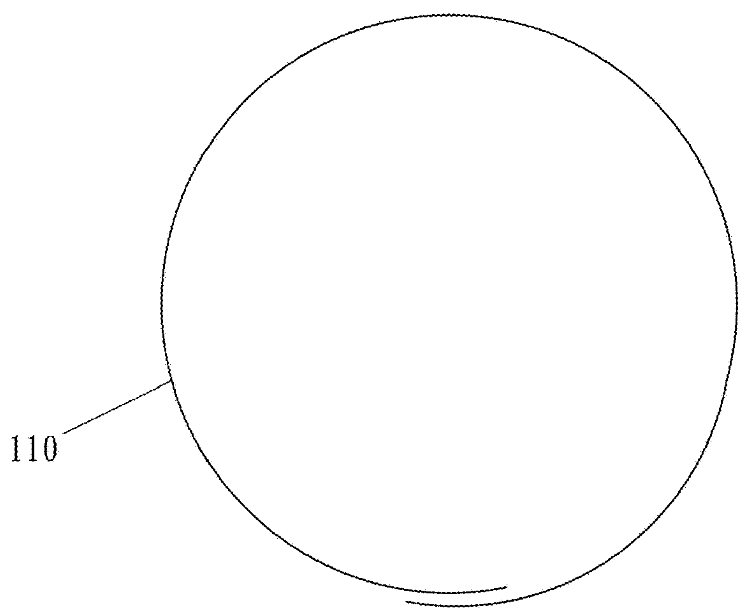

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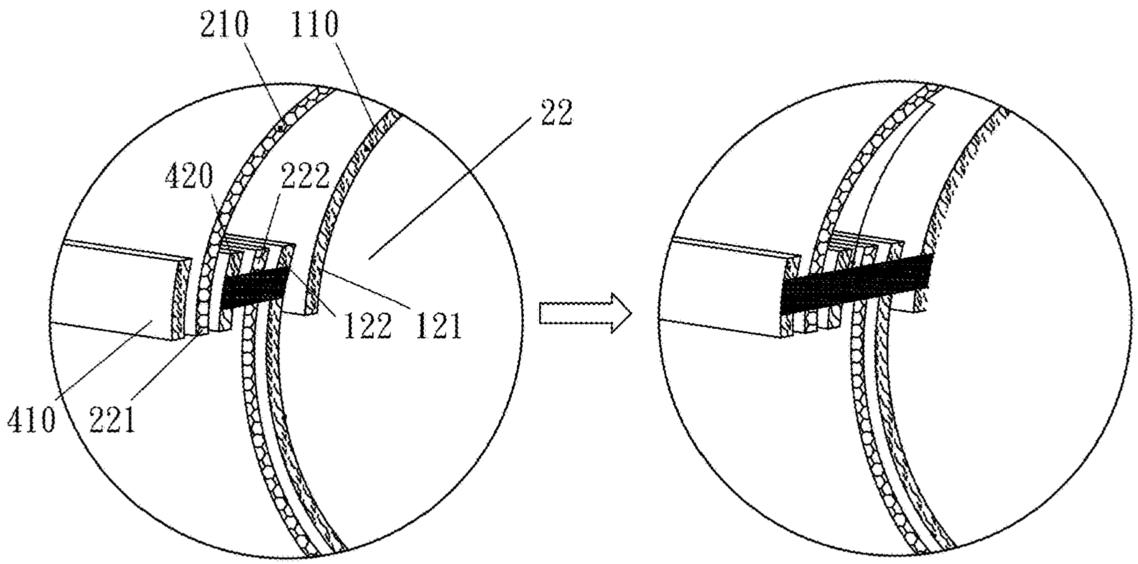

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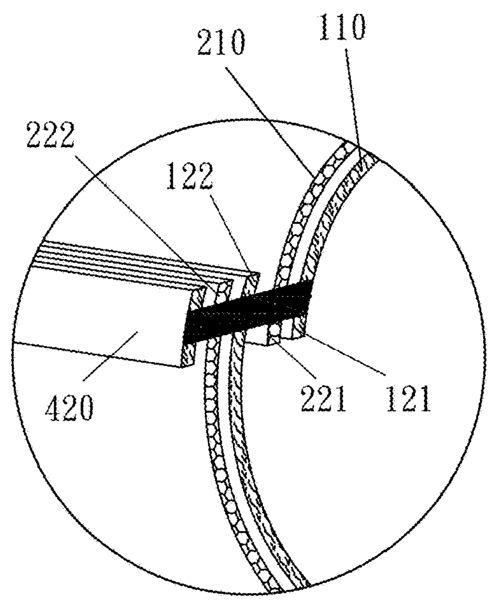

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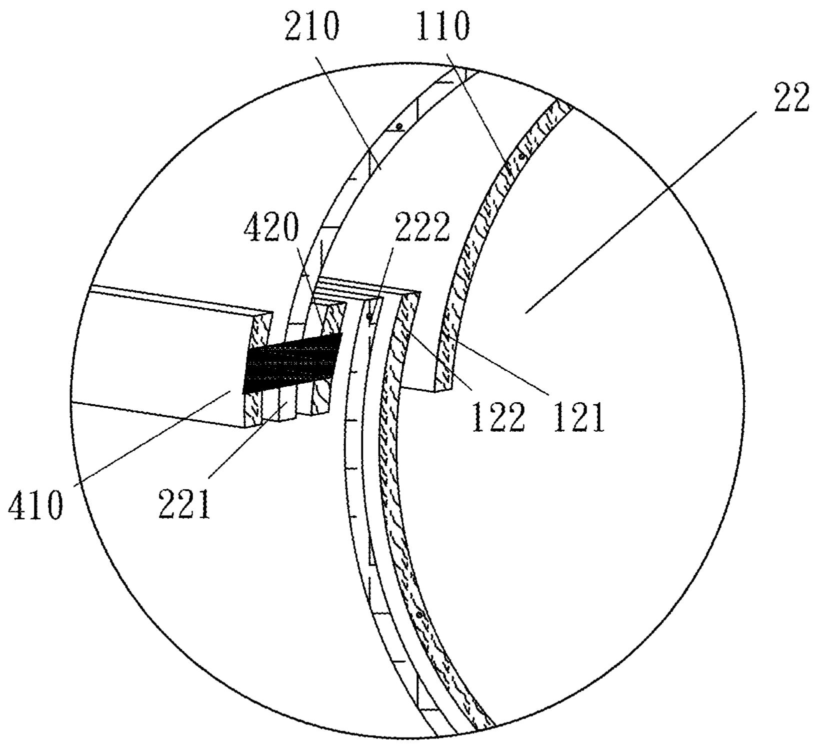

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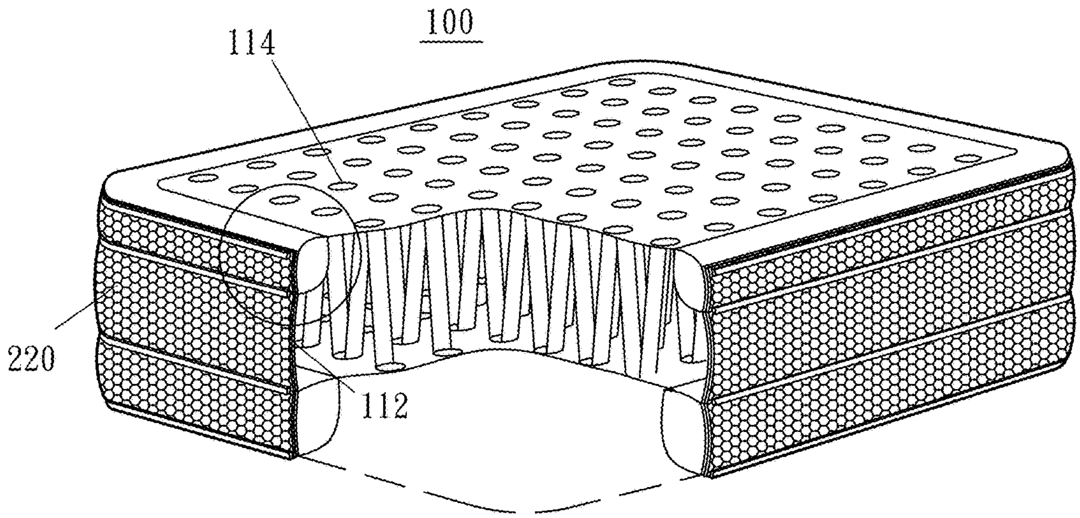

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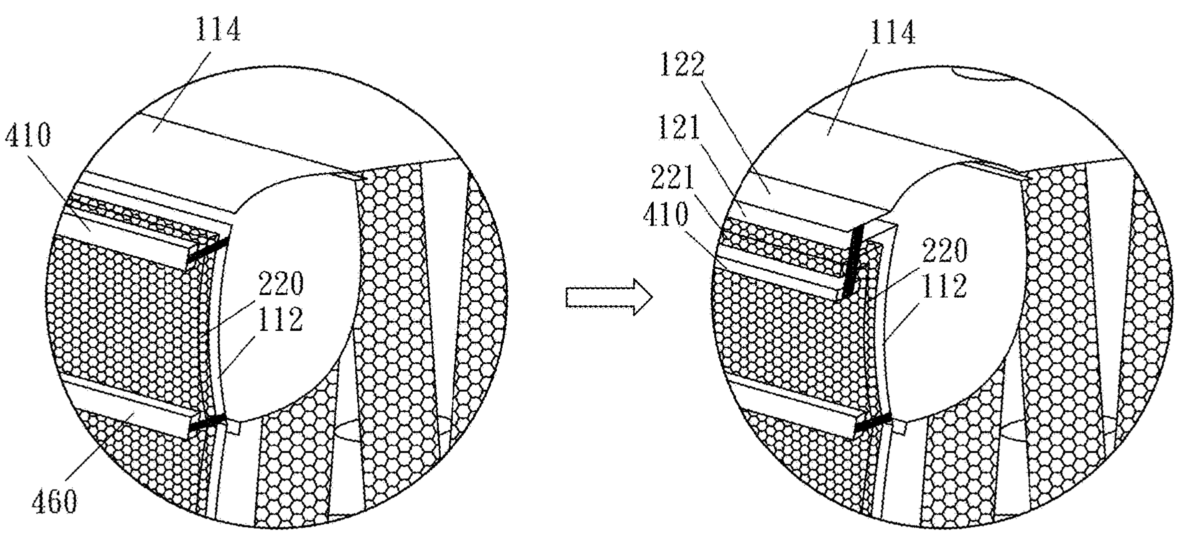

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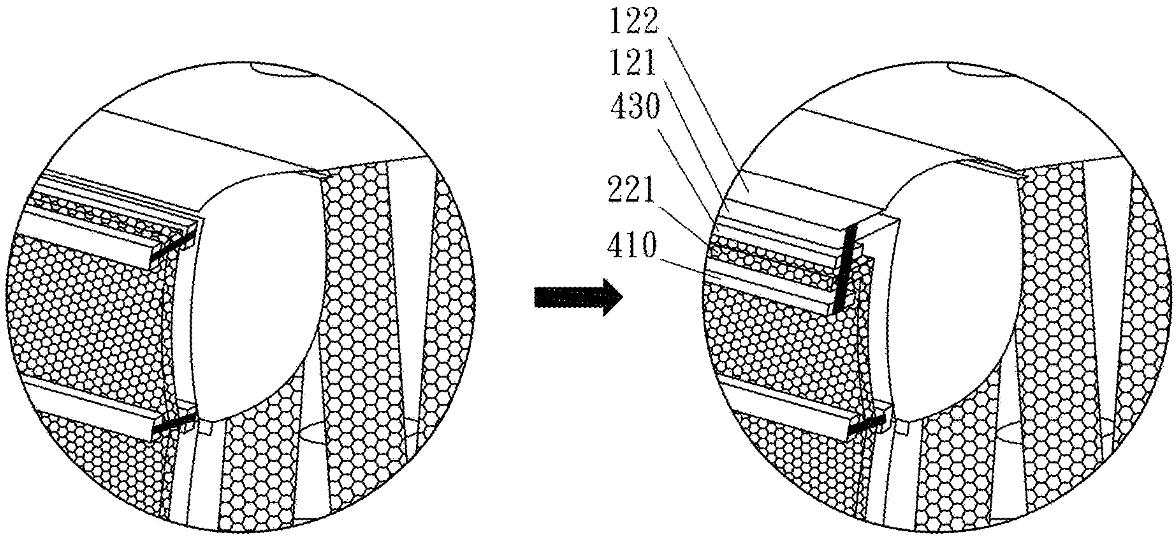

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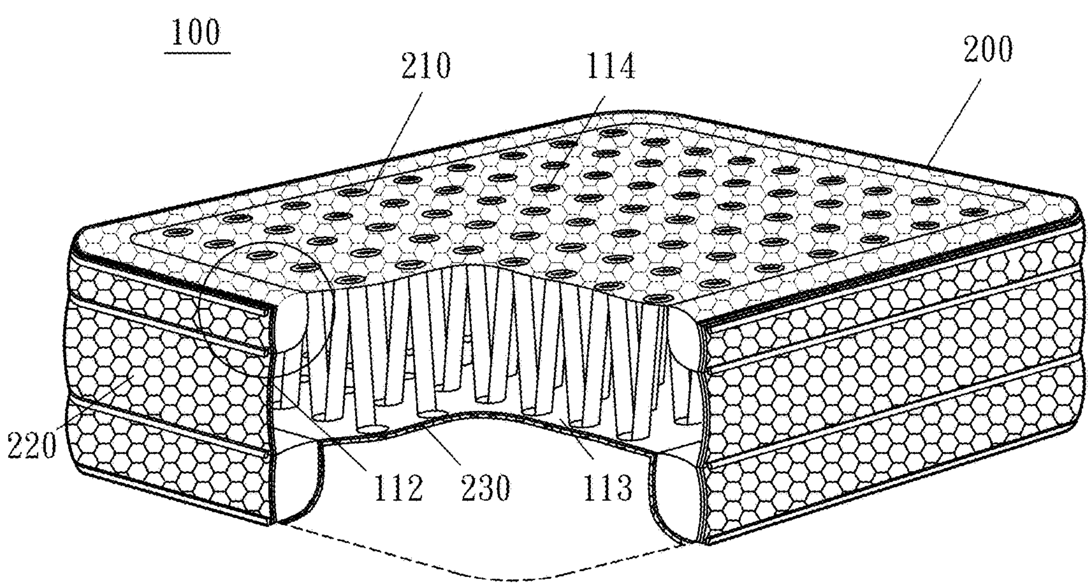

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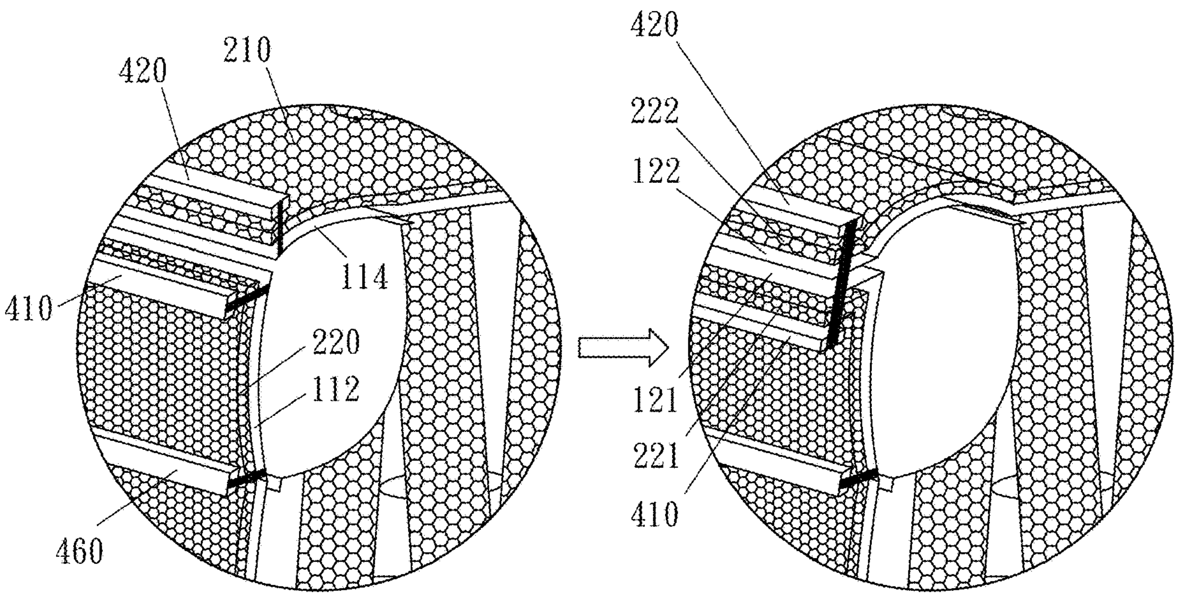
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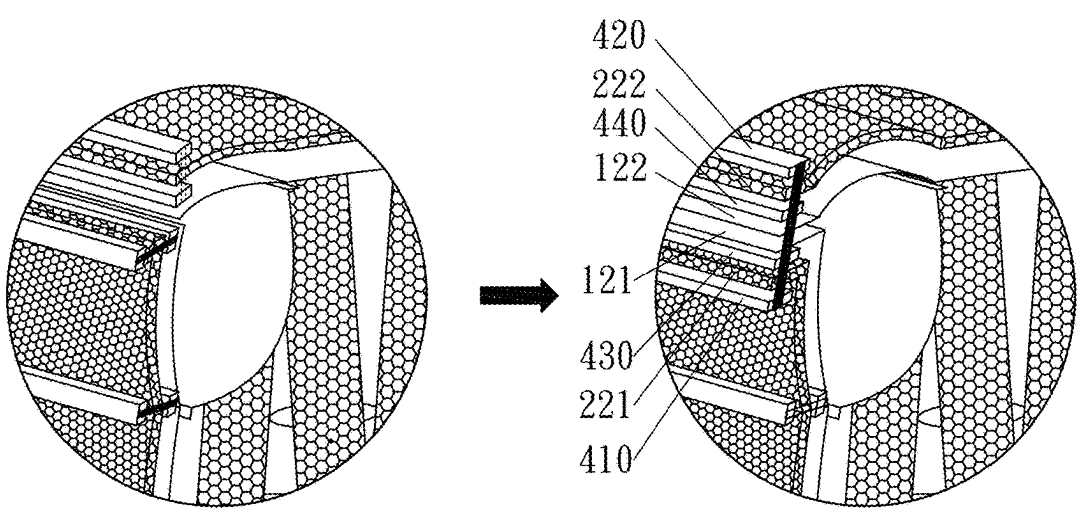
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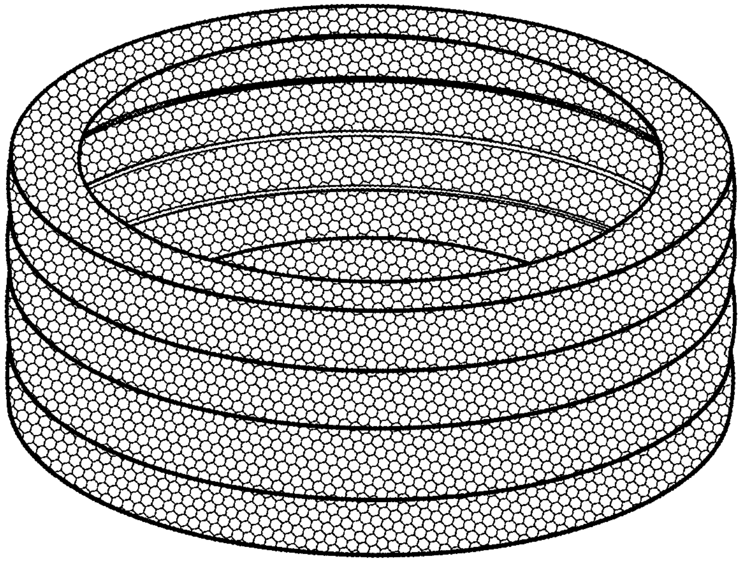
Fig. 36



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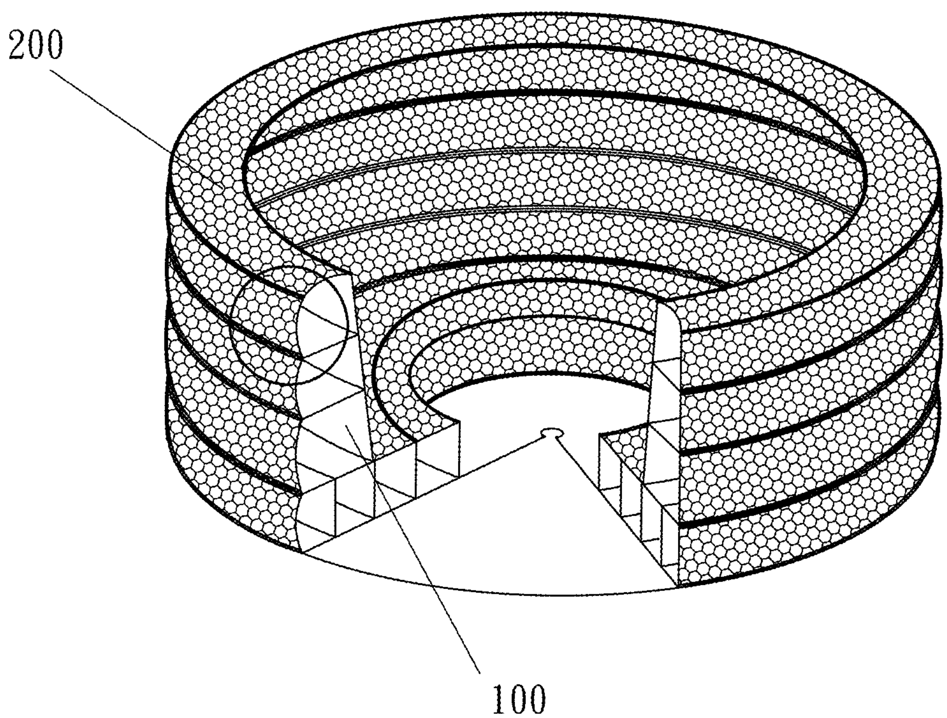

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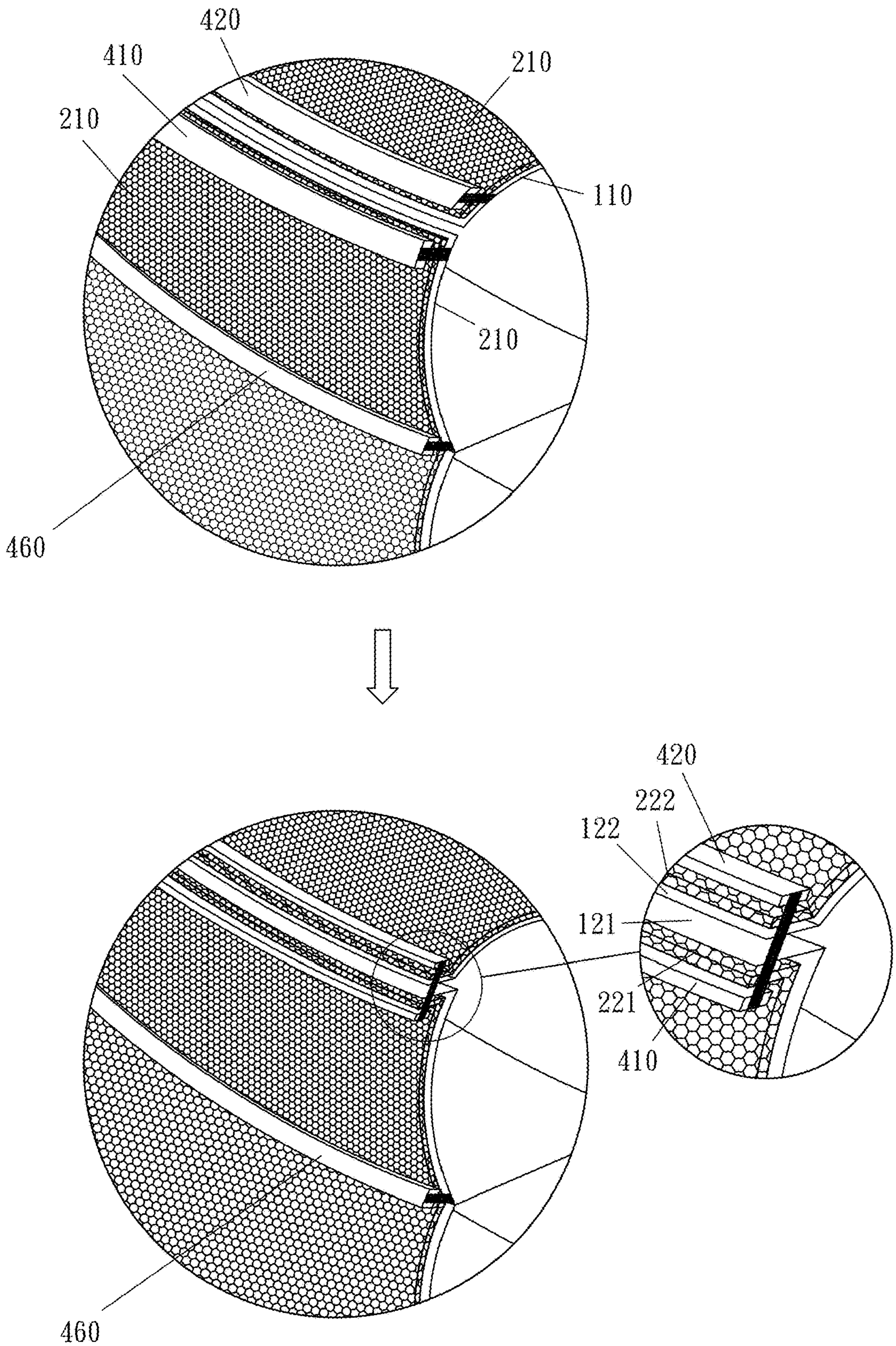

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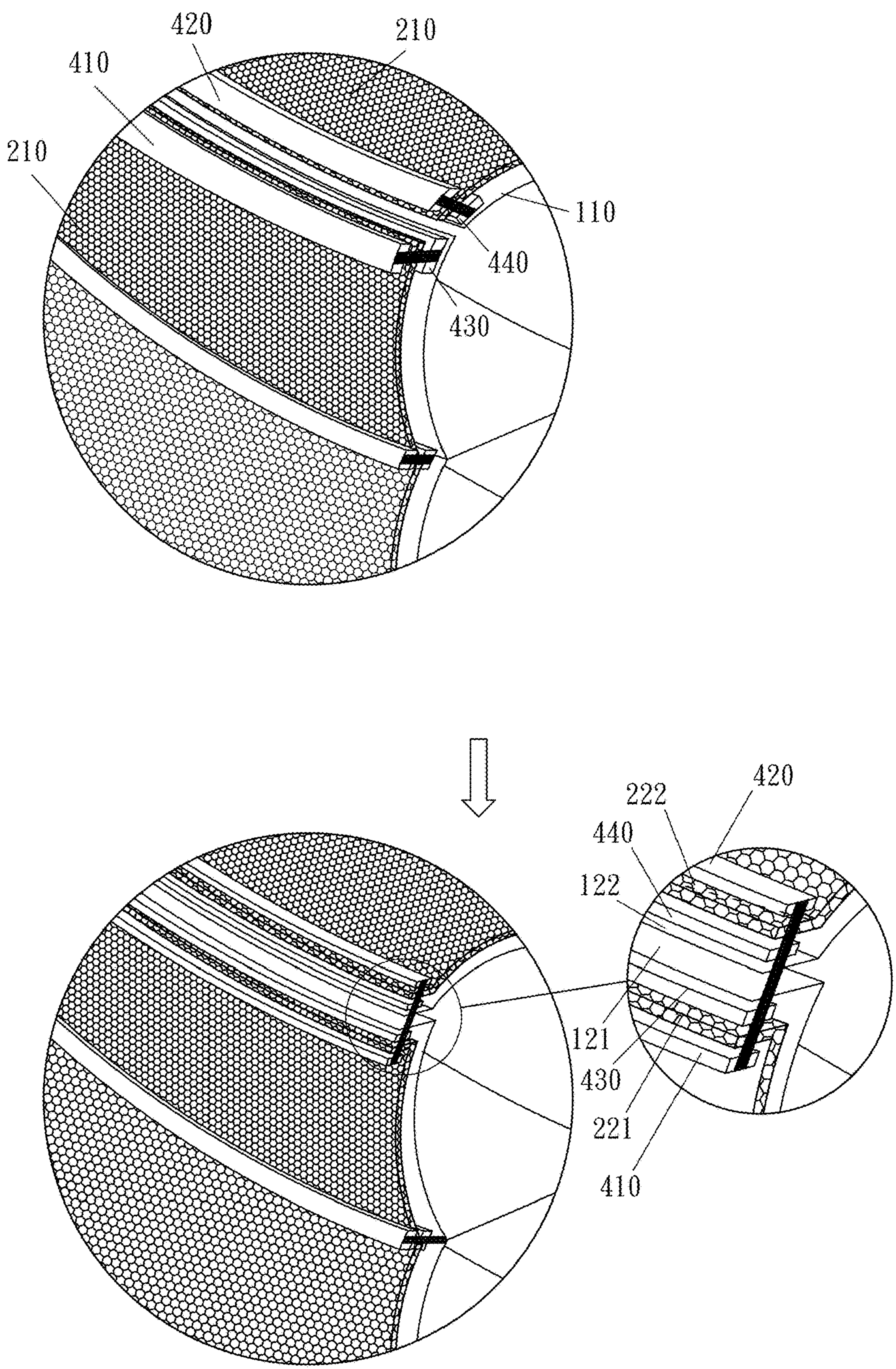

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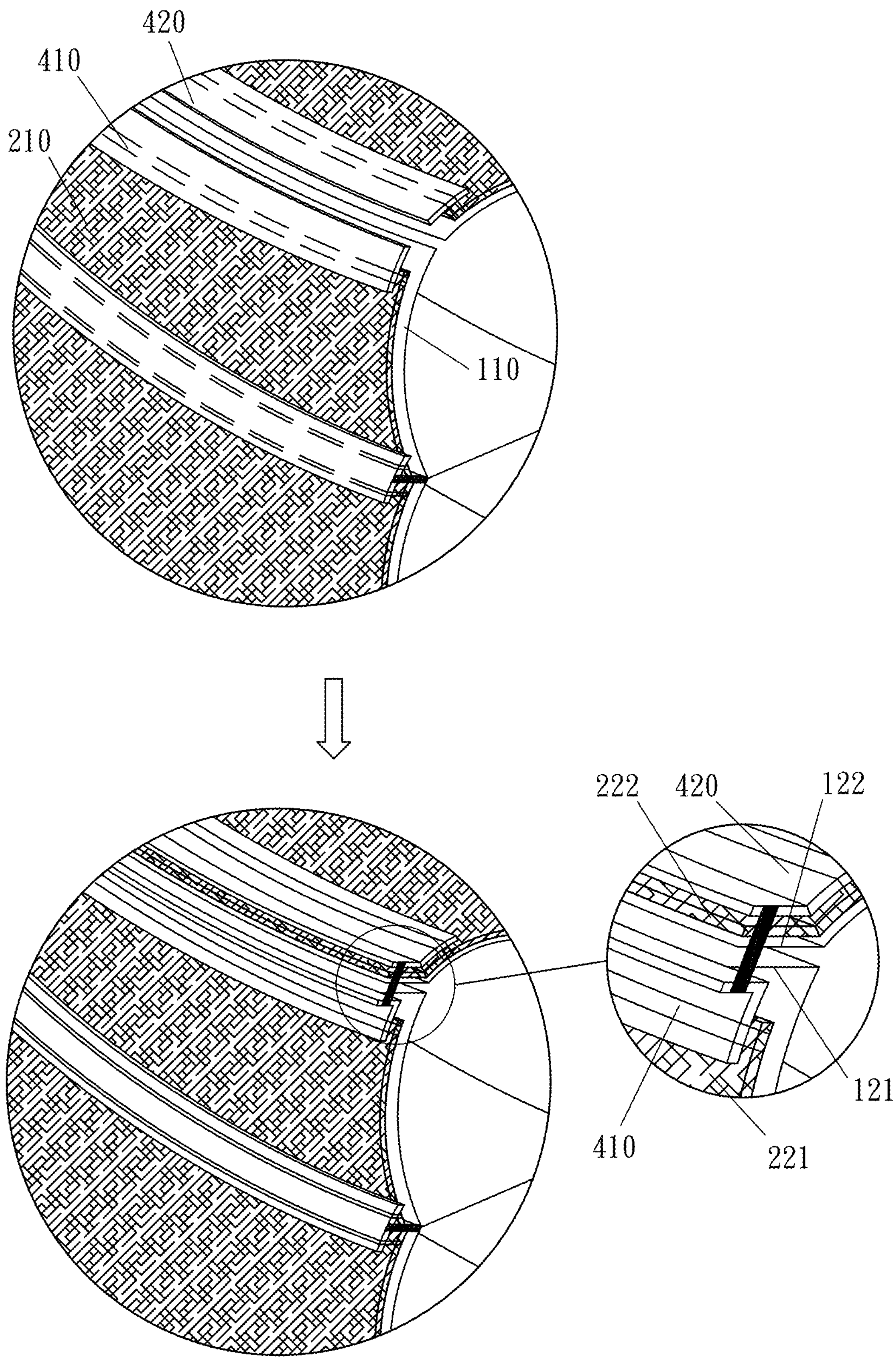

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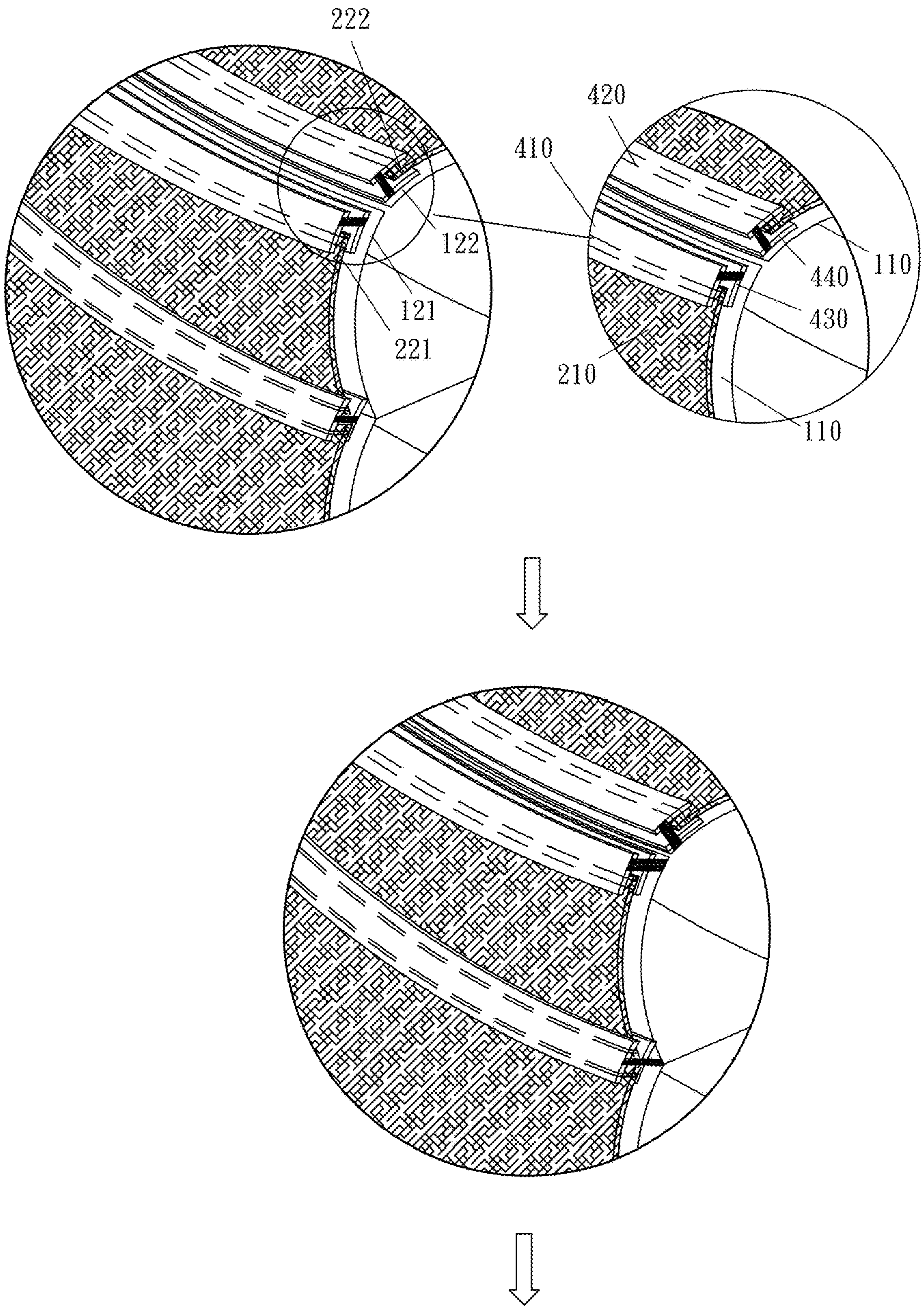

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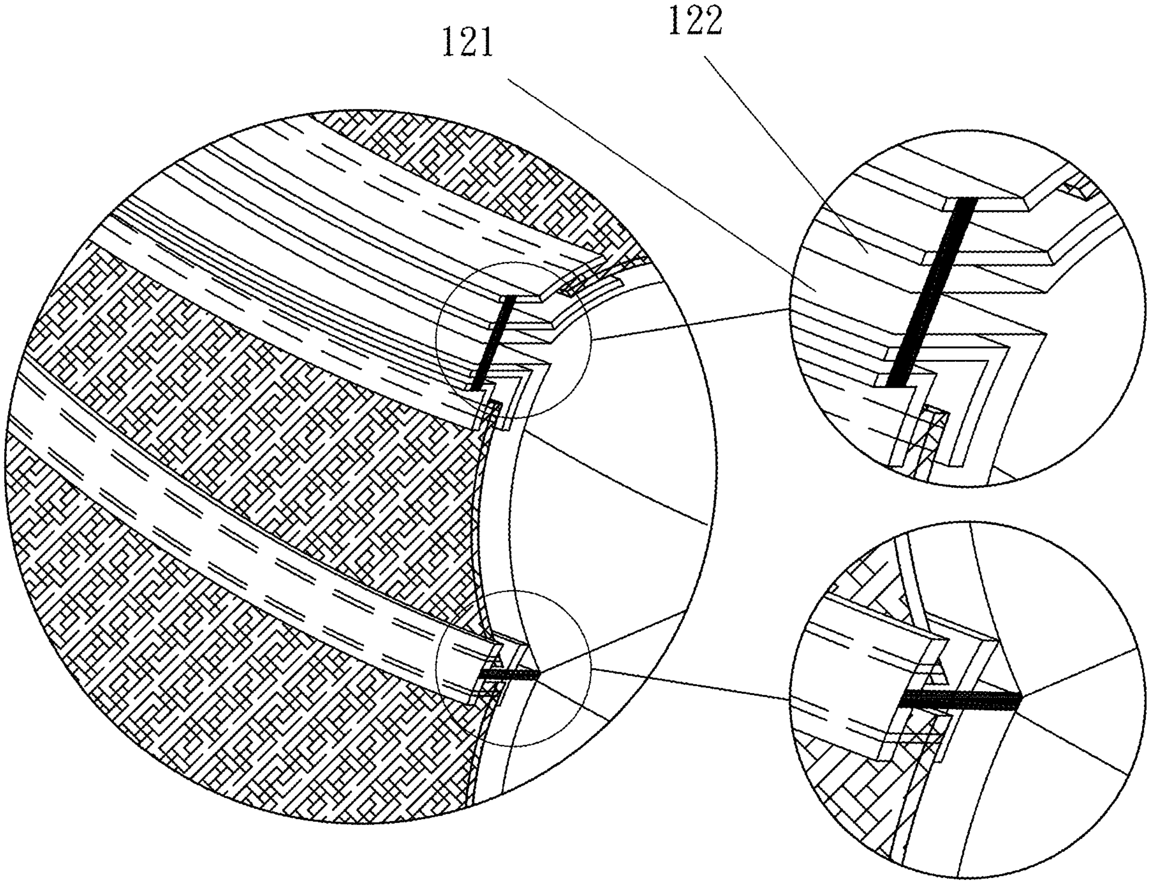

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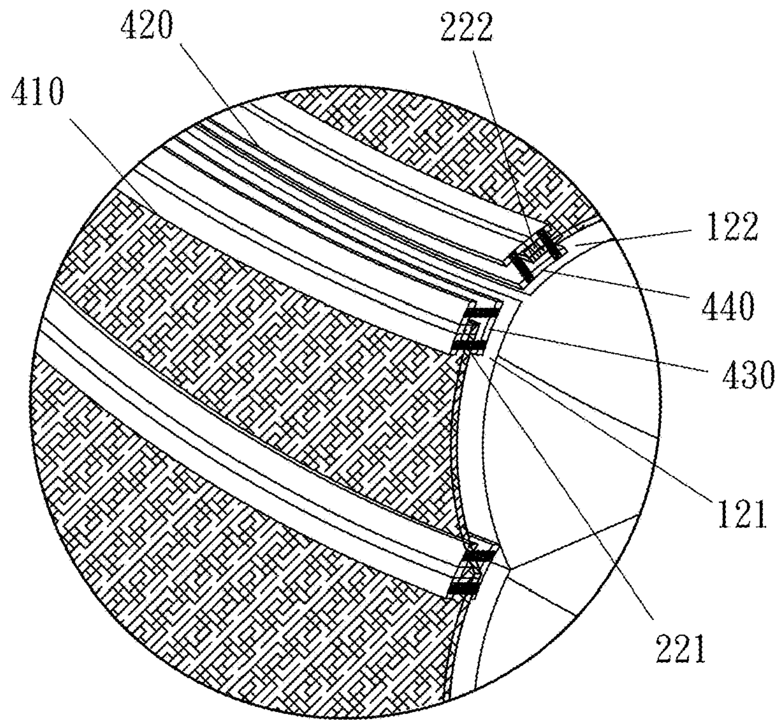

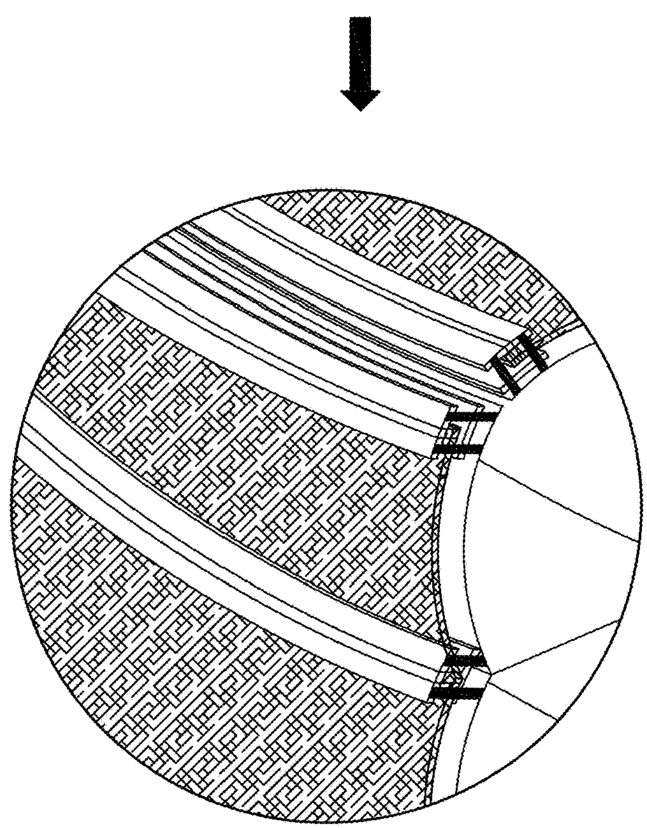

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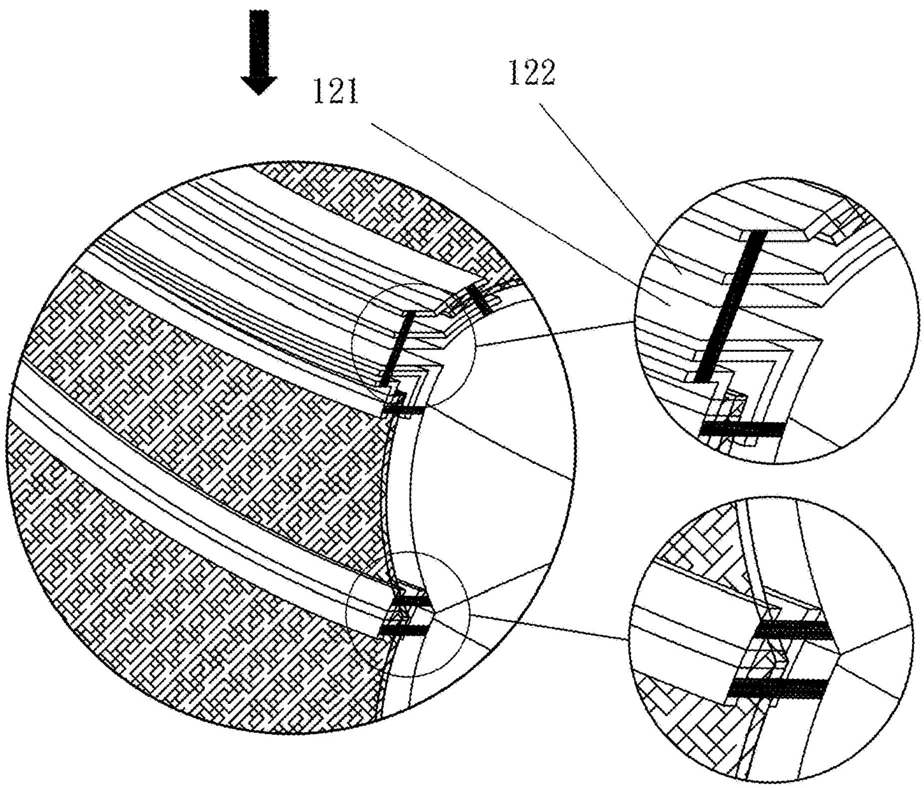

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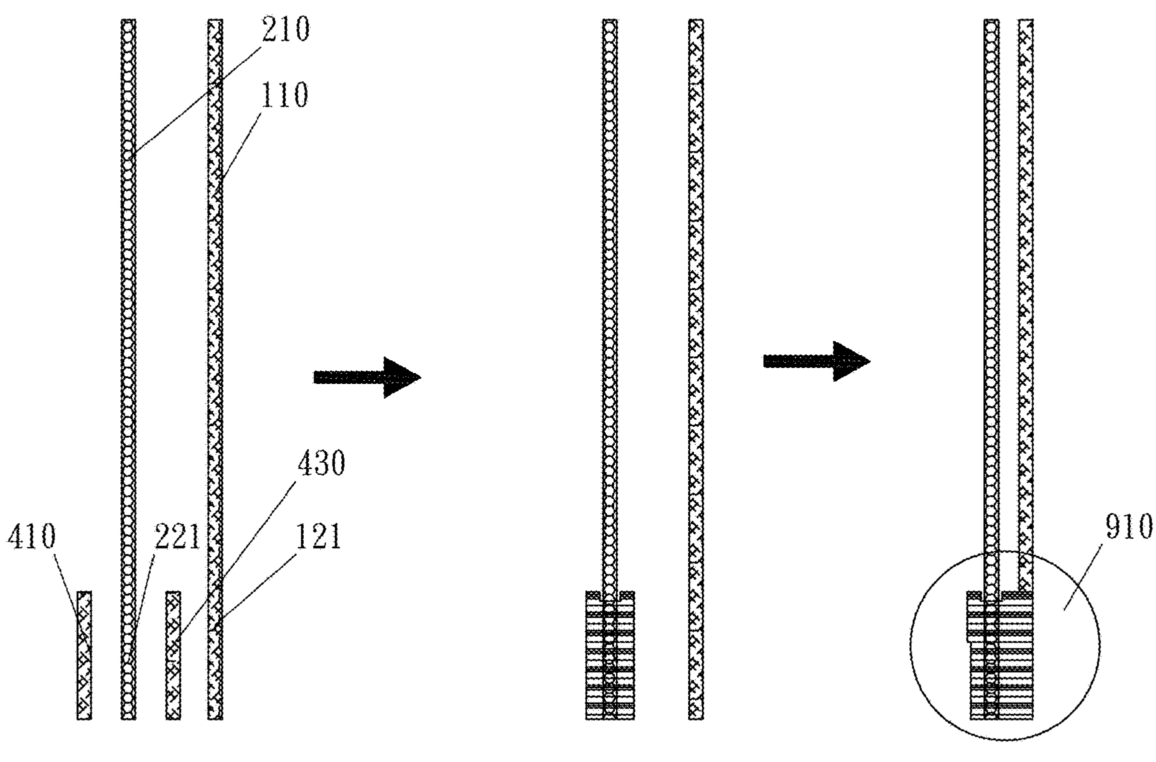

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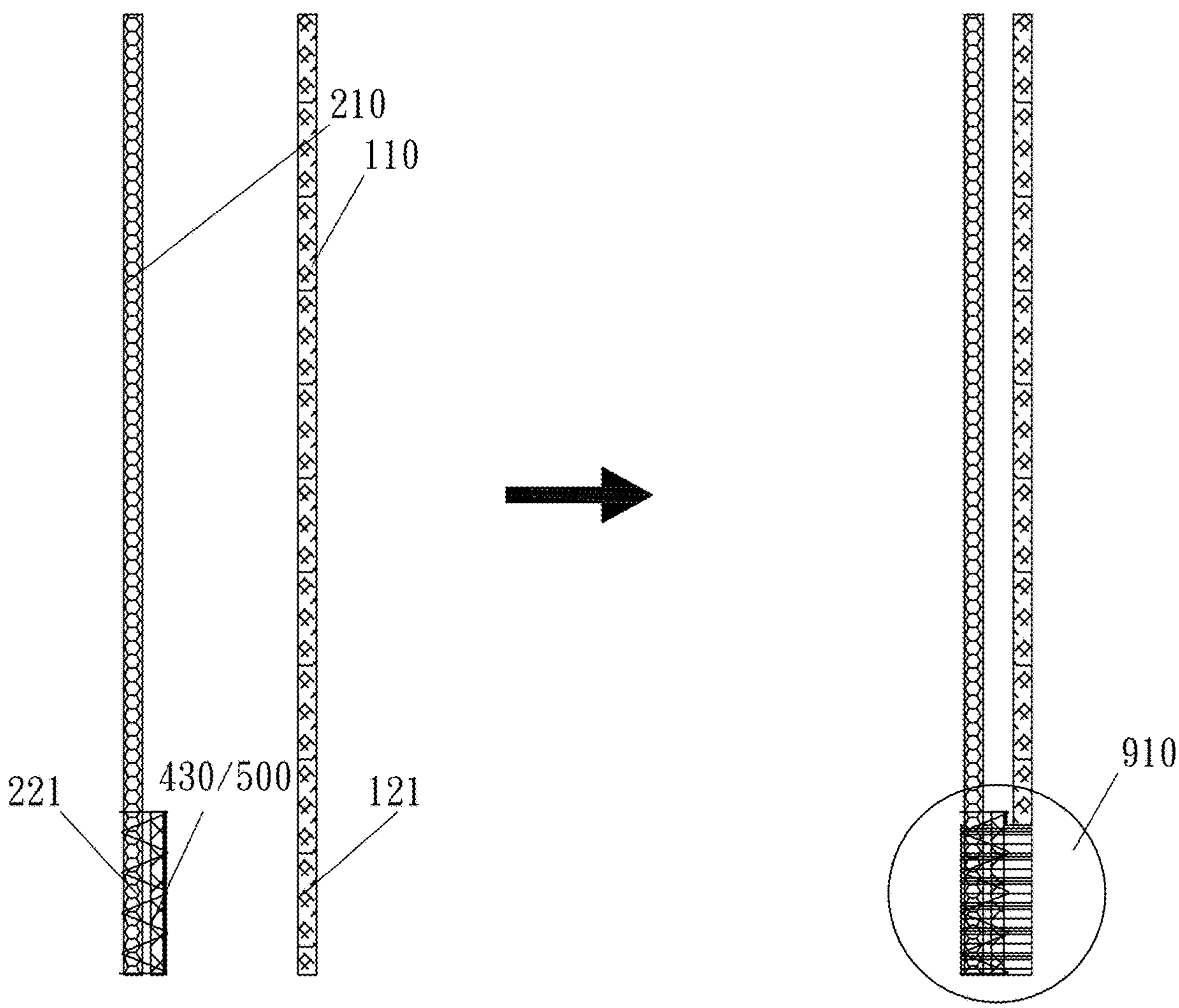

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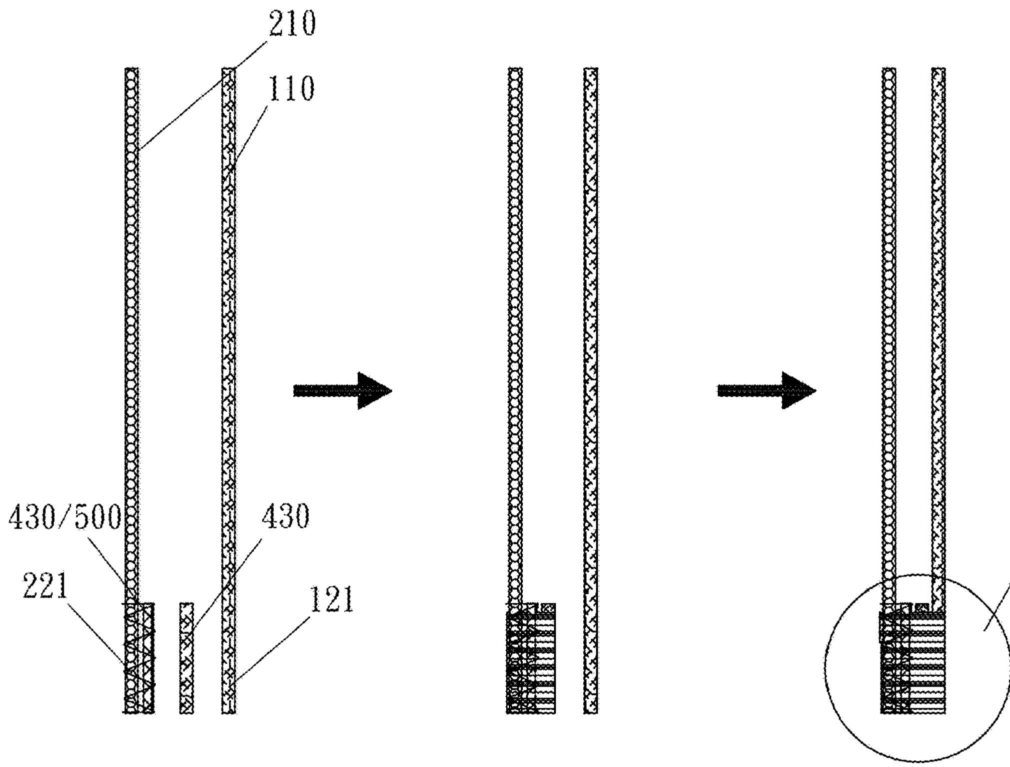

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430
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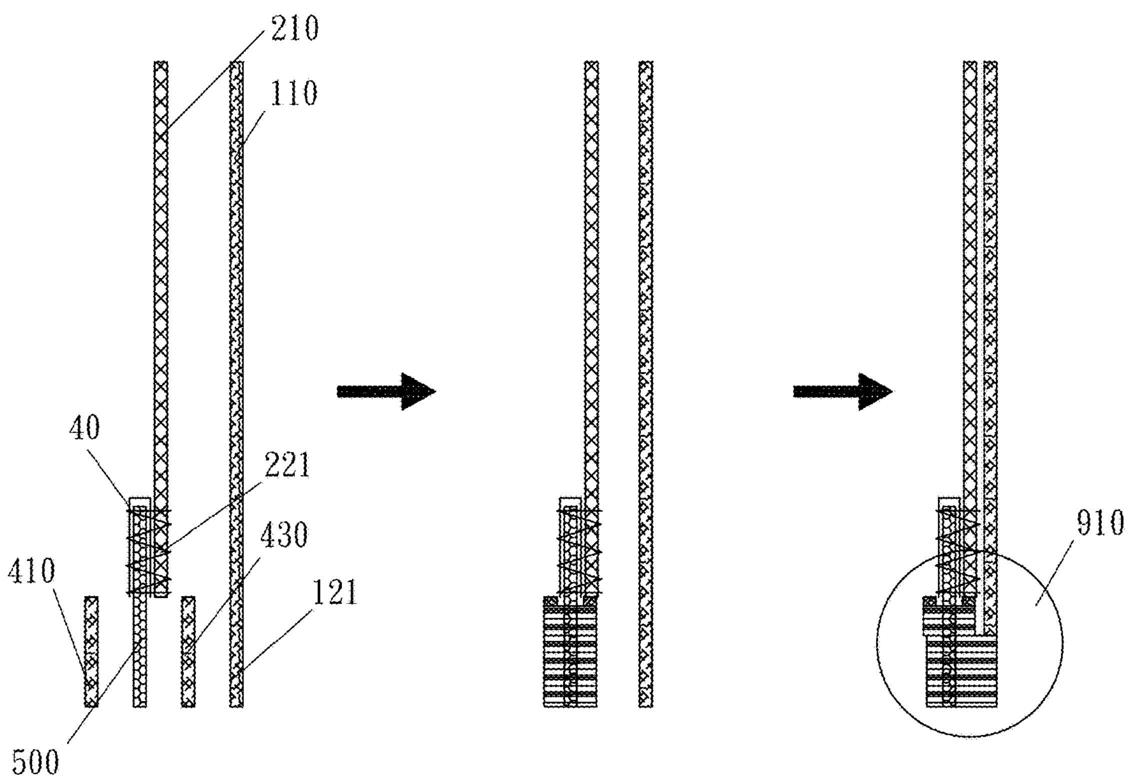
Fig. 46



210
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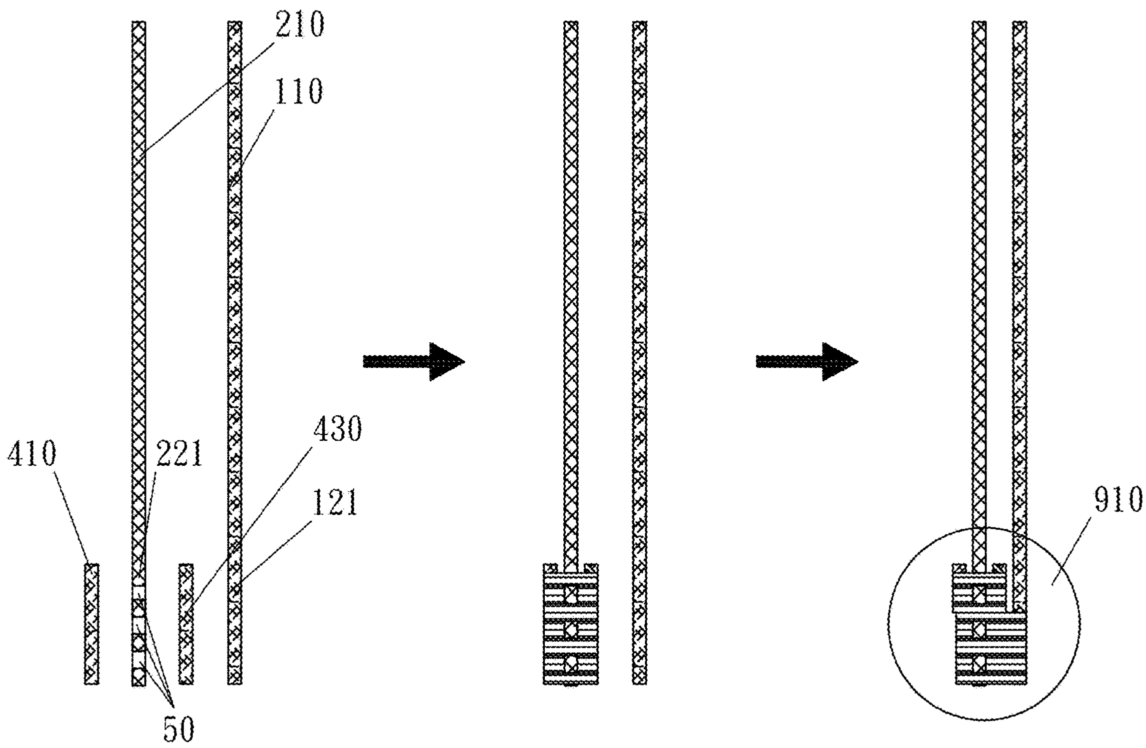
Fig. 47



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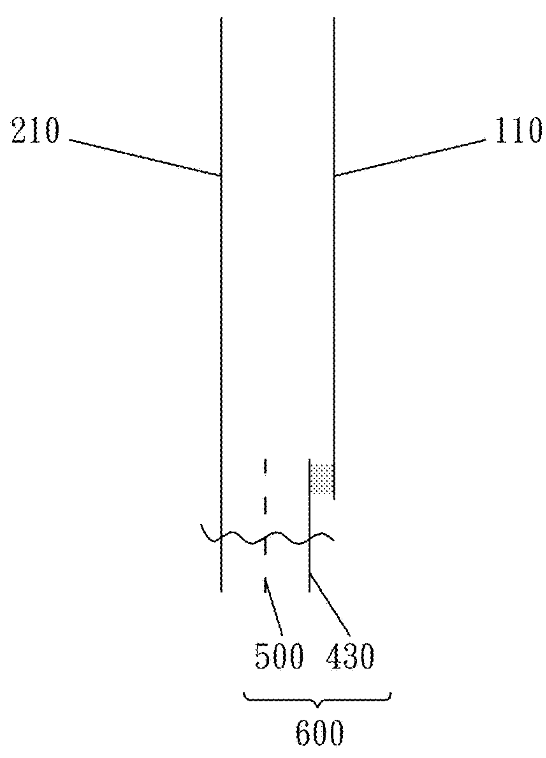

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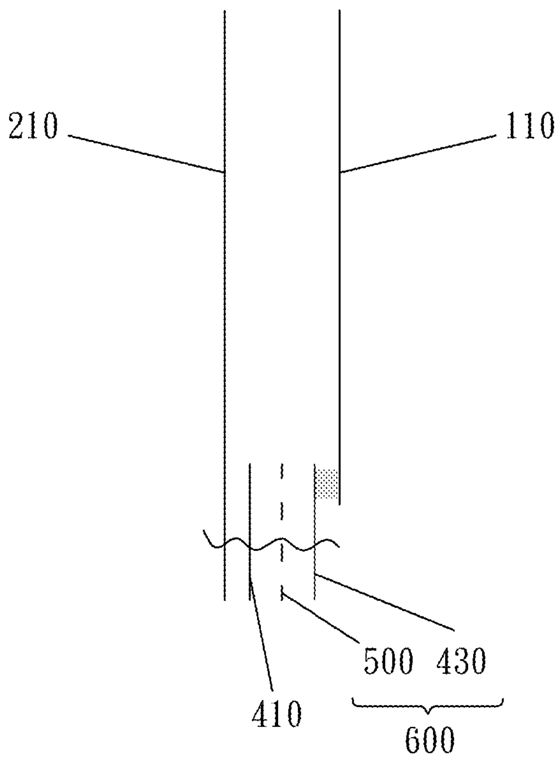

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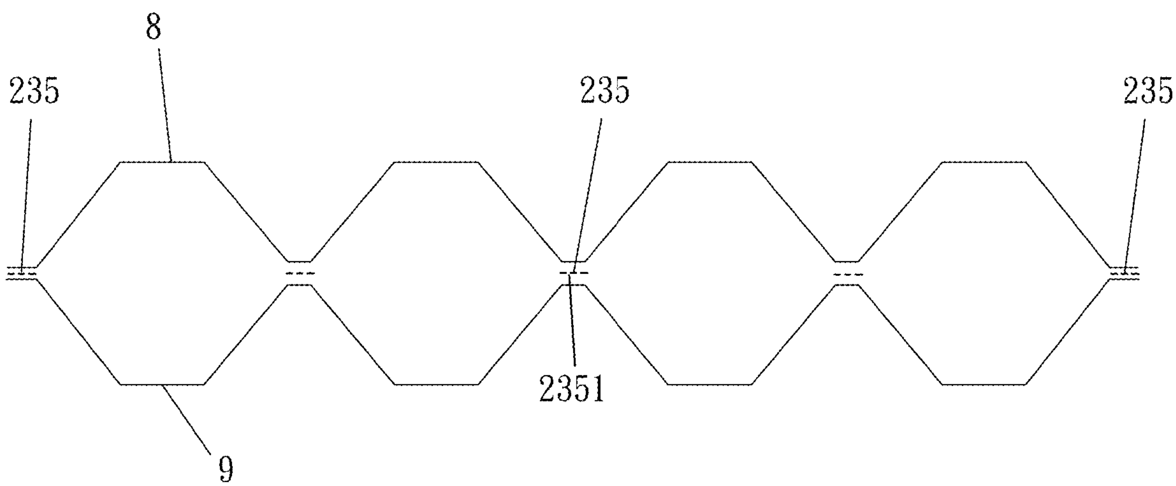

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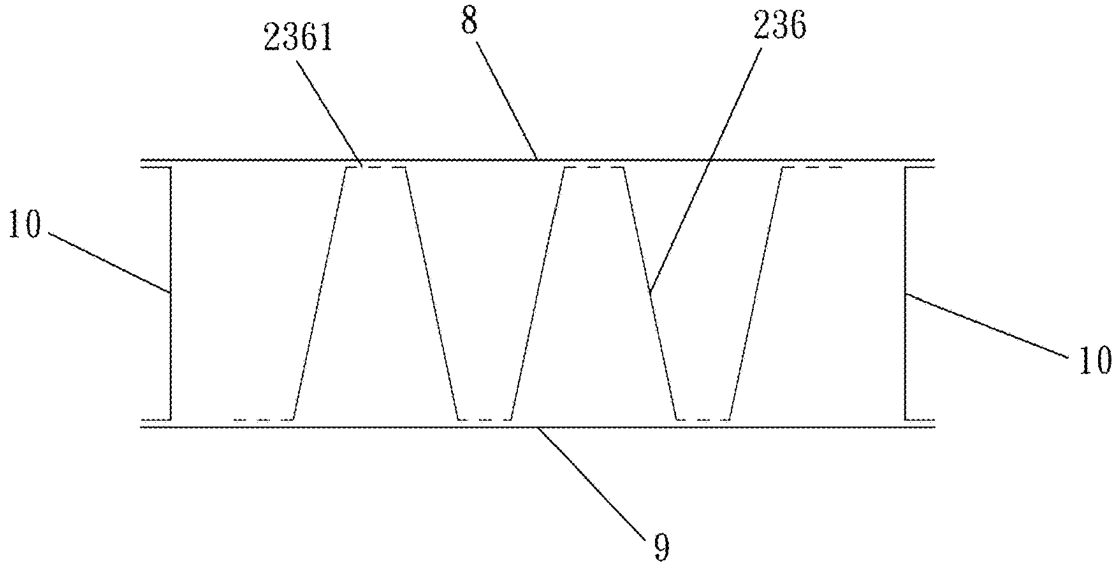

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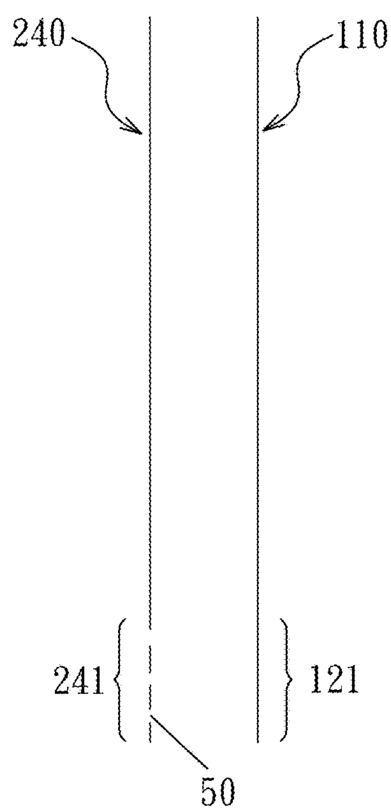

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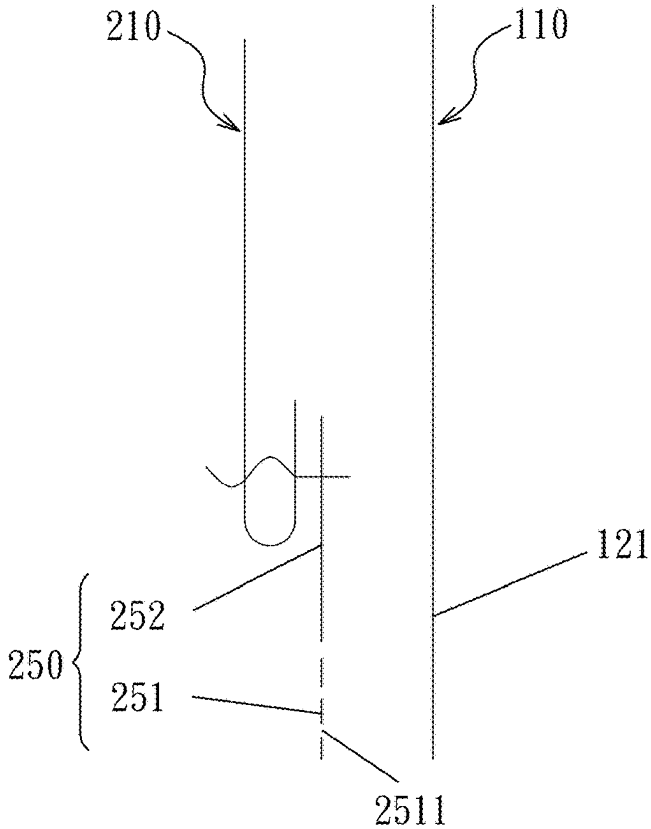

Fig. 54

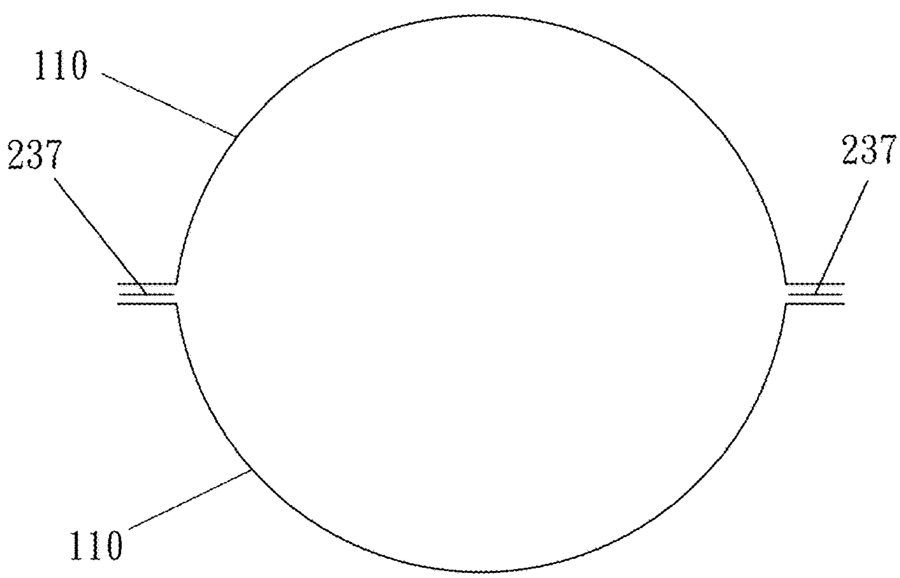

Fig. 55

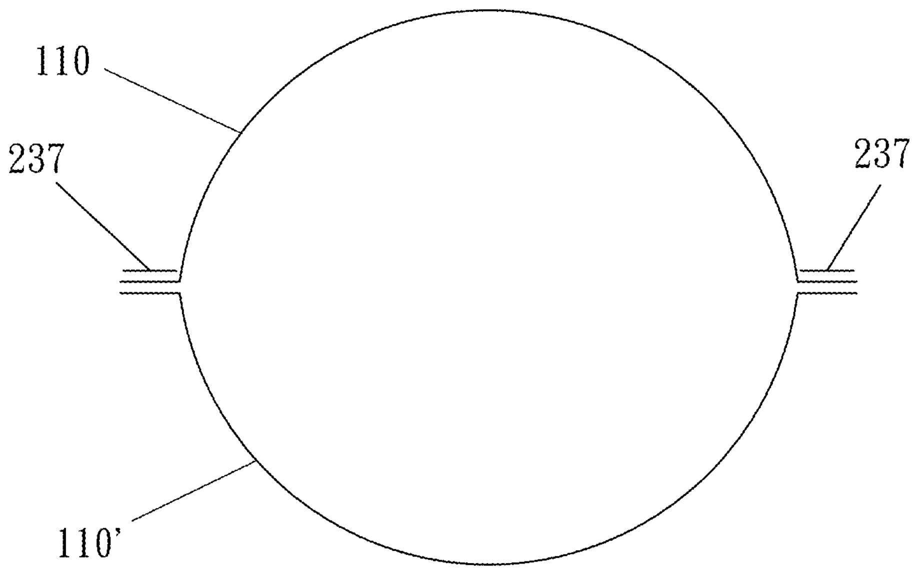

Fig. 56

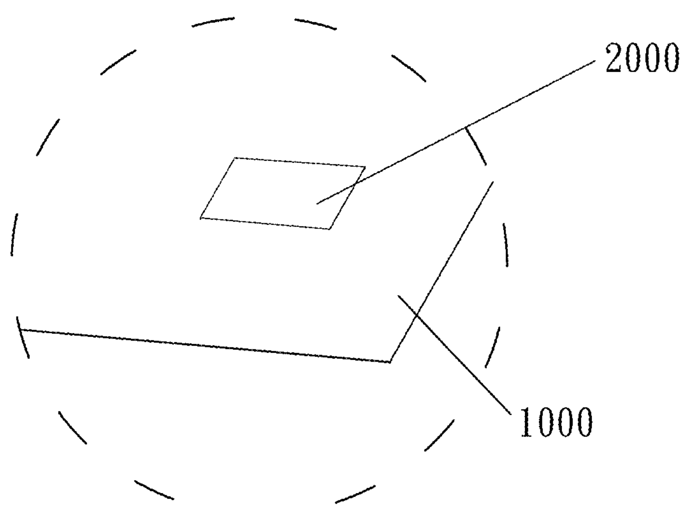

Fig. 57

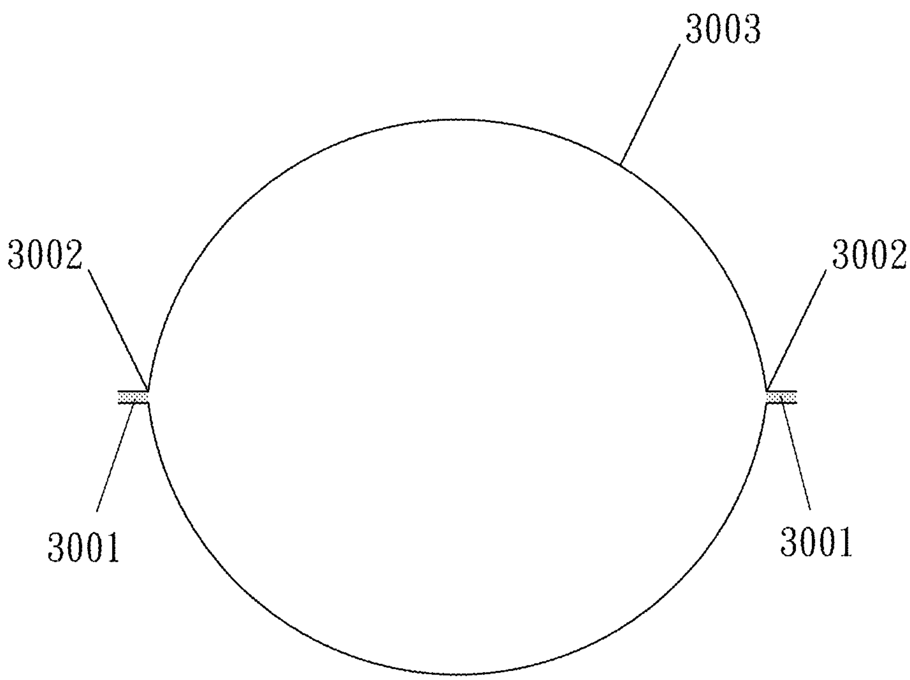

Fig. 58 (PRIOR ART)

FUSION STRUCTURE, INFLATABLE PRODUCT HAVING THE FUSION STRUCTURE, MANUFACTURING METHOD THEREOF, AND RAPIDLY PRODUCED INFLATABLE PRODUCT HAVING COVERING LAYER SIMULTANEOUSLY POSITIONED WITH INFLATABLE BODY

CROSS REFERENCE

[0001] This application is a Continuation-In-Part of application Ser. No. 16/612,353, filed Nov. 8, 2019, which is a national stage application entering the U.S. national phase from an international PCT application No. PCT/CN2018/085922, filed May 8, 2018, which claims the benefit of china application no. 201710321892.9, filed May 9, 2017.

BACKGROUND OF THE INVENTION

Field of the Invention

[0002] The invention relates to an inflatable product, and more particularly to a fusion structure, an inflatable product having the fusion structure, a manufacturing method thereof, and a rapidly produced inflatable product having a covering layer simultaneously positioned with an inflatable body.

Description of the Related Art

[0003] An inflatable product is lightweight, easy to store and portable. Therefore, the application of an inflatable product is wide (not only outdoor products and toys, but also household products can be inflatable products). However, the user can seldom expect a good durability of the inflatable product since an inflatable product is mostly manufactured by way of fusion. Damage to the inflatable product generally occurs at the fusion location, except that the inflatable product is cut or pierced. As shown in FIG. **60**, when inflated, an inflatable product **3003** is stretched, and the fusion portions **3001** become thinner. Generally, rupture of the inflatable product **3003** occurs at the location **3002** where the fusion portions **3001** are thinnest.

[0004] In order to improve the durability, resistance of the inflatable product to rupture in the fusion zone must be enhanced. Further, cutting or piercing the inflatable product becomes easier when the inflatable product bears a great pressure. Therefore, how an inflatable product is improved is a significant issue.

BRIEF SUMMARY OF THE INVENTION

[0005] To address at least one shortcoming of above-described prior art, the invention provides a fusion structure, an inflatable product having the fusion structure, a manufacturing method thereof, and a rapidly produced inflatable product having a covering layer simultaneously positioned with an inflatable body. The invention is simple in structure and is convenient to use. Further, the invention is able to greatly enhance the resistance of the inflatable product to rupture in the fusion zone, to increase stability of use of the inflatable product, to reduce weight of the inflatable product, and to reduce the cost.

[0006] To solve the above-described problems, the invention provides the following technical solutions:

[0007] A fusion structure for an inflatable product includes at least one fusion unit, and the fusion unit includes a target fusion layer, at least one first fusion reinforcement layer and at least one first auxiliary fusion layer which are fused together. The fusion reinforcement layer and the auxiliary fusion layer are alternately disposed at a side of the target fusion layer, while outermost layers of two sides of the fusion unit are the target fusion layer and the auxiliary fusion layer. Alternatively, the fusion reinforcement layer and the auxiliary fusion layer are alternately disposed at two sides of the target fusion layer, while outermost layers of two sides of the fusion unit are the auxiliary fusion layers. A material of which the target fusion layer is made is the same as or compatible with a material of which the auxiliary fusion layer is made. The fusion reinforcement layer is made of a flexible anti-stretch material having a plurality of through holes, and the way to produce the anti-stretch material includes punching, braiding, weaving and fiber pressing. Preferably, the fusion reinforcement layer is a mesh fabric. In such design, the target fusion layer and at least one auxiliary fusion layer hold at least one fusion reinforcement layer therebetween. During the fusion, the target fusion layer and the auxiliary fusion layer are melted into the through holes of the fusion reinforcement layer and are restricted by anti-stretch material which is provided at edges of the through holes, and the direction of restriction is the same as the direction of a pulling force produced by expansion of the inflatable product. It functions similar to reinforcement steels in concrete. Therefore, the ability of the fusion site to resist stretch is greatly enhanced, the ability of the inflatable product to resist pressure of inflation is relatively enhanced, and the durability of the inflatable product is improved.

[0008] The above-described fusion structure can be made into a plurality of preferred forms. In a first form, the fusion structure includes a fusion unit, the fusion unit comprises a target fusion layer, a first fusion reinforcement layer and a first auxiliary fusion layer which are sequentially arranged, and the target fusion layer, the first fusion reinforcement layer and the first auxiliary fusion layer are simultaneously connected by fusion.

[0009] In a second form, the fusion structure includes a fusion unit, the fusion unit includes a target fusion layer, a second auxiliary fusion layer, a first fusion reinforcement layer and a first auxiliary fusion layer which are sequentially arranged, and the target fusion layer, the second auxiliary fusion layer, the first fusion reinforcement layer and the first auxiliary fusion layer are simultaneously connected by fusion. The fusion structure of this form is formed by additionally providing the second auxiliary fusion layer between the target fusion layer and the first fusion reinforcement layer which are based on the fusion structure of the first form. Because the second auxiliary fusion layer is provided, the fusion structure of this form is several times higher than the fusion structure of the first form in strength.

[0010] In a third form, the fusion structure includes a fusion unit, the fusion unit includes a target fusion layer, a second auxiliary fusion layer, a first fusion reinforcement layer and a first auxiliary fusion layer which are sequentially arranged, a periphery of the second auxiliary fusion layer is connected to the target fusion layer by a first fusion, the first fusion reinforcement layer and the first auxiliary fusion layer are connected to the second auxiliary fusion layer and the target fusion layer by a second fusion, and the fusion portions formed by the first fusion and the second fusion are not overlapped. The fusion structure of this form is similar to that of the second form in structure, and the difference therebetween is that the fusion structure of this form is formed by fusion twice. The first fusion is used for connecting the periphery of the second auxiliary fusion layer and the target fusion layer together, while the second fusion is used for connecting the first fusion reinforcement layer, the first auxiliary fusion layer, the second auxiliary fusion layer and the target fusion layer.

[0011] In a fourth form, the fusion structure includes a fusion unit, the fusion unit includes a target fusion layer, a second auxiliary fusion layer, a first fusion reinforcement layer and a first auxiliary fusion layer which are sequentially arranged, the second auxiliary fusion layer is connected to the target fusion layer by a first fusion, the first fusion reinforcement layer and the first auxiliary fusion layer are connected to the second auxiliary fusion layer and the target fusion layer by a second fusion, the fusion portion formed by the first fusion is greater than that formed by the second fusion in area, and the fusion portion formed by the second fusion does not extend beyond that formed by the first fusion. The fusion structure of this form is the same as the fusion structure of the third form in structure, times of fusion and sequence of fusion, and the difference therebetween is that in the fusion structure of the third form, the fusion portions formed by the first fusion and the second fusion are not overlapped while in the fusion structure of this form, the fusion portion formed by the first fusion is greater than that formed by the second fusion in area, and the fusion portion formed by the second fusion does not extend beyond that formed by the first fusion.

[0012] In a fifth form, the fusion structure includes a fusion unit, the fusion unit includes a target fusion layer, a second fusion reinforcement layer, a second auxiliary fusion layer, a first fusion reinforcement layer and a first auxiliary fusion layer which are sequentially arranged, the second fusion reinforcement layer and the second auxiliary fusion layer are connected to the target fusion layer by a first fusion, the first fusion reinforcement layer and the first auxiliary fusion layer are connected to the second fusion reinforcement layer, the second auxiliary fusion layer and the target fusion layer by a second fusion, the fusion portion formed by the first fusion is greater than that formed by the second fusion in area, and the fusion portion formed by the second fusion does not extend beyond that formed by the first fusion. The fusion structure of this form is formed by additionally providing the second fusion reinforcement layer between the target fusion layer and the second auxiliary fusion layer based on the fusion structure of the fourth form. Because the second fusion reinforcement layer is provided, the fusion structure of this form is at least five times higher than the fusion structure of the fourth form in strength.

[0013] In a sixth form, the fusion structure includes a fusion unit, and the fusion unit includes a third auxiliary fusion layer, a third fusion reinforcement layer, a target fusion layer, a second auxiliary fusion layer, a first fusion reinforcement layer and a first auxiliary fusion layer which are sequentially arranged. The third auxiliary fusion layer, the third fusion reinforcement layer and the second auxiliary fusion layer are connected to the target fusion layer by a first fusion, the first fusion reinforcement layer and the first auxiliary fusion layer are connected to the third auxiliary fusion layer, the third fusion reinforcement layer, the second auxiliary fusion layer and the target fusion layer by a second fusion, the fusion portion formed by the first fusion is greater than that formed by the second fusion in area, and the fusion portion formed by the second fusion does not extend beyond that formed by the first fusion.

[0014] In a seventh form, the fusion structure includes a first fusion unit and a second fusion unit. The first fusion unit includes a target fusion layer, a first fusion reinforcement layer and a first auxiliary fusion layer which are sequentially arranged. The target fusion layer, the first fusion reinforcement layer and the first auxiliary fusion layer are connected by a first fusion. The second fusion unit is the same as the first fusion unit in structure. The first fusion unit and the second fusion unit are connected by a second fusion. The first auxiliary fusion layer of the first fusion unit is attached to the first auxiliary fusion layer of the second fusion unit. The fusion portion formed by the first fusion is greater than that formed by the second fusion in area. The fusion portion formed by the second fusion does not extend beyond the fusion portion formed by the first fusion. Actually, the fusion structure of this form is formed by two fusion structures of the first form fused together. The fusion structure of this form has the greatest strength among the above-described fusion structures.

[0015] The invention further provides various inflatable products including the above-described fusion structures, described as follows.

[0016] In a first form, an inflatable product includes a top sheet and a bottom sheet, and peripheries of the top sheet and the bottom sheet are connected by any of the fusion structures of the first, second, third, fourth, fifth and sixth forms. An air chamber is formed between the top sheet and the bottom sheet. The top sheet is the target fusion layer of the fusion structure, and the bottom sheet is the first auxiliary fusion layer of the fusion structure.

[0017] In a second form, an inflatable product includes a top sheet and a bottom sheet, and peripheries of the top sheet and the bottom sheet are connected by the fusion structure of the seventh form. An air chamber is formed between the top sheet and the bottom sheet. The top sheet is the target fusion layer of the first fusion unit, and the bottom sheet is the target fusion layer of the second fusion unit.

[0018] In a third form, an inflatable product includes a top sheet and a bottom sheet, and peripheries of the top sheet and the bottom sheet are directly fused. A plurality of linear fusion sites formed by any of the fusion structures of the first, second, third, fourth, fifth and sixth forms are provided between the top sheet and the bottom sheet. The fusion sites are arranged to form a plurality of spaces between the top sheet and the bottom sheet into air chambers with a predetermined shape. The top sheet is the target fusion layer of the fusion structure, and the bottom sheet is the first auxiliary fusion layer of the fusion structure.

[0019] In a fourth form, an inflatable product includes a top sheet and a bottom sheet, and peripheries of the top sheet and the bottom sheet are directly fused. A plurality of linear fusion sites which are formed by the fusion structure of the seventh form are provided between the top sheet and the bottom sheet. The fusion sites are arranged to form a plurality of spaces between the top sheet and the bottom sheet into air chambers with a predetermined shape. The top sheet is the target fusion layer of the first fusion unit, and the bottom sheet is the first auxiliary fusion layer of the second fusion unit.

[0020] In a fifth form, an inflatable product includes a top sheet, a plurality of tensioning strips and a bottom sheet. Peripheries of the top sheet and the bottom sheet are directly fused, and the tensioning strips are connected to inner surfaces of the top sheet and the bottom sheet by any of the fusion structures of the first, second, third, fourth, fifth and sixth forms. The top sheet and the bottom sheet are the target fusion layer of the fusion structure, and the tensioning strips are the first fusion reinforcement layer of the fusion structure.

[0021] In a sixth form, an inflatable product includes a top sheet, a periphery sheet, a plurality of tensioning strips and a bottom sheet. Two sides of the periphery sheet are directly connected to peripheries of the top sheet and the bottom sheet by forming an air chamber, and the tensioning strips are connected to inner surfaces of the top sheet and the bottom sheet by any of the fusion structures of the first, second, third, fourth, fifth and sixth forms. The top sheet and the bottom sheet are the target fusion layer of the fusion structure, and the tensioning strips are the first fusion reinforcement layer of the fusion structure.

[0022] In a seventh form, an inflatable product includes a top sheet, a periphery sheet, a plurality of tensioning strips and a bottom sheet. Two sides of the periphery sheet are connected to peripheries of the top sheet and the bottom sheet by any of the fusion structures of the first, second, third, fourth, fifth and sixth forms for forming an air chamber. The top sheet and the bottom sheet are the target fusion layer of the fusion structure, and the periphery sheet is the first auxiliary fusion layer of the fusion structure. The tensioning strips are connected to inner surfaces of the top sheet and the bottom sheet by fusion.

[0023] In an eighth form, an inflatable product includes a top sheet, a periphery sheet, a plurality of tensioning strips and a bottom sheet. Two sides of the periphery sheet are connected to peripheries of the top sheet and the bottom sheet by the fusion structure of the seventh form for forming an air chamber. The top sheet and the bottom sheet are the target fusion layer of the first fusion unit of the fusion structure, and the periphery sheet is the target fusion layer of the second fusion unit of the fusion structure. The tensioning strips are connected to inner surfaces of the top sheet and the bottom sheet by fusion.

[0024] Further, the tensioning strips of the inflatable products of the third and fourth forms can be classified into two categories according to the structure thereof and the fusion way that the tensioning strips are connected to the inner surfaces of the top sheet and the bottom sheet. A first category is that the tensioning strips are connected to the inner surfaces of the top sheet and the bottom sheet by any of the fusion structures of the first, second, third, fourth, fifth and sixth forms. The top sheet and the bottom sheet are the target fusion layer of the fusion structure, and the tensioning strips are the first fusion reinforcement layer of the fusion structure.

[0025] A second category is that each of the tensioning strips includes a strip body, and the strip body is fused with the inner surfaces of the top sheet and the bottom sheet at a location where the first auxiliary fusion piece and the second auxiliary fusion piece are provided to hold and fuse with the strip body. The strip body is made of a flexible anti-stretch material having a plurality of through holes, and the strip body is a mesh fabric. The first auxiliary fusion piece and the second auxiliary fusion piece are made of the same material as the auxiliary fusion layer of the above-described fusion structure. Alternatively, the first auxiliary fusion piece or the second auxiliary fusion piece is fixed to a side of the strip body.

[0026] Further, the tensioning strips are arranged between the top sheet and the bottom sheet in a serration pattern, and most of the teeth of the serration pattern are trapezoidal or triangular.

[0027] Further, the inner surfaces of the top sheet and the bottom sheet can be directly connected to the tensioning strips by fusion. Alternatively, a variety of auxiliary structures can be provided at the locations where the inner surfaces of the top sheet and the bottom sheet are connected to the tensioning strips. A first auxiliary structure is a third auxiliary fusion piece which is provided at the location where at least one of the outer surfaces and the inner surfaces of the top sheet and the bottom sheet fuses with the tensioning strips. A second auxiliary structure is a first auxiliary reinforcement piece and a third auxiliary fusion piece which are provided at the location where at least one of the outer surfaces and inner surfaces of the top sheet and the bottom sheet fuses with the tensioning strips. The first auxiliary reinforcement piece is attached to at least one of the outer surfaces and the inner surfaces of the top sheet and the bottom sheet, and the third auxiliary fusion piece is placed over the first auxiliary reinforcement piece. The auxiliary structure can directly fuse with at least one of the outer surfaces and the inner surfaces of the top sheet and the bottom sheet, and then the tensioning strips are connected thereto by fusion. Alternatively, the auxiliary structure is kept in position and then the tensioning strips are fused with the auxiliary structure and at least one of the outer surfaces and the inner surfaces of the top sheet and the bottom sheet. A third auxiliary structure is a fiber glue layer which has a plurality of through holes and is applied to the location where at least one of outer surfaces and the inner surfaces of the top sheet and the bottom sheet fuses with the tensioning strips. The third auxiliary fusion piece is made of a material which is the same as or compatible with the auxiliary fusion layer of the above-described fusion structure. The first auxiliary reinforcement piece is made of a material which is the same as or compatible with the fusion reinforcement layer of the above-described fusion structure.

[0028] Further, a thickness of the top sheet and the bottom sheet without the fusion structure is equal to or is smaller than one-third of a thickness of the top sheet and the bottom sheet with the fusion structure which is formed by including the auxiliary fusion layer but excluding the fusion reinforcement layer. The fusion structure can greatly enhance the ability of the inflatable product to resist stretch at the fusion locations. Thus, there is no need to thicken the top sheet and the bottom sheet of the inflatable product. Also, the cost and the weight of the inflatable product can be decreased.

[0029] The invention further provides a method for manufacturing the tensioning strips of the second category described above, wherein the method includes the following steps:

[0030] In step **S1**, a roll of mesh fabric is loaded and unrolled to a predetermined length.

[0031] In step **S2**, a roll of the first auxiliary fusion piece and a roll of the second auxiliary fusion piece are cut into strips and the strips are sent and positioned at predetermined locations on an upper side and a lower side of the mesh fabric. Alternatively, a roll of the first auxiliary fusion piece or a roll of the second auxiliary fusion piece is cut into strips and the strips are sent and positioned on a side of the mesh fabric.

[0032] In step **S3**, the strips cut from the first auxiliary fusion piece and the second auxiliary fusion piece are placed to hold the mesh fabric and fuse with the mesh fabric. Alternatively, the strips cut from the first auxiliary fusion piece or the second auxiliary fusion piece are fixed to a side of the mesh fabric.

[0033] In step **S4**, the mesh fabric which fuses with the strips cut from the first auxiliary fusion piece and the second auxiliary fusion piece is cut into strips for obtaining tensioning strips. Alternatively, the mesh fabric, a side of which has the strips cut from the first auxiliary fusion piece or the second auxiliary fusion piece attached thereto, is cut into strips for obtaining the tensioning strips.

[0034] An alternative method for manufacturing the tensioning strips includes the following steps:

[0035] In step **S1**, a roll of mesh fabric is loaded and unrolled.

[0036] In step **S2**, the mesh fabric is cut into strips for obtaining the strip body.

[0037] In step **S3**, the first auxiliary fusion piece and the second auxiliary fusion piece are respectively sent and positioned at predetermined locations on an upper side and a lower side of the strip body. Alternatively, the first auxiliary fusion piece or the second auxiliary fusion piece is sent and positioned at predetermined location on a side of the strip body.

[0038] In step **S4**, the first auxiliary fusion piece and the second auxiliary fusion piece are placed to hold the strip body and fuse with the strip body for obtaining the tensioning strips. Alternatively, the first auxiliary fusion piece or the second auxiliary fusion piece is fixed to a side of the strip body for obtaining the tensioning strips.

[0039] The present embodiment further provide a method for manufacturing the above-described inflatable product of the seventh or eighth form including the tensioning strips of the second category. The method includes following steps.

[0040] In step **S1**, rolls of the top sheet and the bottom sheet are loaded, unrolled and kept in position.

[0041] In step **S2**, the tensioning strips, which are manufactured by the above-described method, are sent and positioned at predetermined locations between the top sheet and the bottom sheet.

[0042] In step **S3**, the parts of the tensioning strips which have the first auxiliary fusion pieces and the second auxiliary fusion pieces thereon are fused with the inner surfaces of the top sheet and the bottom sheet. Alternatively, the parts of the tensioning strips, a side of which has the first auxiliary fusion pieces or the second auxiliary fusion pieces fixed thereto, are fused with the inner surfaces of the top sheet and the bottom sheet.

[0043] In step **S4**, the sides of the periphery sheet are connected to the peripheries of the top sheet and the bottom sheet by any of the fusion structures described in the first, second, third, fourth, fifth and sixth forms for forming the air chamber and obtaining the inflatable product.

[0044] Alternatively, the method for manufacturing the inflatable product includes following steps:

[0045] In step **S1**, rolls of the top sheet and the bottom sheet are loaded and unrolled and kept in position.

[0046] In step **S2**, the third auxiliary fusion pieces are fused with the outer surfaces or the inner surfaces of the top sheet and the bottom sheet at predetermined locations where the tensioning strips are going to fuse with the top sheet and the bottom sheet.

[0047] In step **S3**, the tensioning strips which are manufactured by the above-described method are sent and kept at the predetermined locations between the top sheet and the bottom sheet.

[0048] In step **S4**, the parts of the tensioning strips having the first auxiliary fusion pieces and the second auxiliary fusion pieces thereon are fused with the parts of the inner surfaces of the top sheet and the bottom sheet having the third auxiliary fusion pieces thereon. Alternatively, the parts of the tensioning strips, each of which has the first auxiliary fusion piece or the second auxiliary fusion piece fixed to a side thereof, are fused with the parts of the inner surfaces of the top sheet and the bottom sheet which have the third auxiliary fusion pieces fixed thereto.

[0049] In step **S5**, the sides of the periphery sheet are connected to the peripheries of the top sheet and the bottom sheet by any of the fusion structures described in the first, second, third, fourth, fifth, sixth and seventh forms for forming the air chamber and obtaining the inflatable product.

[0050] Further alternatively, a method for manufacturing the inflatable product includes the following steps:

[0051] In step **S1**, rolls of the top sheet and the bottom sheet are loaded, unrolled and kept in position.

[0052] In step **S2**, the first auxiliary reinforcement pieces and the third auxiliary fusion pieces are placed on the outer surfaces or the inner surfaces of the top sheet and the bottom sheet at predetermined locations where the top sheet and the bottom sheet are going to fuse with the tensioning strips, wherein the first auxiliary reinforcement pieces and the third auxiliary fusion pieces are only kept in position without fusing with the top sheet and the bottom sheet.

[0053] In step **S3**, the tensioning strips which are manufactured by the above-described method are sent and positioned at predetermined locations between the top sheet and the bottom sheet.

[0054] In step **S4**, the parts of the tensioning strips which have the first auxiliary fusion pieces and the second auxiliary fusion pieces thereon are fused with the parts of the inner surfaces of the top sheet and the bottom sheet which have the first auxiliary reinforcement pieces and the third auxiliary fusion pieces thereon. Alternatively, the parts of the tensioning strips, each of which has the first auxiliary fusion piece or the second auxiliary fusion piece fixed to a side thereof, are fused with the parts of the inner surfaces of the top sheet and the bottom sheet which have the third auxiliary fusion pieces fixed thereto.

[0055] In step **S5**, the sides of the periphery sheet are connected to the peripheries of the top sheet and the bottom sheet by any of the fusion structures described in the first, second, third, fourth, fifth, sixth and seventh forms for forming the air chamber and obtaining the inflatable product.

[0056] Yet further alternatively, a method for manufacturing the inflatable product includes following steps:

[0057] In step **S1**, rolls of the top sheet and the bottom sheet are loaded, unrolled and kept in position.

[0058] In step **S2**, the first auxiliary reinforcement pieces and the third auxiliary fusion pieces are placed on the outer surfaces or the inner surfaces of the top sheet and the bottom sheet at predetermined locations where the top sheet and the bottom sheet are going to fuse with the tensioning strips, and the first auxiliary reinforcement pieces and the third auxiliary fusion pieces are fused with the inner surfaces of the top sheet and the bottom sheet.

[0059] In step S**3**, the tensioning strips which are manufactured by the above-described method are sent and positioned at predetermined locations between the top sheet and the bottom sheet.

[0060] In step S**4**, the parts of the tensioning strips which have the first auxiliary fusion pieces and the second auxiliary fusion pieces thereon are fused with the parts of the inner surfaces of the top sheet and the bottom sheet which have the first auxiliary reinforcement pieces and the third auxiliary fusion pieces thereon. Alternatively, the parts of the tensioning strips, each of which has the first auxiliary fusion piece or the second auxiliary fusion piece fixed to a side thereof, are fused with the parts of the inner surfaces of the top sheet and the bottom sheet which have the first auxiliary reinforcement pieces and the third auxiliary fusion pieces thereon.

[0061] In step S**5**, the sides of the periphery sheet **10** are connected to the peripheries of the top sheet **8** and the bottom sheet **9** by any of the fusion structures described in the first, second, third, fourth, fifth, sixth and seventh forms for forming the air chamber and obtaining the inflatable product.

[0062] Further alternatively, a method for manufacturing the inflatable product includes the following steps:

[0063] In step S**1**, rolls of the top sheet and the bottom sheet are loaded, unrolled and kept in position.

[0064] In step S**2**, a fiber glue layer which has a plurality of through holes is formed on the outer surfaces or the inner surfaces of the top sheet and the bottom sheet at predetermined locations where the top sheet and the bottom sheet are going to fuse with the tensioning strips.

[0065] In step S**3**, the tensioning strips which are manufactured by the above-described method are sent and positioned at predetermined locations between the top sheet and the bottom sheet.

[0066] In step S**4**, the parts of the tensioning strips which have the first auxiliary fusion pieces and the second auxiliary fusion pieces thereon are fused with the parts of the inner surfaces of the top sheet and the bottom sheet which have the fiber glue layer formed thereon, wherein the fiber glue layer has through holes. Alternatively, the parts of the tensioning strips, each of which has the first auxiliary fusion piece or the second auxiliary fusion piece fixed to a side thereof, are fused with the parts of the inner surfaces of the top sheet and the bottom sheet which have the fiber glue layer formed thereon, wherein the fiber glue layer has through holes.

[0067] In step S**5**, the sides of the periphery sheet are connected to the peripheries of the top sheet and the bottom sheet by any of the fusion structures described in the first, second, third, fourth, fifth, sixth and seventh forms for forming the air chamber and obtaining the inflatable product.

[0068] The invention further provides various inflatable products rapidly produced and provided with covering layers simultaneously positioned with inflatable bodies, described as follows.

[0069] The inflatable product of a first form includes at least one inflatable body, and the inflatable body is provided with a fusion portion formed thereon. An anti-stretch sheet is disposed outside or within the inflatable body for covering the inflatable body or being covered by the inflatable body. The anti-stretch sheet is provided with a fusion portion formed thereon. The fusion portion of the inflatable body and the fusion portion of the anti-stretch sheet are fixed to each other.

[0070] The inflatable body is made of an airtight polymer material, such as plastic film (different kinds of plastic films can be selected according to different products, e.g. polyvinyl chloride, PVC). The anti-stretch sheet is made of an anti-stretch material, such as fabric, net (mesh fabric), and threads (weaving mesh). When the anti-stretch sheet covers the inflatable body, the inflatable body is prevented from being excessively expanded because of excessive pressure. When the anti-stretch sheet covers the inflatable body, the load bearing capacity of the inflatable body is improved. It is understood that the anti-stretch sheet is a material that can prevent the inflatable body from rupture because of excessive pressure, and the inflatable body is restricted thereby.

[0071] The inflatable body is capable of sealing air therein, and the anti-stretch sheet is capable of restricting the inflatable body. Generally, the materials of the inflatable body and the anti-stretch sheet are different. When the fusion portions of the inflatable body and the anti-stretch sheet are fused together, the connection therebetween is liable to failure. Therefore, the fusion portion of the anti-stretch sheet has penetrability, and the material of the inflatable body is melted into through holes of the anti-stretch sheet during the fusion and is restricted by the through holes. The direction of restriction is the same as the direction of pulling force produced by expansion of the inflatable product during inflation. It is similar to reinforcement steels in concrete, and therefore ability of the fusion portion to resist stretch is greatly enhanced.

[0072] The inflatable body can be entirely or partly covered by the anti-stretch sheet. Similarly, the anti-stretch sheet body can be entirely or partly covered by the inflatable body.

[0073] "To fix" or "to be fixed" described in this specification is for purposes of formation of an airtight inflatable body after the fusion portions are firmly connected. "To fix" or "to be fixed" can be done by various ways which include fusion, joining, pressing, hot pressing, gluing, sewing, means of Velcro, latching and so on.

[0074] For improving the ability of the fusion portion to resist stretch, the inflatable product further includes at least one airtight auxiliary fusion layer configured to fix to the fusion portions of the inflatable body and the anti-stretch sheet. The material of the airtight auxiliary fusion layer is the same as or compatible with the material of the inflatable body. During the fusion, material of the airtight auxiliary fusion layer is melted into the through holes, so that the ability of the fusion portion to resist stretch is further improved.

[0075] The fusion portions of the airtight auxiliary fusion layer and the anti-stretch sheet are placed in contact with each other and firmly connected. The airtight auxiliary fusion layer is adjacent to the anti-stretch sheet, so that the material thereof is easier to enter the through holes.

[0076] The inflatable product further includes at least one fusion reinforcement layer configured to fix to the fusion portions of the inflatable body and the anti-stretch sheet. The fusion reinforcement layer is capable of improving the strength of the fusion portion, so as to prevent rupture.

[0077] The fusion reinforcement layer is made of an anti-stretch material, and/or the structure of the fusion reinforcement layer is the same as that of the anti-stretch sheet. The fusion reinforcement layer is placed in contact with and is fixed to the fusion portion of the inflatable product and/or the airtight auxiliary fusion layer. The fusion reinforcement layer is adjacent to the fusion portion of the inflatable body or the airtight auxiliary fusion layer. The material of the inflatable body is easily melted into the through holes of the fusion reinforcement layer and kept therein.

[0078] Formation of penetrability can be done by various ways which includes punching, weaving, threads and forming spaces between fibers.

[0079] The fusion portions can be formed to be right-angled, bent, curved, beveled, stacked, annular, circular, elliptical, triangular or flat.

[0080] The inflatable body can include an assembly of columns, a bed body, a seat, a deckchair, a cushion body, a concave body, a box-shaped body, a boat-shaped body, a skateboard body or an airtight body.

[0081] The inflatable product of a second form includes at least one inflatable body, and the inflatable body is provided with a fusion structure formed thereon. The fusion structure includes a first fusion unit and a second fusion unit which are fixed to each other. The first fusion unit and/or the second fusion unit is fixed in a way that two airtight auxiliary fusion layers are placed to hold a fusion portion of an anti-stretch sheet and are then fixed to a fusion potion of the inflatable body.

[0082] The inflatable product of a third form includes at least one inflatable body, and the inflatable body is provided with a fusion structure formed thereon. The fusion structure includes a first fusion unit and a second fusion unit which are fixed to each other. The first fusion unit and/or the second fusion unit is fixed in a way that an airtight auxiliary fusion layer or a fusion reinforcement layer is fixed to a fusion portion of an anti-stretch sheet by sewing and is then fixed to a fusion potion of the inflatable body. The fusion potion of the inflatable body is in contact with the airtight auxiliary fusion layer or the fusion reinforcement layer.

[0083] The inflatable product of a fourth form includes at least one inflatable body, and the inflatable body is provided with a fusion structure formed thereon. The fusion structure includes a first fusion unit and a second fusion unit which are fixed to each other. The first fusion unit and/or the second fusion unit is fixed in a way that an airtight auxiliary fusion layer or a fusion reinforcement layer is fixed to a fusion portion of an anti-stretch sheet by sewing, and the airtight auxiliary fusion layer or the fusion reinforcement layer is fixed to another airtight auxiliary fusion layer and then fixed to a fusion potion of the inflatable body. The fusion potion of the inflatable body is in contact with the airtight auxiliary fusion layer.

[0084] The inflatable product of a fifth form includes at least one inflatable body, and the inflatable body is provided with a fusion structure formed thereon. The fusion structure includes a first fusion unit and a second fusion unit which are fixed to each other. The first fusion unit and/or the second fusion unit is fixed in a way that a portion of a fusion reinforcement layer is wrapped by a cloth and is then fixed to a fusion portion of an anti-stretch sheet by sewing. Two airtight auxiliary fusion layers are placed to hold the unwrapped portion of the fusion reinforcement layer (not wrapped by the cloth) and are then fixed to a fusion portion of the inflatable body. The fusion potion of the inflatable body is in contact with the airtight auxiliary fusion layers.

[0085] Comparing to the prior art, the invention has following advantages:

- **[0086]** 1. The fusion structure of the invention is that the target fusion layer and at least one auxiliary fusion layer are fused together with at least one fusion reinforcement layer held therebetween. During the fusion, the target fusion layer and the auxiliary fusion layer are melted into the through holes of the fusion reinforcement layer and are restricted by an anti-stretch material at an edge of the through holes, and the direction of restriction is the same as the direction of pulling force produced by expansion of the inflatable body during inflation. It is similar to reinforcement steels in concrete. Therefore, ability of the inflatable body to resist stretch at the fusion site is greatly enhanced. That is, ability of the inflatable body to resist pressure of inflation is relatively enhanced, so as to improve the durability of the inflatable product.
- **[0087]** 2. It is required that an inflatable product is not liable to rupture at the fusion location. Therefore, the top sheet and the bottom sheet of the conventional inflatable product are thickened, so that the ability of the inflatable product to resist stretch at the fusion location can meet the requirement. The fusion structure of the invention can greatly improve the ability of the inflatable product to resist stretch at the fusion location, so that thickening the top sheet and the bottom sheet of the inflatable product is not necessary. Therefore, the cost can be saved, and the weight of the inflatable product can be decreased.
- **[0088]** 3. The anti-stretch sheet of the application covers or is covered by the inflatable body. When the pressure in the air chamber formed by the inflatable body is excessive, the anti-stretch sheet restricts the inflatable body, thereby improving the ability of the inflatable body to resist pressure and preventing rupture.
- **[0089]** 4. In the product of the invention, the fusion portion of the inflatable body, the fusion portion of the anti-stretch sheet, and the fusion portion of the airtight auxiliary fusion layer (or the fusion portion of the fusion reinforcement layer) can be simultaneously fixed at the same location, and the inflatable body and the covering layer (one of them covers or is covered by the other) are positioned with respect to each other by fusion.

[0090] The invention further provides an inflatable product that includes an inflatable body and a fiber-contained reinforcement element. The fiber-contained reinforcement element is connected to the inflatable body so as to increase mechanical strength of the inflatable product at a location where the fiber-contained reinforcement element is connected to the inflatable body. The fiber-contained reinforcement element includes fibers or a derivative thereof generated by processing fibers.

[0091] The fiber-contained reinforcement element may be made of fusible material with fibers added therein or applied thereto. Further, the fiber-contained reinforcement element is connected to the inflatable body by fusion.

[0092] The fiber-contained reinforcement element may include a portion. The portion has through holes. The portion is connected to the inflatable body by fusion. The through holes are configured to receive a melted portion of the inflatable body.

[0093] The fiber-contained reinforcement element comprises a portion. The portion has no fibers added therein or applied thereto. The portion is connected to the inflatable body by fusion.

[0094] The inflatable product may further include an anti-stretch sheet which has an edge sewed with the fiber-contained reinforcement element.

[0095] The inflatable product may further include an anti-stretch sheet which has an edge folded and sewed with the fiber-contained reinforcement element.

[0096] The inflatable body may include a first portion and a second portion. The fiber-contained reinforcement element is disposed between the first portion and the second portion. The first portion and the second portion are connected through the fiber-contained reinforcement element by fusion.

[0097] The fiber-contained reinforcement element may be disposed in the inflatable body. The inflatable body includes a first portion and a second portion. The fiber-contained reinforcement element is connected to the first portion and the second portion by fusion, thereby controlling expansion of the inflatable body when the inflatable body is inflated.

[0098] The inflatable body includes a first portion and a second portion connected to each other by fusion. The fiber-contained reinforcement element may include glue with fibers added therein. Further, the fiber-contained reinforcement element is disposed between the first portion and the second portion.

[0099] The inflatable body includes a first portion and a second portion. The fiber-contained reinforcement element comprises fusible material with fibers added therein or applied thereto. The fiber-contained reinforcement element is disposed between the first portion and the second portion. The first portion, the fiber-contained reinforcement element and the second portion are connected by fusion.

[0100] The inflatable body includes a first portion and a second portion connected to each other by fusion. The fiber-contained reinforcement element may include glue with fibers added therein. Further, the fiber-contained reinforcement element is externally disposed on the first portion and/or the second portion.

[0101] The inflatable body includes a first portion and a second portion. The fiber-contained reinforcement element comprises fusible material with fibers added therein or applied thereto. The fiber-contained reinforcement element is externally disposed on the first portion and/or the second portion. The first portion, the second portion and the fiber-contained reinforcement element are connected by fusion.

[0102] The inflatable product may further include an anti-stretch layer configured to cover the inflatable body. The fiber-contained reinforcement element includes a first auxiliary fusion sheet connected to the inflatable body by fusion, and a fusion reinforcement sheet connected to the first auxiliary fusion sheet by sewing.

[0103] The fiber-contained reinforcement element may further include a second auxiliary fusion sheet wherein the second auxiliary fusion sheet, the fusion reinforcement sheet and the first auxiliary fusion sheet are connected by sewing.

BRIEF DESCRIPTION OF THE DRAWINGS

[0104] FIG. **1**A is a schematic view of a first embodiment;

[0105] FIG. **1**B is a partial view of the first embodiment;

[0106] FIG. **2** is a schematic view of a second embodiment;

[0107] FIG. **3** is a schematic view of a third embodiment;

[0108] FIG. **4** is a schematic view of a fourth embodiment;

[0109] FIG. **5** is a schematic view of a fifth embodiment;

[0110] FIG. **6** is a schematic view of a sixth embodiment;

[0111] FIG. **7** is a schematic view of a seventh embodiment;

[0112] FIG. **8**A is a schematic view of an eighth embodiment wherein an inflatable product includes a top sheet and a bottom sheet, the peripheries of which are fused by the fusion structure described in the first embodiment;

[0113] FIG. **8**B is a schematic view of the eighth embodiment wherein an inflatable product includes a top sheet and a bottom sheet, the peripheries of which are fused by the fusion structure described in the second embodiment;

[0114] FIG. **8**C is a schematic view of the eighth embodiment wherein an inflatable product includes a top sheet and a bottom sheet, the peripheries of which are fused by the fusion structure described in the third embodiment;

[0115] FIG. **8**D is a schematic view of the eighth embodiment wherein an inflatable product includes a top sheet and a bottom sheet, the peripheries of which are fused by the fusion structure described in the fourth embodiment;

[0116] FIG. **8**E is a schematic view of the eighth embodiment wherein an inflatable product includes a top sheet and a bottom sheet, the peripheries of which are fused by the fusion structure described in the fifth embodiment;

[0117] FIG. **8**F is a schematic view of the eighth embodiment wherein an inflatable product includes a top sheet and a bottom sheet, the peripheries of which are fused by the fusion structure described in the sixth embodiment;

[0118] FIG. **9** is a schematic view of a ninth embodiment;

[0119] FIG. **10** is a schematic view of a tenth embodiment;

[0120] FIG. **11** is a schematic view of an eleventh embodiment;

[0121] FIG. **12** is a schematic view of a twelfth embodiment;

[0122] FIG. **13** is a schematic view of a thirteenth embodiment;

[0123] FIG. **14** is a schematic view of a sixteenth embodiment;

[0124] FIG. **15** is a schematic view showing the process for manufacturing tensioning strips of the sixteenth embodiment;

[0125] FIG. **16** is a schematic view showing the process for manufacturing tensioning strips of the sixteenth embodiment;

[0126] FIG. **17** is an enlarged view of portion "B" of FIG. **16**;

[0127] FIG. **18** is a schematic view showing the process for manufacturing inflatable product of the sixteenth embodiment;

[0128] FIG. **19** is an enlarged view of portion "C" of FIG. **18** showing the outer surface or inner surface of a top sheet and a bottom sheer of the sixteenth embodiment has a third auxiliary fusion piece thereon;

[0129] FIG. **20** is an enlarged view of portion "C" of FIG. **18** showing the outer surface or inner surface of a top sheet and a bottom sheer of the sixteenth embodiment has a third auxiliary fusion piece and a first auxiliary reinforcement piece thereon;

[0130] FIG. **21** is an enlarged view of portion "C" of FIG. **18** showing the outer surface or inner surface of a top sheet and a bottom sheer of the sixteenth embodiment has a fiber glue layer thereon;

[0131] FIG. **22** is a schematic view of an eighteenth embodiment;

[0132] FIG. **23** is a schematic view of a nineteenth embodiment;

[0133] FIG. **24** is a schematic view of a twentieth embodiment;

[0134] FIG. **25**A is a schematic view of a portion of FIG. **24**;

[0135] FIG. **25**B depicts the fusion of the portion of FIG. **25**A;

[0136] FIG. **26** is a schematic view of a twenty-first embodiment;

[0137] FIG. **27**A is a schematic view of a twenty-second embodiment;

[0138] FIG. **27**B is a schematic view of back sealing by fusion of FIG. **27**A;

[0139] FIG. **28** is a schematic view of a twenty-third embodiment;

[0140] FIG. **29** is a schematic view of a twenty-fourth embodiment;

[0141] FIG. **30** is a schematic view of a twenty-fifth embodiment;

[0142] FIG. **31** is a schematic view of a twenty-sixth embodiment;

[0143] FIG. **32** is a schematic view of a portion of FIG. **31**;

[0144] FIG. **33** is a schematic view of a twenty-seventh embodiment;

[0145] FIG. **34** is a schematic view of a twenty-eighth embodiment;

[0146] FIG. **35** is a schematic view of a portion of FIG. **34**;

[0147] FIG. **36** is a schematic view of a twenty-ninth embodiment;

[0148] FIG. **37** is a schematic view of a thirtieth embodiment;

[0149] FIG. **38** is a schematic view of the thirtieth embodiment;

[0150] FIG. **39** is a schematic view of a portion of FIG. **38**;

[0151] FIG. **40** is a schematic view of a thirty-first embodiment;

[0152] FIG. **41** is a schematic view of a thirty-second embodiment;

[0153] FIG. **42**A is a schematic view of a thirty-third embodiment wherein the sewing and fusion at an early stage is shown;

[0154] FIG. **42**B shows all the fusion at a late stage of FIG. **42**A;

[0155] FIG. **43**A is a schematic view of a thirty-fourth embodiment wherein the sewing and fusion at an early stage is shown;

[0156] FIG. **43**B shows all the fusion at a late stage of FIG. **43**A;

[0157] FIG. **44** is a schematic view of a thirty-fifth embodiment;

[0158] FIG. **45** is a schematic view of a thirty-sixth embodiment;

[0159] FIG. **46** is a schematic view of a thirty-seventh embodiment;

[0160] FIG. **47** is a schematic view of a thirty-eighth embodiment;

[0161] FIG. **48** is a schematic view of a thirty-ninth embodiment.

[0162] FIG. **49** is a schematic view of a fortieth embodiment.

[0163] FIG. **50** is a schematic view of a forty-first embodiment.

[0164] FIG. **51** is a schematic view of a forty-second embodiment.

[0165] FIG. **52** is a schematic view of a forty-third embodiment.

[0166] FIG. **53** is a schematic view of a forty-fourth embodiment.

[0167] FIG. **54** is a schematic view of a forty-fifth embodiment.

[0168] FIG. **55** is a schematic view of a forty-sixth embodiment or a forty-seventh embodiment.

[0169] FIG. **56** is a schematic view of a forty-eighth embodiment or a forty-ninth embodiment.

[0170] FIG. **57** is a schematic view of a fiftieth embodiment.

[0171] FIG. **58** depicts a conventional inflatable product and the fusion portions thereof.

DETAILED DESCRIPTION OF THE INVENTION

[0172] The drawings are for purposes of illustration only and are not to be construed as limiting the scope of the invention. For better description of the embodiments, some parts of the drawings may be omitted, enlarged or reduced, and do not represent the actual product size. It will be apparent to those skilled in the art that some known structures in the drawings and the descriptions thereof may be omitted. The position relationship shown in the drawings are for purposes of illustration only and are not to be construed as limiting the scope of the invention.

First Embodiment

[0173] Referring to FIGS. **1**A and **1**B, a fusion structure of an inflatable product includes a fusion unit, and the fusion unit includes a target fusion layer **1**, a first fusion reinforcement layer **2** and a first auxiliary fusion layer **3** which are sequentially arranged. The target fusion layer **1**, the first fusion reinforcement layer **2** and the first auxiliary fusion layer **3** are connected together by a single fusion. The target fusion layer **1** is made of an airtight polymer material (e.g. plastic sheet) and can be used as surface material (e.g. top sheet, bottom sheet or periphery sheet) of the inflatable product. The first fusion reinforcement layer **2** is made of a flexible anti-stretch material (e.g. mesh fabric) having a plurality of through holes and is configured to restrict or resist the pulling forces produced by expansion of the inflatable product. It is worth noting that since the materials of the target fusion layer **1** and the first fusion reinforcement layer **2** are different, the target fusion layer **1** and the first fusion reinforcement layer **2** cannot be directly fused together. The first auxiliary fusion layer **3** is made of the same material as the target fusion layer **1**, and therefore the first auxiliary fusion layer **3** and the target fusion layer **1** can be fused together. In other words, the first auxiliary fusion layer **3** not only can be used as a surface material (e.g. top sheet, bottom sheet or periphery sheet) of the inflatable product, but also enables connection of the first fusion reinforcement layer **2** to the target fusion layer **1** during the fusion.

Second Embodiment

[0174] Referring to FIG. **2**, a fusion structure of an inflatable product includes a fusion unit, and the fusion unit includes a target fusion layer **1**, a second auxiliary fusion layer **4**, a first fusion reinforcement layer **2** and a first auxiliary fusion layer **3** which are sequentially arranged. The target fusion layer **1**, the second auxiliary fusion layer **4**, the first fusion reinforcement layer **2** and the first auxiliary fusion layer **3** are connected together by a single fusion. The fusion structure of this form is formed by additionally providing the second auxiliary fusion layer **4** between the target fusion layer **1** and the first fusion reinforcement layer **2** of the fusion structure of the first embodiment. Because the second auxiliary fusion layer **4** is provided, the fusion structure of this form is several times higher than the fusion structure of the first embodiment in strength. The second auxiliary fusion layer **4** functions the same as the first auxiliary fusion layer **3** described in the first embodiment.

Third Embodiment

[0175] Referring to FIG. **3**, a fusion structure of an inflatable product includes a fusion unit, and the fusion unit includes a target fusion layer **1**, a second auxiliary fusion layer **4**, a first fusion reinforcement layer **2** and a first auxiliary fusion layer **3** which are sequentially arranged. A periphery of the second auxiliary fusion layer **4** is connected to the target fusion layer **1** by a first fusion, and the first fusion reinforcement layer **2** and the first auxiliary fusion layer **3** are connected to the second auxiliary fusion layer **4** and the target fusion layer **1** by a second fusion, wherein the fusion portions formed by the first fusion and the second fusion are not overlapped. The fusion structure of this form is similar to the fusion structure described in the second embodiment, and the difference therebetween is that the fusion structure of this form is formed by twice fusion. The first fusion is done for connecting the periphery of the second auxiliary fusion layer **4** and the target fusion layer **1**, and the second fusion is done for connecting the first fusion reinforcement layer **2**, the first auxiliary fusion layer **3**, the second auxiliary fusion layer **4** and the target fusion layer **1**.

Fourth Embodiment

[0176] Referring to FIG. **4**, a fusion structure of an inflatable product includes a fusion unit, and the fusion unit includes a target fusion layer **1**, a second auxiliary fusion layer **4**, a first fusion reinforcement layer **2** and a first auxiliary fusion layer **3** which are sequentially arranged. The second auxiliary fusion layer **4** is connected to the target fusion layer **1** by a first fusion, the first fusion reinforcement layer **2** and the first auxiliary fusion layer **3** are connected to the second auxiliary fusion layer **4** and the target fusion layer **1** by a second fusion, wherein a fusion portion formed by the first fusion is greater than that of the second fusion in area, and the fusion portion formed by the second fusion does not extend beyond that formed by the first fusion. The fusion structure of this form is the same as the fusion structure described in the third embodiment in structure, times of fusion and sequence of fusion, and the difference therebetween is that in the fusion structure of the third embodiment, the fusion portions formed by the first fusion and the second fusion are not overlapped while in the fusion structure of this form, the fusion portion formed by the first fusion is greater than that formed by the second fusion in area, and the fusion portion formed by the second fusion does not extend beyond that formed by the first fusion.

Fifth Embodiment

[0177] Referring to FIG. **5**, a fusion structure of an inflatable product includes a fusion unit, and the fusion unit includes a target fusion layer **1**, a second fusion reinforcement layer **5**, a second auxiliary fusion layer **4**, a first fusion reinforcement layer **2** and a first auxiliary fusion layer **3** which are sequentially arranged. The second fusion reinforcement layer **5** and the second auxiliary fusion layer **4** are connected to the target fusion layer **1** by a first fusion, and the first fusion reinforcement layer **2** and the first auxiliary fusion layer **3** are connected to the second fusion reinforcement layer **5**, the second auxiliary fusion layer **4** and the target fusion layer **1** by a second fusion, wherein a fusion portion formed by the first fusion is greater than that formed by the second fusion in area, and the fusion portion formed by the second fusion does not extend beyond that formed by the first fusion. The fusion structure of this form is formed by additionally providing the second fusion reinforcement layer **5** between the target fusion layer **1** and the second auxiliary fusion layer **4** of the fusion structure of the fourth embodiment. Because the second fusion reinforcement layer **5** is provided, the fusion structure of this form is at least five times higher than the fusion structure of the fourth embodiment in strength. The second fusion reinforcement layer **5** functions the same as the first fusion reinforcement layer **2** described in the first embodiment.

Sixth Embodiment

[0178] Referring to FIG. **6**, a fusion structure of an inflatable product includes a fusion unit, and the fusion unit includes a third auxiliary fusion layer **6**, a third fusion reinforcement layer **7**, a target fusion layer **1**, a second auxiliary fusion layer **4**, a first fusion reinforcement layer **2** and a first auxiliary fusion layer **3** which are sequentially arranged. The third auxiliary fusion layer **6**, the third fusion reinforcement layer **7** and the second auxiliary fusion layer **4** are connected to the target fusion layer **1** by a first fusion, and the first fusion reinforcement layer **2** and the first auxiliary fusion layer **3** are connected to the third auxiliary fusion layer **6**, the third fusion reinforcement layer **7**, the second auxiliary fusion layer **4** and the target fusion layer **1** by a second fusion, wherein a fusion portion formed by the first fusion is greater than that formed by the second fusion in area, and the fusion portion formed by the second fusion does not extend beyond that formed by the first fusion. The third auxiliary fusion layer **6** functions the same as the first auxiliary fusion layer **3** described in the first embodiment, and the third fusion reinforcement layer **7** functions the same as the first fusion reinforcement layer **2** described in the first embodiment.

Seventh Embodiment

[0179] Referring to FIG. **7**, a fusion structure of an inflatable product includes a first fusion unit and a second fusion unit. The first fusion unit includes a target fusion layer **1**, a first fusion reinforcement layer **2** and a first auxiliary fusion layer **3** which are sequentially arranged. The target fusion layer **1**, the first fusion reinforcement layer **2** and the first auxiliary fusion layer **3** are connected by a first fusion. The second fusion unit is same as the first fusion unit in structure, and the first fusion unit and the second fusion unit are connected by a second fusion. The first auxiliary fusion layer **3** of the first fusion unit is attached to the first auxiliary fusion layer **3** of the second fusion unit. A fusion portion formed by the first fusion is greater than that formed by the second fusion in area, and the fusion portion formed by the second fusion does not extend beyond that formed by the first fusion. Actually, the fusion structure of this form is formed by fusing two fusion structures described in the first embodiment. The fusion structure of this form has the greatest strength among the above-described fusion structures.

Eighth Embodiment

[0180] An inflatable product includes a top sheet **8** and a bottom sheet **9**, and peripheries of the top sheet **8** and the bottom sheet **9** are fused by any of the fusion structures described in the first, second, third, fourth, fifth and sixth embodiments, respectively shown in FIGS. **8**A, **8**B, **8**C, **8**D, **8**E and **8**F. An air chamber is formed between the top sheet **8** and the bottom sheet **9**. The top sheet **8** can be regarded as the target fusion layer **1** of the fusion structure, and the bottom sheet **9** can be regarded as the first auxiliary fusion layer **3** of the fusion structure.

Ninth Embodiment

[0181] Referring to FIG. **9**, an inflatable product includes a top sheet **8** and a bottom sheet **9**, and peripheries of the top sheet **8** and the bottom sheet **9** are fused by the fusion structure described in the seventh embodiment. An air chamber is formed between the top sheet **8** and the bottom sheet **9**. The top sheet **8** can be regarded as the target fusion layer **1** of the first fusion unit, and the bottom sheet **9** can be regarded as the target fusion layer **1** of the second fusion unit.

Tenth Embodiment

[0182] Referring to FIG. **10**, an inflatable product includes a top sheet **8** and a bottom sheet **9**, and peripheries of the top sheet **8** and the bottom sheet **9** are directly fused to form a plurality of linear fusion sites where the fusion structures described in the first, second, third, fourth, fifth and sixth embodiments are formed. The fusion sites are arranged to form a plurality of spaces between the top sheet **8** and the bottom sheet **9** into air chambers with a predetermined shape. The top sheet **8** can be regarded as the target fusion layer **1** of the fusion structure, and the bottom sheet **9** can be regarded as the first auxiliary fusion layer **3** of the fusion structure.

Eleventh Embodiment

[0183] Referring to FIG. **11**, an inflatable product includes a top sheet **8** and a bottom sheet **9**, and peripheries of the top sheet **8** and the bottom sheet **9** are directly fused to form a plurality of linear fusion sites where the fusion structure described in the seventh embodiment are formed. The fusion sites are arranged so that a plurality of spaces formed between the top sheet **8** and the bottom sheet **9** are air chambers with a predetermined shape. The top sheet **8** can be regarded as the target fusion layer **1** of the first fusion unit, and the bottom sheet **9** can be regarded as the first auxiliary fusion layer **3** of the second fusion unit.

Twelfth Embodiment

[0184] Referring to FIG. **12**, an inflatable product includes a top sheet **8**, a plurality of tensioning strips **11** and a bottom sheet **9**. Peripheries of the top sheet **8** and the bottom sheet **9** are directly fused, and the tensioning strips **11** are connected to inner surfaces of the top sheet **8** and the bottom sheet **9** by any of the fusion structures described in the first, second, third, fourth, fifth and sixth embodiments. The top sheet **8** and the bottom sheet **9** can be regarded as the target fusion layer **1** of the fusion structure, and the tensioning strips **11** can be regarded as the first fusion reinforcement layer **2** of the fusion structure.

Thirteenth embodiment

[0185] Referring to FIG. **13**, an inflatable product includes a top sheet **8**, a periphery sheet **10**, a plurality of tensioning strips **11** and a bottom sheet **9**. Two sides of the periphery sheet **10** are directly welded to peripheries of the top sheet **8** and the bottom sheet **9** for forming an air chamber, and the tensioning strips **11** are welded to inner surfaces of the top sheet **8** and the bottom sheet **9** by any of the fusion structures described in the first, second, third, fourth, fifth and sixth embodiments. The top sheet **8** and the bottom sheet **9** can be regarded as the target fusion layer **1** of the fusion structure, and the tensioning strips **11** can be regarded as the first fusion reinforcement layer **2** of the fusion structure.

Fourteenth Embodiment

[0186] An inflatable product includes a top sheet **8**, a periphery sheet **10**, a plurality of tensioning strips **11** and a bottom sheet **9**. Two sides of the periphery sheet **10** are connected to peripheries of the top sheet **8** and the bottom sheet **9** by any of the fusion structures described in the first, second, third, fourth, fifth and sixth embodiments for forming an air chamber. The top sheet **8** and the bottom sheet **9** can be regarded as the target fusion layer **1** of the fusion structure, and the periphery sheet **10** can be regarded as the first auxiliary fusion layer **3** of the fusion structure. The tensioning strips **11** are connected to inner surfaces of the top sheet **8** and the bottom sheet **9** by fusion.

[0187] The tensioning strips **11** are connected to the inner surfaces of the top sheet **8** and the bottom sheet **9** by any of the fusion structures described in the first, second, third, fourth, fifth and sixth embodiments. The top sheet **8** and the bottom sheet **9** can be regarded as the target fusion layer **1** of the fusion structure, and the tensioning strips **11** can be regarded as the first fusion reinforcement layer **2** of the fusion structure.

Fifteenth Embodiment

[0188] The fifteenth embodiment is similar to the fourteenth embodiment, and the difference therebetween is that two sides of the periphery sheet **10** is connected to peripheries of the top sheet **8** and the bottom sheet **9** by the fusion structure described in the seventh embodiment for forming an air chamber. The top sheet **8** and the bottom sheet **9** can be regarded as the target fusion layer **1** of the first fusion unit of the fusion structure, and the periphery sheet **10** can be regarded as the target fusion layer **1** of the second fusion unit of the fusion structure. The tensioning strips **11** are connected to inner surfaces of the top sheet **8** and the bottom sheet **9** by fusion.

Sixteenth Embodiment

[0189] Referring to FIGS. **14-21**, an inflatable product includes a top sheet **8**, a periphery sheet **10**, a plurality of tensioning strips **11** and a bottom sheet **9**. Two sides of the periphery sheet **10** are connected to peripheries of the top sheet **8** and the bottom sheet **9** by any of the fusion structures described in the first, second, third, fourth, fifth and sixth embodiments for forming an air chamber. The top sheet **8** and the bottom sheet **9** can be regarded as the target fusion layer **1** of the fusion structure, and the periphery sheet **10** can be regarded as the first auxiliary fusion layer **3** of the fusion structure. The tensioning strips **11** are connected to inner surfaces of the top sheet **8** and the bottom sheet **9** by fusion.

[0190] Each of the tensioning strips **11** includes a strip body **111**, a first auxiliary fusion piece **12** and a second auxiliary fusion piece **13**. The strip body **111** is connected to the inner surfaces of the top sheet **8** and the bottom sheet **9** by fusion where the first auxiliary fusion piece **12** and the second auxiliary fusion piece **13** are provided to hold and fuse with the strip body **111**. Alternatively, the first auxiliary fusion piece **12** or the second auxiliary fusion piece **13** is fixed to a side of the strip body **111**. The strip body **111** is made of a flexible anti-stretch material (e.g. mesh fabric) having a plurality of through holes and is configured to restrict or resist the pulling force produced by expansion of the inflatable product. The first auxiliary fusion piece **12** and the second auxiliary fusion piece **13** are made of the same material (that is, airtight polymer material) (e.g. plastic sheet) as the top sheet **8** and the bottom sheet **9** and enable connection of the strip body **111** to the inner surfaces of the top sheet **8** and the bottom sheet **9** during the fusion.

[0191] The inner surfaces of the top sheet **8** and the bottom sheet **9** can be directly connected to the tensioning strips **11** by fusion. Alternatively, a variety of auxiliary structures can be provided at the locations where the inner surfaces of the top sheet **8** and the bottom sheet **9** are connected to the tensioning strips **11**. The auxiliary structure can be a third auxiliary fusion piece **14** which is provided at the location where at least one of the inner surfaces of the top sheet **8** and the bottom sheet **9** fuses with the tensioning strips **11**. The auxiliary structure can also be a first auxiliary reinforcement piece **15** and a third auxiliary fusion piece **14** which are provided at the location where at least one of the outer surfaces and inner surfaces of the top sheet **8** and the bottom sheet **9** fuses with the tensioning strips **11**. The first auxiliary reinforcement piece **15** is attached to at least one of the outer surfaces and the inner surfaces of the top sheet **8** and the bottom sheet **9**, and the third auxiliary fusion piece **14** is placed over the first auxiliary reinforcement piece **15**. The auxiliary structure can directly fuse with at least one of the outer surfaces and the inner surfaces of the top sheet **8** and the bottom sheet **9**, and then the tensioning strips **11** are connected thereto by fusion. Alternatively, the auxiliary structure is kept in position and then the tensioning strips **11** are fused with the auxiliary structure and at least one of the outer surfaces and the inner surfaces of the top sheet **8** and the bottom sheet **9**. Further, the auxiliary structure can be a fiber glue layer **16** which has a plurality of through holes and is applied to the location where at least one of outer surfaces and the inner surfaces of the top sheet **8** and the bottom sheet **9** fuses with the tensioning strips **11**. The first auxiliary reinforcement piece **15** is made of a flexible anti-stretch material (e.g. mesh fabric) having a plurality of through holes and is configured to restrict or resist pulling force produced by expansion of the inflatable product during inflation. The fiber glue layer **16** is formed by applying fiber glue (glue with fiber added therein) to the location where at least one of the outer surfaces and the inner surfaces of the top sheet **8** and the bottom sheet **9** fuses with the tensioning strips **11**, for restricting or resisting the pulling force produced by expansion of the inflatable product during inflation. The third auxiliary fusion piece **14** is made of the same material as the top sheet **8** and the bottom sheet **9** that facilitates connection of the first auxiliary reinforcement piece **15** to at least one of the outer surfaces and the inner surfaces of the top sheet **8** and the bottom sheet **9** by fusion.

[0192] As shown in FIG. **15**, in the present embodiment, a method for manufacturing the tensioning strips **11** includes following steps:

[0193] In step S**1**, a roll of mesh fabric is loaded and unrolled.

[0194] In step S**2**, a roll of the first auxiliary fusion piece **12** and a roll of the second auxiliary fusion piece **13** are cut into strips and the strips are placed at predetermined locations on an upper side and a lower side of the mesh fabric. Alternatively, a roll of the first auxiliary fusion piece **12** or a roll of the second auxiliary fusion piece **13** is cut into strips and the strips are placed on a side of the mesh fabric.

[0195] In step S**3**, the strips cut from the first auxiliary fusion piece **12** and the second auxiliary fusion piece **13** are placed to hold the mesh fabric and fuse with the mesh fabric. Alternatively, the strips cut from the first auxiliary fusion piece **12** or the second auxiliary fusion piece **13** are fixed to a side of the mesh fabric.

[0196] In step S**4**, the mesh fabric which fuses with the strips cut from the first auxiliary fusion piece **12** and the second auxiliary fusion piece **13** is cut into strips for obtaining tensioning strips **11**. Alternatively, the mesh fabric, a side of which has the strips cut from the first auxiliary fusion piece **12** or the second auxiliary fusion piece **13** attached thereto, is cut into strips for obtaining the tensioning strips **11**.

[0197] As shown in FIGS. **16-17**, an alternative method for manufacturing the tensioning strips **11** includes the following steps:

[0198] In step S**1'**, a roll of mesh fabric is loaded and unrolled.

[0199] In step S**2'**, the mesh fabric is cut into strips for obtaining the strip body **111**.

[0200] In step S**3'**, the first auxiliary fusion piece **12** and the second auxiliary fusion piece **13** are respectively placed and positioned at predetermined locations on an upper side and a lower side of the strip body **111**. Alternatively, the first auxiliary fusion piece **12** or the second auxiliary fusion piece **13** is placed and positioned at predetermined location on a side of the strip body **111**.

[0201] In step S**4'**, the first auxiliary fusion piece **12** and the second auxiliary fusion piece **13** are placed to hold the strip body **111** and fuse with the strip body **111** for obtaining the tensioning strips **11**. Alternatively, the first auxiliary fusion piece **12** or the second auxiliary fusion piece **13** is fixed to a side of the strip body **111** for obtaining the tensioning strips **11**.

[0202] In the present embodiment, a method for manufacturing the inflatable product includes the following steps:

[0203] In step S5, rolls of the top sheet 8 and the bottom sheet 9 are loaded, unrolled and kept in position.

[0204] In step S6, the tensioning strips 11, which are manufactured by the above-described method, are placed at predetermined locations between the top sheet 8 and the bottom sheet 9, and the parts of the tensioning strips 11 which have the first auxiliary fusion pieces 12 and the second auxiliary fusion pieces 13 thereon are fused with the inner surfaces of the top sheet 8 and the bottom sheet 9. Alternatively, the parts of the tensioning strips 11, a side of which has the first auxiliary fusion pieces 12 or the second auxiliary fusion pieces 13 fixed thereto, are fused with the inner surfaces of the top sheet 8 and the bottom sheet 9.

[0205] In step S7, the sides of the periphery sheet 10 are connected to the peripheries of the top sheet 8 and the bottom sheet 9 by any of the fusion structures described in the first, second, third, fourth, fifth and sixth embodiments for forming the air chamber and obtaining the inflatable product.

[0206] As shown in FIGS. 18-19, an alternative method for manufacturing the inflatable product includes the following steps:

[0207] In step S5, rolls of the top sheet 8 and the bottom sheet 9 are loaded and unrolled and kept in position, and the third auxiliary fusion pieces 14 are fused with the outer surfaces or the inner surfaces of the top sheet 8 and the bottom sheet 9 at predetermined locations where the tensioning strips 11 are going to fuse with the top sheet 8 and the bottom sheet 9.

[0208] In step S6, the tensioning strips 11 which are manufactured by the above-described method are sent and kept at the predetermined locations between the top sheet 8 and the bottom sheet 9, and the parts of the tensioning strips 11 having the first auxiliary fusion pieces 12 and the second auxiliary fusion pieces 13 thereon are fused with the parts of the inner surfaces of the top sheet 8 and the bottom sheet 9 having the third auxiliary fusion pieces 14 thereon. Alternatively, the parts of the tensioning strips 11, each of which has the first auxiliary fusion piece 12 or the second auxiliary fusion piece 13 fixed to a side thereof, are fused with the parts of the inner surfaces of the top sheet 8 and the bottom sheet 9 which have the third auxiliary fusion pieces 14 fixed thereto.

[0209] In step S7, the sides of the periphery sheet 10 are connected to the peripheries of the top sheet 8 and the bottom sheet 9 by any of the fusion structures described in the first, second, third, fourth, fifth and sixth embodiments for forming the air chamber and obtaining the inflatable product.

[0210] As shown in FIGS. 18 and 20, an alternative method for manufacturing the inflatable product includes the following steps:

[0211] In step S5, rolls of the top sheet 8 and the bottom sheet 9 are loaded, unrolled and kept in position, and the first auxiliary reinforcement pieces 15 and the third auxiliary fusion pieces 14 are placed on the outer surfaces or the inner surfaces of the top sheet 8 and the bottom sheet 9 at predetermined locations where the top sheet 8 and the bottom sheet 9 are going to fuse with the tensioning strips 11, wherein the first auxiliary reinforcement pieces 15 and the third auxiliary fusion pieces 14 are only kept in position without fusing with other elements.

[0212] In step S6, the tensioning strips 11 which are manufactured by the above-described method are sent and positioned at predetermined locations between the top sheet 8 and the bottom sheet 9, and the parts of the tensioning strips 11 which have the first auxiliary fusion pieces 12 and the second auxiliary fusion pieces 13 thereon are fused with the parts of the inner surfaces of the top sheet 8 and the bottom sheet 9 which have the first auxiliary reinforcement pieces 15 and the third auxiliary fusion pieces 14 thereon. Alternatively, the parts of the tensioning strips 11, each of which has the first auxiliary fusion piece 12 or the second auxiliary fusion piece 13 fixed to a side thereof, are fused with the parts of the inner surfaces of the top sheet 8 and the bottom sheet 9 which have the third auxiliary fusion pieces 14 fixed thereto.

[0213] In step S7, the sides of the periphery sheet 10 are connected to the peripheries of the top sheet 8 and the bottom sheet 9 by any of the fusion structures described in the first, second, third, fourth, fifth and sixth embodiments for forming the air chamber and obtaining the inflatable product.

[0214] As shown in FIGS. 18 and 20, a further alternative method for manufacturing the inflatable product includes the following steps:

[0215] In step S5, rolls of the top sheet 8 and the bottom sheet 9 are loaded, unrolled and kept in position, the first auxiliary reinforcement pieces 15 and the third auxiliary fusion pieces 14 are placed on the outer surfaces or the inner surfaces of the top sheet 8 and the bottom sheet 9 at predetermined locations where the top sheet 8 and the bottom sheet 9 are going to fuse with the tensioning strips 11, and the first auxiliary reinforcement pieces 15 and the third auxiliary fusion pieces 14 are fused with the inner surfaces of the top sheet 8 and the bottom sheet 9.

[0216] In step S6, the tensioning strips 11 which are manufactured by the above-described method are placed at predetermined locations between the top sheet 8 and the bottom sheet 9, and the parts of the tensioning strips 11 which have the first auxiliary fusion pieces 12 and the second auxiliary fusion pieces 13 thereon are fused with the parts of the inner surfaces of the top sheet 8 and the bottom sheet 9 which have the first auxiliary reinforcement pieces 15 and the third auxiliary fusion pieces 14 thereon. Alternatively, the parts of the tensioning strips 11, each of which has the first auxiliary fusion piece 12 or the second auxiliary fusion piece 13 fixed to a side thereof, are fused with the parts of the inner surfaces of the top sheet 8 and the bottom sheet 9 which have the first auxiliary reinforcement pieces 15 and the third auxiliary fusion pieces 14 thereon.

[0217] In step S7, the sides of the periphery sheet 10 are connected to the peripheries of the top sheet 8 and the bottom sheet 9 by any of the fusion structures described in the first, second, third, fourth, fifth and sixth embodiments for forming the air chamber and obtaining the inflatable product.

[0218] As shown in FIGS. 18 and 21, a yet further alternative method for manufacturing the inflatable product includes the following steps:

[0219] In step S5, rolls of the top sheet 8 and the bottom sheet 9 are loaded, unrolled and kept in position, and a fiber glue layer 16 which has a plurality of through holes is formed on the outer surfaces or the inner surfaces of the top sheet 8 and the bottom sheet 9 at predetermined locations where the top sheet **8** and the bottom sheet **9** are going to fuse with the tensioning strips **11**.

[0220] In step S**6**, the tensioning strips **11** which are manufactured by the above-described method are sent and positioned at predetermined locations between the top sheet **8** and the bottom sheet **9**, and the parts of the tensioning strips **11** which have the first auxiliary fusion pieces **12** and the second auxiliary fusion pieces **13** thereon are fused with the parts of the inner surfaces of the top sheet **8** and the bottom sheet **9** which have the fiber glue layer **16** formed thereon, wherein the fiber glue layer **16** has through holes. Alternatively, the parts of the tensioning strips **11**, each of which has the first auxiliary fusion piece **12** or the second auxiliary fusion piece **13** fixed to a side thereof, are fused with the parts of the inner surfaces of the top sheet **8** and the bottom sheet **9** which have the fiber glue layer **16** formed thereon, wherein the fiber glue layer **16** has through holes.

[0221] In step S7, the sides of the periphery sheet **10** are connected to the peripheries of the top sheet **8** and the bottom sheet **9** by any of the fusion structures described in the first, second, third, fourth, fifth and sixth embodiments for forming the air chamber and obtaining the inflatable product.

Seventeenth Embodiment

[0222] The seventeenth embodiment is similar to the sixteenth embodiment, and the difference therebetween is that two sides of the periphery sheet **10** are connected to peripheries of the top sheet **8** and the bottom sheet **9** by the fusion structure described in the seventh embodiment for forming an air chamber. The top sheet **8** and the bottom sheet **9** can be regarded as the target fusion layer **1** of the first fusion unit of the fusion structure, and the periphery sheet **10** can be regarded as the target fusion layer **1** of the second fusion unit of the fusion structure. The tensioning strips **11** are connected to inner surfaces of the top sheet **8** and the bottom sheet **9** by fusion.

Eighteenth Embodiment

[0223] Referring to FIG. **22**, an inflatable product includes an inflatable body **100**. The inflatable body **100** includes two target fusion layers **110** which are connected by fusion, thereby forming a fusion portion. The fusion portion includes a first fusion portion **121** and a second fusion portion **122**.

[0224] An anti-stretch sheet **200** is disposed outside the inflatable body **100** for covering the inflatable body **100**. The anti-stretch sheet **200** includes two fusion reinforcement layers **210**, a fusion portion is formed thereon, and the fusion portion includes a first fusion portion **221** and a second fusion portion **222**.

[0225] The inflatable product further includes a plurality of airtight auxiliary fusion layers **400** which are fused with the fusion portion of the inflatable body and the fusion portion of the anti-stretch sheet. The airtight auxiliary fusion layers **400** include auxiliary fusion layers **410**, **420**. The target fusion layers **110** and the auxiliary fusion layers **410**, **420** are made of polyvinyl chloride (PVC), and the fusion reinforcement layers **210** are mesh fabric.

[0226] The first fusion portion **121** and the second fusion portion **122** of the target fusion layers **110** are placed in contact with each other. The first fusion portion **221** and the second fusion portion **222** of the fusion reinforcement layers **210** are respectively placed in contact with the first fusion portion **121** and the second fusion portion **122** of the target fusion layers **110**. The auxiliary fusion layers **410**, **420** are placed in contact with the first fusion portion **221** and the second fusion portion **222** of the fusion reinforcement layers **210**. The inflatable product provided with an air chamber is formed by fusion and has an angle at the fusion location after inflation of the inflatable body.

Nineteenth Embodiment

[0227] Referring to FIG. **23**, the nineteenth embodiment is similar to the eighteenth embodiment, and the difference therebetween is that an anti-stretch sheet **300** is provided within the inflatable body **100** for being covered by the inflatable body **100**. The anti-stretch sheet **300** includes two fusion reinforcement layers **310**, a fusion portion is formed thereon, and the fusion portion includes a first fusion portion **321** and a second fusion portion **322**. An auxiliary fusion layer **450** is provided between the first fusion portion **321** and the second fusion portion **322**. The inflatable product having an air chamber is formed by fusion and has an angle at the fusion location.

Twentieth Embodiment

[0228] Referring to FIGS. **24** and **25**A, an inflatable tent includes an inflatable tube **20** and a tent cloth **30**. The inflatable tube **20** includes a tube body **22** and a tube end (not shown) disposed on an end portion of the tube body, and the tube body **22** and the tube end are connected by fusion to form an air chamber. The tube body **22** includes an inflatable body. Referring to FIG. **25**B, the inflatable body is formed by two target fusion layers **110** which are arranged symmetrically and connected by fusion to form a fusion portion thereon. The fusion portion includes a first fusion portion **121** and a second fusion portion **122** (FIG. **25**A only shows a side of fusion structure, while the other side of fusion structure is omitted).

[0229] An anti-stretch sheet is disposed outside the inflatable body for covering the inflatable body. The anti-stretch sheet includes two fusion reinforcement layers **210**, a fusion portion is formed thereon, and the fusion portion includes a first fusion portion **221** and a second fusion portion **222**.

[0230] The inflatable tube **20** further includes a plurality of airtight auxiliary fusion layers which are fused with the fusion portion of the inflatable body and the fusion portion of the anti-stretch sheet. The airtight auxiliary fusion layers include auxiliary fusion layers **410**, **420**. The target fusion layers **110** and the auxiliary fusion layers **410**, **420** are made of polyvinyl chloride (PVC), and the fusion reinforcement layers **210** are mesh fabric.

[0231] The first fusion portion **121** and the second fusion portion **122** of the target fusion layers **110** are placed in contact with each other. The first fusion portion **221** and the second fusion portion **222** of the fusion reinforcement layers **210** are respectively placed in contact with the first fusion portion **121** and the second fusion portion **122** of the target fusion layers **110**. The auxiliary fusion layers **410**, **420** are placed in contact with the first fusion portion **221** and the second fusion portion **222** of the fusion reinforcement layers **210**. The tube body **22** having the air chamber is formed by fusion and has an angle at the fusion location after inflation of the inflatable body. During the fusion, the first fusion portion **121** of the target fusion layers **110**, the first fusion portion **221** of the fusion reinforcement layers **210** and the auxiliary fusion layer **410** are fused, the second fusion portion **122** of the target fusion layers **110**, the second fusion portion **222** of the fusion reinforcement layers **210** and the auxiliary fusion layer **420** are fused, and then the two fusion units are fused.

Twenty-First Embodiment

[0232] Referring to FIG. **26**, the twenty-first embodiment is similar to the twentieth embodiment, and the difference therebetween is that an auxiliary fusion layer **430** is additionally provided between the first fusion portion **121** of the target fusion layers **110** and the first fusion portion **221** of the fusion reinforcement layers **210**, and an auxiliary fusion layer **440** is additionally provided between the second fusion portion **122** of the target fusion layers **110** and the second fusion portion **222** of the fusion reinforcement layers **210**. The tube body **22** having the air chamber is formed by fusion and has an angle at the fusion location after inflation of the inflatable body. During the fusion, the first fusion portion **121** of the target fusion layers **110**, the auxiliary fusion layer **430**, the first fusion portion **221** of the fusion reinforcement layers **210** and the auxiliary fusion layer **410** are fused, the second fusion portion **122** of the target fusion layers **110**, the auxiliary fusion layer **440**, the second fusion portion **222** of the fusion reinforcement layers **210** and the auxiliary fusion layer **420** are fused, and then the two fusion units are fused.

Twenty-Second Embodiment

[0233] Referring to FIG. **27**A, the twenty-second embodiment is similar to the twentieth embodiment, and the difference therebetween is that the first fusion portion **121** of the target fusion layer **110** is placed in contact with the first fusion portion **221** of the fusion reinforcement layer **210**, the second fusion portion **122** of the target fusion layer **110** is placed in contact with the first fusion portion **221** of the fusion reinforcement layer **210**, the second fusion portion **222** of the fusion reinforcement layer **210** is placed in contact with the second fusion portion **122** of the target fusion layer **110**, and the auxiliary fusion layer **420** is placed in contact with the second fusion portion **222** of the fusion reinforcement layer **210**. Referring to FIG. **27**B, the target fusion layer **110** is back sealed by fusion to form the tube body **22** having the air chamber. During the fusion, the first fusion portion **121** of the target fusion layer **110**, the first fusion portion **221** of the fusion reinforcement layer **210**, and the second fusion portion **122** of the target fusion layer **110** are simultaneously fused, and then are fused with the second fusion portion **222** of the fusion reinforcement layer **210** and the auxiliary fusion layer **420**.

Twenty-Third Embodiment

[0234] Referring to FIG. **28**, the twenty-third embodiment is similar to the twenty-second embodiment, and the difference therebetween is that the first fusion portion **121** of the target fusion layer **110** is placed in contact with the second fusion portion **122** of the target fusion layer **110**, the second fusion portion **122** of the target fusion layer **110** is placed in contact with the second fusion portion **222** of the fusion reinforcement layer **210**, the auxiliary fusion layer **420** is placed in contact with the second fusion portion **222** of the fusion reinforcement layer **210**, the first fusion portion **221** of the fusion reinforcement layer **210** is placed in contact with the auxiliary fusion layer **420**, and the auxiliary fusion layer **410** is placed in contact with the first fusion portion **221** of the fusion reinforcement layer **210**. The tube body **22** having the air chamber is formed by fusion and has a curved shape at the fusion location. During the fusion, the auxiliary fusion layer **420**, the second fusion portion **222** of the fusion reinforcement layer **210** and the second fusion portion **122** of the target fusion layer **110** are simultaneously fused, and then are fused with the first fusion portion **121** of the target fusion layer **110**, the first fusion portion **221** of the fusion reinforcement layer **210** and the auxiliary fusion layer **410**.

Twenty-Fourth Embodiment

[0235] Referring to FIG. **29**, the twenty-fourth embodiment is similar to the twenty-second embodiment, and the difference therebetween is that the first fusion portion **121** of the target fusion layer **110**, the first fusion portion **221** of the fusion reinforcement layer **210**, the second fusion portion **122** of the target fusion layer **110**, the second fusion portion **222** of the fusion reinforcement layer **210** and the auxiliary fusion layer **420** are simultaneously fused.

Twenty-Fifth Embodiment

[0236] Referring to FIG. **30**, the twenty-fifth embodiment is similar to the twenty-third embodiment, and the difference therebetween is that the auxiliary fusion layer **410**, the first fusion portion **221** of the fusion reinforcement layer **210** and the auxiliary fusion layer **420** are simultaneously welded, and then are fused with the second fusion portion **222** of the fusion reinforcement layer **210** and the second fusion portion **122** and the first fusion portion **121** of the target fusion layer **110**.

[0237] Twenty-Sixth Embodiment

[0238] Referring to FIGS. **31**-**32**, an inflatable bed includes a top sheet **114**, a periphery sheet **112** and a bottom sheet, and peripheries of the three sheets are fused to form an inflatable body **100**. Fusion units are formed between the top sheet **114** and the periphery sheet **112** as well as between the periphery sheet **112** and the bottom sheet. The fusion units includes a first fusion portion **121** formed on the periphery of the periphery sheet, a second fusion portion **122** formed on the periphery of the top sheet, and a second fusion portion (not shown) formed on the periphery of the bottom sheet.

[0239] The periphery sheet **112** is covered by an anti-stretch sheet, and the anti-stretch sheet includes a fusion reinforcement layer **220** with a first fusion portion **221** formed on the periphery thereof.

[0240] The inflatable bed further includes a plurality of airtight auxiliary fusion layers, and the airtight auxiliary fusion layers include an auxiliary fusion layer **410**. The top sheet **114**, the periphery sheet **112**, the bottom sheet and the auxiliary fusion layer **410** are made of polyvinyl chloride (PVC), and the fusion reinforcement layer **220** is mesh fabric.

[0241] The first fusion portion **121** of the periphery sheet **112** is placed in contact with the first fusion portion **221** of the fusion reinforcement layer **220**, the auxiliary fusion layer **410** is placed in contact with the first fusion portion **221** of the fusion reinforcement layer **220**, and the first fusion portion **121** of the periphery sheet **112** is placed in contact with the second fusion portion **122** of the top sheet **114**, so that they can be are firmly connected by fusion to form an angle at the fusion location. During the fusion, the first fusion portion **121** of the periphery sheet **112**, the first fusion portion **221** of the fusion reinforcement layer **220** and the auxiliary fusion layer **410** are simultaneously fused and then are fused with the second fusion portion **122** of the top sheet **114**.

[0242] The way that the first fusion portion of the periphery sheet **112** is fused with the second fusion portion of the bottom sheet and the first fusion portion of the fusion reinforcement layer **220** is similar to the above-mentioned way, and therefore the descriptions thereof are omitted.

[0243] Furthermore, middle of the fusion reinforcement layer **220** is provided with a plurality of auxiliary fusion layers **460** which are spaced and are firmly connected to the fusion reinforcement layer **220** and the periphery sheet **112** by fusion for improving the pressure bearing ability.

Twenty-Seventh Embodiment

[0244] Referring to FIG. **33**, the twenty-seventh embodiment is similar to the twenty-sixth embodiment, and the difference therebetween is that an auxiliary fusion layer **430** is additionally provided between the first fusion portion **121** of the periphery sheet and the first fusion portion **221** of the fusion reinforcement layer. During the fusion, the first fusion portion **121** of the periphery sheet, the auxiliary fusion layer **430**, the first fusion portion **221** of the fusion reinforcement layer and the auxiliary fusion layer **410** are simultaneously fused and then are fused with the second fusion portion **122** of the top sheet.

Twenty-Eighth Embodiment

[0245] Referring to FIGS. **34-35**, an inflatable bed includes a top sheet **114**, a periphery sheet **112** and a bottom sheet **113**, and peripheries of the three sheets are fused to form an inflatable body **100**. Fusion portions are formed between the top sheet **114** and the periphery sheet **112** as well as between the periphery sheet **112** and the bottom sheet **113**. The fusion portions includes a first fusion portion **121** formed on the periphery of the periphery sheet **112**, a second fusion portion **122** formed on the periphery of the top sheet **114** and a second fusion portion (not shown) formed on the periphery of the bottom sheet **113**.

[0246] The inflatable bed further includes an anti-stretch sheet **200**, and the anti-stretch sheet **200** includes a fusion reinforcement layer **220** covering the periphery sheet **112**, a fusion reinforcement layer **210** covering the top sheet **114** and a fusion reinforcement layer **230** covering the bottom sheet **113**. A second fusion portion **222** is formed on periphery of the fusion reinforcement layer **210**, a first fusion portion **221** is formed on periphery of the fusion reinforcement layer **220**, and a second fusion portion is formed on periphery of the fusion reinforcement layer **230**.

[0247] The inflatable bed further includes a plurality of airtight auxiliary fusion layers, and the airtight auxiliary fusion layers include an auxiliary fusion layer **410** and an auxiliary fusion layer **420**. The top sheet **114**, the periphery sheet **112**, the bottom sheet **113** and the auxiliary fusion layers **410**, **420** are made of polyvinyl chloride (PVC), and the fusion reinforcement layers **210**, **220**, **230** are mesh fabric.

[0248] The first fusion portion **121** of the periphery sheet **112** is placed in contact with the first fusion portion **221** of the fusion reinforcement layer **220**, and the auxiliary fusion layer **410** is placed in contact with the first fusion portion **221** of the fusion reinforcement layer **220**, so as to be connected by a first fusion. The second fusion portion **122** of the top sheet **114** is placed in contact with the second fusion portion **222** of the fusion reinforcement layer **210**, and the auxiliary fusion layer **420** is placed in contact with the second fusion portion **222** of the fusion reinforcement layer **210**, so as to be connected by a second fusion. The first fusion portion **121** of the periphery sheet **112** is placed in contact with the second fusion portion **122** of the top sheet **114**. Then, the first fusion portion **121** and the second fusion portion **122** are firmly connected by the second fusion and have an angle at the fusion location after inflation of the inflatable body.

[0249] The way to fuse the first fusion portion of the periphery sheet **112**, the second fusion portion of the bottom sheet **113**, the first fusion portion **221** of the fusion reinforcement layer **220** and the second fusion portion of the fusion reinforcement layer **230** is similar to the above-mentioned way, and therefore the descriptions thereof are omitted.

[0250] Furthermore, middle of the fusion reinforcement layer **220** is provided with a plurality of auxiliary fusion layers **460** which are spaced and are firmly connected to the fusion reinforcement layer **220** and the periphery sheet **112** by fusion for improving the pressure bearing ability.

Twenty-Ninth Embodiment

[0251] Referring to FIG. **36**, the twenty-ninth embodiment is similar to the twenty-eighth embodiment, and the difference therebetween is that an auxiliary fusion layer **430** is additionally provided between the first fusion portion **121** of the periphery sheet and the first fusion portion **221** of the fusion reinforcement layer, an auxiliary fusion layer **440** is additionally provided between the second fusion portion **122** of the top sheet and the second fusion portion **222** of the fusion reinforcement layer, and an auxiliary fusion layer (not shown) is additionally provided between the second fusion portion of the bottom sheet and the second fusion portion of the fusion reinforcement layer.

Thirtieth Embodiment

[0252] Referring to FIGS. **37-39**, an inflatable pool includes an inflatable body **100**. The inflatable body **100** is formed by fusing a plurality of target fusion layers **110**, and a fusion portion is formed thereon. The fusion portion includes a first fusion portion **121** and a second fusion portion **122**.

[0253] An anti-stretch sheet **200** is disposed outside the inflatable body **100** for covering the inflatable body **100**. The anti-stretch sheet **200** includes a plurality of fusion reinforcement layers **210**, and a fusion portion is formed thereon. The fusion portion includes a first fusion portion **221** and a second fusion portion **222**.

[0254] The inflatable pool further includes a plurality of airtight auxiliary fusion layers which are firmly connected to the fusion portion of the inflatable body and the fusion portion of the anti-stretch sheet by fusion. The airtight auxiliary fusion layers include auxiliary fusion layers **410**, **420**. The target fusion layers **110** and the auxiliary fusion layers **410**, **420** are made of polyvinyl chloride (PVC), and the fusion reinforcement layers **210** are mesh fabric.

[0255] The first fusion portion **121** and the second fusion portion **122** of the target fusion layers **110** are placed in contact with each other, the first fusion portion **221** and the second fusion portion **222** of the fusion reinforcement layers **210** are respectively placed in contact with the first fusion portion **121** and the second fusion portion **122** of the target fusion layers **110**, and the auxiliary fusion layers **410**, **420** are respectively placed in contact with the first fusion portion **221** and the second fusion portion **222** of the fusion reinforcement layers **210**, so that they can be firmly connected by fusion and have an angle at the fusion locations. During the fusion, the first fusion portion **121** of the target fusion layers **110**, the first fusion portion **221** of the fusion reinforcement layers **210** and the auxiliary fusion layer **410** are simultaneously fused, the second fusion portion **122** of the target fusion layers **110**, the second fusion portion **222** of the fusion reinforcement layers **210** and the auxiliary fusion layer **420** are simultaneously fused, and then the above-mentioned structures which are individually fused have their fused portions firmly connected by further fusion.

[0256] Furthermore, middle of the fusion reinforcement layer **210** is provided with a plurality of auxiliary fusion layers **460** which are spaced and are firmly connected to the fusion reinforcement layer **210** and the target fusion layer **110** by fusion for improving the pressure bearing ability.

Thirty-First Embodiment

[0257] Referring to FIG. **40**, the thirty-first embodiment is similar to the thirtieth embodiment, and the difference therebetween is that an auxiliary fusion layer **430** is additionally provided between the first fusion portion **121** of the target fusion layer **110** and the first fusion portion **221** of the fusion reinforcement layer **210**, and an auxiliary fusion layer **440** is additionally provided between the second fusion portion **122** of the target fusion layer **110** and first fusion portion **222** of the fusion reinforcement layer **210**.

Thirty-Second Embodiment

[0258] Referring to FIG. **41**, the thirty-second embodiment is similar to the thirtieth embodiment, and the difference therebetween is that during the fusion, the first fusion portion **221** of the fusion reinforcement layer **210** and the auxiliary fusion layer **410** are fixed by sewing, the first fusion portion **222** of the fusion reinforcement layer **210** and the auxiliary fusion layer **420** are fixed by sewing, and the first fusion portion **121** and the second fusion portion **122** of the target fusion layer **110** are placed in contact with each other and then fused together to fix the whole fused portion.

Thirty-Third Embodiment

[0259] Referring to FIG. **42**A, the thirty-third embodiment is similar to the thirty-first embodiment, and the difference therebetween is that during the fusion, the first fusion portion **221** of the fusion reinforcement layer **210** and the auxiliary fusion layer **410** are fixed by sewing, the auxiliary fusion layer **410** and the auxiliary fusion layer **430** are firmly connected by fusion and are then firmly connected to the first fusion portion **121** of the target fusion layer **110** by further fusion, the second fusion portion **222** of the fusion reinforcement layer **210** and the auxiliary fusion layer **420** are fixed by sewing, and the auxiliary fusion layer **420** and the auxiliary fusion layer **440** are firmly connected by fusion and are then firmly connected to the second fusion portion **122** of the target fusion layer **110** by further fusion. Referring to FIG. **42**B, the first fusion portion **121** and the second fusion portion **122** of the target fusion layer **110** are placed in contact with each other, and then fused together to fix the whole fused portion.

Thirty-Fourth Embodiment

[0260] Referring to FIG. **43**A, the thirty-fourth embodiment is similar to the thirty-third embodiment, and the difference therebetween is that during the fusion, the first fusion portion **221** of the fusion reinforcement layer and the auxiliary fusion layer **410** are fixed by sewing, the auxiliary fusion layer **410** and the auxiliary fusion layer **430** are firmly connected by fusion (fused at two different locations) and are then firmly connected to the first fusion portion **121** of the target fusion layer by further fusion, the second fusion portion **222** of the fusion reinforcement layer and the auxiliary fusion layer **420** are fixed by sewing, and the auxiliary fusion layer **420** and the auxiliary fusion layer **440** are firmly connected by fusion (fused at two different locations) and are then firmly connected to the second fusion portion **122** of the target fusion layer by further fusion. Referring to FIG. **43**B, the first fusion portion **121** and the second fusion portion **122** of the target fusion layer are placed in contact with each other to fix the whole fused portion.

Thirty-Fifth Embodiment

[0261] Referring to FIG. **44**, fusion portions of an inflatable product are formed as a fusion structure, and the fusion structure includes a first fusion unit **910** and a second fusion unit (FIG. **44** shows the first fusion unit **910** but does not show the second fusion unit) which are welded to each other. As shown in FIG. **44**, the way to produce the first fusion unit is that an auxiliary fusion layer **410** and an auxiliary fusion layer **430** are placed to firmly hold a fusion portion **221** of an anti-stretch sheet and are then firmly connected to a fusion portion **121** of an inflatable body by fusion. The anti-stretch sheet includes a target fusion layer **210**, and the inflatable body includes a target fusion layer **110**. The way to produce the second fusion unit is same as that to produce the first fusion unit.

Thirty-Sixth Embodiment

[0262] Referring to FIG. **45**, fusion portions of an inflatable product are formed as a fusion structure, and the fusion structure includes a first fusion unit **910** and a second fusion unit (FIG. **45** shows the first fusion unit **910** but does not show the second fusion unit) which are fused with each other. As shown in FIG. **45**, the way to produce the first fusion unit is that an auxiliary fusion layer **430** or a fusion reinforcement layer **500** is fixed to a fusion portion **221** of an anti-stretch sheet by sewing and is then firmly connected to a fusion portion **121** of an inflatable body by fusion. The fusion portion **121** of the inflatable body is placed in contact with the auxiliary fusion layer **430** or the fusion reinforcement layer **500**. The anti-stretch sheet includes a target fusion layer **210**, and the inflatable body includes a target fusion layer **110**. The way to produce the second fusion unit is same as that to produce the first fusion unit.

Thirty-Seventh Embodiment

[0263] Referring to FIG. **46**, fusion portions of an inflatable product are formed as a fusion structure, and the fusion structure includes a first fusion unit **910** and a second fusion unit (FIG. **46** shows the first fusion unit **910** but does not show the second fusion unit) which are fused with each other. The way to produce the first fusion unit is that an auxiliary fusion layer **430** or a fusion reinforcement layer **500** is fixed to a fusion portion **221** of an anti-stretch sheet by sewing, is fixed to another auxiliary fusion layer **430**, and is firmly connected to a fusion portion **121** of an inflatable body by fusion. The fusion portion **121** of the inflatable body is placed in contact with another auxiliary fusion layer **430**. The anti-stretch sheet includes a target fusion layer **210**, and the inflatable body includes a target fusion layer **110**. The way to produce the second fusion unit is same as that to produce the first fusion unit.

Thirty-Eighth Embodiment

[0264] Referring to FIG. **47**, fusion portions of an inflatable product are formed as a fusion structure, and the fusion structure includes a first fusion unit **910** and a second fusion unit (FIG. **46** shows the first fusion unit **910** but does not show the second fusion unit) which are fused with each other. The way to produce the first fusion unit is that a portion of a fusion reinforcement layer **500** is wrapped by a cloth **40** and is fixed to a fusion portion **221** of an anti-stretch sheet by sewing, and another portion of the fusion reinforcement layer **500** which is not wrapped by the cloth **40** is firmly held between two auxiliary fusion layers **410**, **430** and is firmly connected to a fusion portion **121** of an inflatable body by fusion. The fusion portion **121** of the inflatable body is placed in contact with the auxiliary fusion layer **430**. The anti-stretch sheet includes a target fusion layer **210**, and the inflatable body includes a target fusion layer **110**. The way to produce the second fusion unit is same as that to produce the first fusion unit.

Thirty-Ninth Embodiment

[0265] Referring to FIG. **48**, the structure of a first fusion unit is similar to that of the thirty-fifth embodiment, and the difference therebetween is that the fusion portion **221** of the anti-stretch sheet is provided with a punch hole **50**.

Fortieth Embodiment

[0266] Referring to FIG. **49**, an inflatable product has an inflatable body which includes a target fusion layer **110**, an anti-stretch layer **210** for covering the target fusion layer **110**, and a fiber-contained reinforcement element **600**. In this embodiment, the target fusion layer **110** is made of fusible material (e.g. polyvinyl chloride, PVC), and the anti-stretch layer **210** is made of cloth for covering the target fusion layer **110** and restricting expansion of the inflatable body. The fiber-contained reinforcement element **600** includes a fusion reinforcement sheet **500** and an auxiliary fusion sheet **430**. The fusion reinforcement sheet **500** is made of a flexible anti-stretch material (e.g. nets or mesh fabric) having a plurality of through holes. The auxiliary fusion sheet **430** is made of fusible material (e.g. polyvinyl chloride, PVC). A portion of the auxiliary fusion sheet **430** is connected to the target fusion layer **110** by fusion, and another portion of the auxiliary fusion sheet **430** is connected to the fusion reinforcement sheet **500** and the anti-stretch layer **210** by sewing.

Forty-First Embodiment

[0267] Referring to FIG. **50**, the differences between the fortieth embodiment and the forty-first embodiment are that the inflatable product of the forty-first embodiment further includes another auxiliary fusion sheet **410** disposed between the anti-stretch layer **210** and the fusion reinforcement sheet **500**. The anti-stretch layer **210**, the auxiliary fusion sheet **410**, the fusion reinforcement sheet **500** and the auxiliary fusion sheet **430** are connected by sewing.

Forty-Second Embodiment

[0268] Referring to FIG. **51**, an inflatable product includes a top sheet **8**, a bottom sheet **9**, and at least one fiber-contained reinforcement element **235**. In this embodiment, the top sheet **8** and the bottom sheet **9** are made of fusible material (e.g. polyvinyl chloride, PVC). The fiber-contained reinforcement element **235** is a fusible sheet which is formed by adding fibers (e.g. glass fibers, carbon fibers or synthetic fibers) into fusible material (e.g. polyvinyl chloride, PVC) or by applying fibers to fusible material. The fiber-contained reinforcement element **235** has plural through holes **2351**.

[0269] The top sheet **8** and the bottom sheet **9** are connected by fusion to form at least one inflatable body. During manufacturing, the fiber-contained reinforcement element **235** is placed at a predetermined location between the top sheet **8** and the bottom sheet **9** where the top sheet **8** and the bottom sheet **9** are heated and pressed by using a mold (not shown). Because the material of the top sheet **8** and the bottom sheet **9** is compatible with that of the fiber-contained reinforcement element **235**, the top sheet **8**, the bottom sheet **9** and the fiber-contained reinforcement element **235** can be fused together. The fiber contained by the fiber-contained reinforcement element **235** can improve the mechanical strength at the fusion site thereby protecting the connection of the top sheet **8** and the bottom sheet **9** from failure.

[0270] As described above, the fiber-contained reinforcement element **235** has plural through holes **2351**. In the fusion process, the melted portions of the top sheet **8** and the bottom sheet **9** can enter the through holes **2351** of the fiber-contained reinforcement element **235**, thereby increasing the area of fusion. By such arrangement, the mechanical strength at the fusion site can be further improved.

[0271] The forty-second embodiment can be modified in a way that the fiber-contained reinforcement element **235** has no through holes **2351**, and part of the fiber-contained reinforcement element **235** has no fibers added therein or applied thereto. A portion of the fiber-contained reinforcement element **235** having no fibers added therein or applied thereto can be fused with the top sheet **8** and the bottom sheet **9**. Another portion of the fiber-contained reinforcement element **235** having fibers can protect the connection of the top sheet **8** and the bottom sheet **9** from failure.

Forty-Third Embodiment

[0272] Referring to FIG. **52**, an inflatable product includes a top sheet **8**, a bottom sheet **9** and a periphery sheet **10**, wherein the periphery sheet **10** are connected to peripheries of the top sheet **8** and the bottom sheet **9** by fusion to form an inflatable body. At least one fiber-contained reinforcement element **236** is disposed in the inflatable body and connected to inner surfaces of the top sheet **8** and the bottom sheet **9** by fusion wherein the fiber-contained reinforcement element **236** has a plurality of through holes **2361** receiving the melted plastic of the top sheet **8** and the bottom sheet **9**. In this embodiment, the fiber-contained reinforcement element **236** is a tensioning strip for controlling expansion of the inflatable product when the inflatable product is inflated. It is understood that the fiber-contained reinforcement element **236** is not limited to a strip in the invention. In some other embodiments, the fiber-contained reinforcement element may be a strap, a band, a string, a cord or in other shapes.

[0273] Similarly, the material of the top sheet **8** and the bottom sheet **9** is compatible with that of the fiber-contained reinforcement element **236**. Therefore, the top sheet **8**, the bottom sheet **9** and the fiber-contained reinforcement element **236** can be fused together. The fiber contained by the fiber-contained reinforcement element **236** can improve the mechanical strength at the fusion site thereby protecting the connection from failure. The through holes **2361** provided for receiving the melted plastic of the top sheet **8** and the bottom sheet **9** in the fusion process can further improve the mechanical strength at the fusion site.

Forty-Fourth Embodiment

[0274] Referring to FIG. **53**, an inflatable body includes at least one plastic sheet **110**. A fiber-contained reinforcement element **240** is disposed outside the inflatable body to cover the inflatable body or disposed inside the inflatable body to be covered by the inflatable body.

[0275] In this embodiment, the fiber-contained reinforcement element **240** is a plastic sheet with fiber added therein. The fiber-contained reinforcement element **240** has a fusion portion **241** while the plastic sheet **110** has another fusion portion **121**. The fiber-contained reinforcement element **240** is connected to the plastic sheet **110** by fusing the fusion portions **121** and **241** together, wherein the fusion portion **241** has a plurality of through holes **50** receiving the melted plastic of the fusion portion **241**. The fiber in the fiber-contained reinforcement element **240** can improve the mechanical strength at the fusion site, thereby protecting the connection from failure. The through holes **50** provided for receiving the melted plastic of the fusion portion **121** of the plastic sheet **110** in the fusion process can further improve the mechanical strength at the fusion site.

Forty-Fifth Embodiment

[0276] Referring to FIG. **54**, an inflatable body includes at least one plastic sheet **110**. An anti-stretch sheet **210** is disposed outside the inflatable body to cover the inflatable body or disposed inside the inflatable body to be covered by the inflatable body.

[0277] The anti-stretch sheet **210** is connected to the inflatable product through a fiber-contained reinforcement element **250**. In this embodiment, the anti-stretch sheet **210** may be made of any anti-stretch material such as fabric, net (mesh fabric), and threads (weaving mesh). The fiber-contained reinforcement element **250** is a plastic sheet with fibers added therein. As shown in FIG. **54**, the plastic sheet **110** has a fusion portion **121**. The fiber-contained reinforcement element **250** has a connecting portion **252** and a fusion portion **251**. The anti-stretch sheet **210** has an edge folded and sewed with the connecting portion **252** of the fiber-reinforced plastic element **250**. The fusion portion **251** of the fiber-contained reinforcement element **250** and the fusion portion **121** of the plastic sheet **110** are fused together, wherein the fusion portion **251** has through holes **2511** receiving the melted plastic of the fusion portion **121**. Similarly, the fibers in the fiber-contained reinforcement element **250** and the through holes **2511** for receiving the melted plastic can improve the mechanical strength at the fusion site, thereby protecting the connection between the plastic sheet **110** and the fiber-contained reinforcement element **250** from failure.

[0278] It is worth noting that the anti-stretch sheet **210** is not directly connected to the plastic sheet **110** because the anti-stretch sheet **210** cannot be fused with the plastic sheet **110**. Instead, the anti-stretch sheet **210** is connected to the plastic sheet **110** through the fiber-contained reinforcement element **250** which can be fused with the plastic sheet **110**.

[0279] The forty-fifth embodiment can be modified in a way that the fiber-contained reinforcement element **250** has no through holes **2511**, and part of the fiber-contained reinforcement element **250** has no fibers added therein or applied thereto. A portion of the fiber-contained reinforcement element **250** having no fibers added therein or applied thereto can be fused with the plastic sheet **110**. Another portion of the fiber-contained reinforcement element **250** having fibers can protect the connection of the plastic sheet **110** and the anti-stretch sheet **210** from failure.

Forty-Sixth Embodiment

[0280] Referring to FIG. **55**, an inflatable body includes at least two plastic sheets **110**, **110'** and at least one fiber-contained reinforcement element **237** disposed therebetween. In this embodiment, the fiber-contained reinforcement element **237** is fiber glue. The fiber glue is produced by adding fibers (e.g. glass fibers, carbon fibers or synthetic fibers) into glue.

[0281] During manufacturing, the fiber glue **237** is applied to an edge (or portion) of the plastic sheet **110'**. The plastic sheet **110** has an edge (or portion) placed corresponding to the edge of the plastic sheet **110'**. Then, the edges of the plastic sheets **110**, **110'** are fused together to form the inflatable body. Similarly, the fiber in the fiber glue **237** can improve the mechanical strength at the fusion site, thereby protecting the connection between the plastic sheets **110** and **110'** from failure.

Forty-Seventh Embodiment

[0282] FIG. **55** can also depict the forty-seventh embodiment, wherein an inflatable body includes at least two plastic sheets **110**, **110'** and at least one fiber-contained reinforcement element **237** disposed therebetween. In this embodiment, the fiber-contained reinforcement element **237** is made of fusible material with fibers (e.g. glass fibers, carbon fibers or synthetic fibers) added therein or applied thereto.

[0283] During manufacturing, the fiber-contained reinforcement element **237** is placed between edges (or portions) of the plastic sheets **110**, **110'**. Then, the edges of the plastic sheets **110**, **110'** are fused together to form the inflatable body. Similarly, the fibers of the fiber-contained reinforcement element **237** can improve the mechanical strength at the fusion site, thereby protecting the connection between the plastic sheets **110** and **110'** from failure.

Forty-Eighth Embodiment

[0284] Referring to FIG. **56**, an inflatable body includes at least two plastic sheets **110** and **110'** connected to each other.

Specifically, the plastic sheets **110**, **110'** have their edges (or portions) connected by fusion to form the inflatable body. A fiber-contained reinforcement element **237** is externally provided at the edges where the plastic sheets **110**, **110'** are connected.

[0285] In this embodiment, the fiber-contained reinforcement element **237** is fiber glue. The fiber glue is produced by adding fibers (e.g. glass fibers, carbon fibers or synthetic fibers) into glue. The fiber in the fiber glue **237** can still improve the mechanical strength at the connection between the plastic sheets **110** and **110'**, thereby protecting the connection from failure.

[0286] In FIG. **56**, the fiber-contained reinforcement element **237** is externally provided at the edges of the plastic sheet **110**. It is understood that fiber-contained reinforcement element **237** may be externally provided at the edges of the plastic sheets **110'**, or the edges of both of the plastic sheets **110**, **110'**.

Forty-Ninth Embodiment

[0287] FIG. **56** can also depict the forty-ninth embodiment, wherein an inflatable body includes at least two plastic sheets **110**, **110'** and at least one fiber-contained reinforcement element **237** disposed therebetween. In this embodiment, the fiber-contained reinforcement element **237** is made of fusible material with fibers (e.g. glass fibers, carbon fibers or synthetic fibers) added therein or applied thereto.

[0288] During manufacturing, the fiber-contained reinforcement element **237** is placed at edges (or portions) of the plastic sheets **110**, **110'**. Then, the edges of the plastic sheets **110**, **110'** are fused together to form the inflatable body. Similarly, the fibers of the fiber-contained reinforcement element **237** can improve the mechanical strength at the fusion site, thereby protecting the connection between the plastic sheets **110** and **110'** from failure.

Fiftieth Embodiment

[0289] FIG. **57** shows a portion **1000** of an inflatable body, and a fiber-contained reinforcement element **2000** connected to the portion **1000**. The inflatable body is made of fusible material (e.g. polyvinyl chloride, PVC). The fiber-contained reinforcement element **2000** is made of fusible material (e.g. plastic) or glue, with fibers (e.g. glass fibers, carbon fibers or synthetic fibers) added therein or applied thereto. Similarly, the fibers in the reinforcement element **2000** can improve the mechanical strength at the area where the portion **1000** and the fiber-contained reinforcement element **2000** are connected.

[0290] It is understood that the fiber-contained reinforcement element **2000** may have through holes (not shown) when made of fusible material.

[0291] In the invention, the fiber-contained reinforcement element may be any objects generated by processing fibers. In addition to those described above, the fiber-contained reinforcement element may be a cloth, a woven fabric, a mesh fabric and others.

[0292] It is apparent that the above-described embodiments of the invention are merely examples for clearly illustrating the invention and does not limit the applications of the invention. For those skilled in the art, variations or modifications of other different forms may be made in light of the above description. There is no need and possibility to exhaust all of the applications. Any modifications, equivalent replacement and improvement without departing from the spirit and principles of the invention are intended to fall within the scope of the appended claims.

What is claimed is:

1. An inflatable product comprising:
an inflatable body;
a fiber-contained reinforcement element connected to the inflatable body so as to increase mechanical strength of the inflatable product at a location where the fiber-contained reinforcement element is connected to the inflatable body;
wherein the fiber-contained reinforcement element comprises fibers or a derivative thereof generated by processing fibers.

2. The inflatable product as claimed in claim **1**, wherein:
the fiber-contained reinforcement element is made of fusible material with fibers added therein or applied thereto;
the fiber-contained reinforcement element is connected to the inflatable body by fusion.

3. The inflatable product as claimed in claim **2**, wherein:
the fiber-contained reinforcement element comprises a portion;
the portion has through holes;
the portion is connected to the inflatable body by fusion;
the through holes are configured to receive a melted portion of the inflatable body.

4. The inflatable product as claimed in claim **2**, wherein:
the fiber-contained reinforcement element comprises a portion;
the portion has no fibers added therein or applied thereto;
the portion is connected to the inflatable body by fusion.

5. The inflatable product as claimed in claim **2**, further comprising:
an anti-stretch sheet comprising an edge sewed with the fiber-contained reinforcement element.

6. The inflatable product as claimed in claim **2**, further comprising:
an anti-stretch sheet comprising an edge folded and sewed with the fiber-contained reinforcement element.

7. The inflatable product as claimed in claim **2**, wherein:
the inflatable body comprises a first portion and a second portion;
the fiber-contained reinforcement element is disposed between the first portion and the second portion;
the first portion and the second portion are connected through the fiber-contained reinforcement element by fusion.

8. The inflatable product as claimed in claim **2**, wherein:
the fiber-contained reinforcement element is disposed in the inflatable body;
the inflatable body comprises a first portion and a second portion;
the fiber-contained reinforcement element is connected to the first portion and the second portion by fusion, so as to increase mechanical strength of the inflatable product at the location where the fiber-contained reinforcement element is connected to the inflatable body.

9. The inflatable product as claimed in claim **1**, wherein:
the inflatable body comprises a first portion and a second portion connected to each other by fusion;
the fiber-contained reinforcement element comprises glue with fibers added therein;

the fiber-contained reinforcement element is disposed between the first portion and the second portion.

10. The inflatable product as claimed in claim **1**, wherein:

the inflatable body comprises a first portion and a second portion;

the fiber-contained reinforcement element comprises fusible material with fibers added therein or applied thereto;

the fiber-contained reinforcement element is disposed between the first portion and the second portion;

the first portion, the fiber-contained reinforcement element and the second portion are connected by fusion.

11. The inflatable product as claimed in claim **1**, wherein:

the inflatable body comprises a first portion and a second portion connected to each other by fusion;

the fiber-contained reinforcement element comprises glue with fibers added therein;

the fiber-contained reinforcement element is externally disposed on the first portion and/or the second portion.

12. The inflatable product as claimed in claim **1**, wherein:

the inflatable body comprises a first portion and a second portion;

the fiber-contained reinforcement element comprises fusible material with fibers added therein or applied thereto;

the fiber-contained reinforcement element is externally disposed on the first portion and/or the second portion;

the first portion, the second portion and the fiber-contained reinforcement element are connected by fusion.

13. The inflatable product as claimed in claim **1**, further comprising:

an anti-stretch layer configured to cover the inflatable body;

wherein the fiber-contained reinforcement element comprises a first auxiliary fusion sheet connected to the inflatable body by fusion, and a fusion reinforcement sheet;

wherein the anti-stretch layer, the fusion reinforcement sheet and the first auxiliary fusion sheet are connected by sewing.

14. The inflatable product as claimed in claim **13**, further comprising:

a second auxiliary fusion sheet;

wherein the anti-stretch layer, the second auxiliary fusion sheet, the fusion reinforcement sheet and the first auxiliary fusion sheet are connected by sewing.

* * * * *